SUPPLEMENTARY MATERIAL

Thermal adaptation spares the catalytic neighbourhood in bacterial serine proteases

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All tables are generated directly from the analysis pipeline and are also available as a spreadsheet and as tab-separated files in the public repository, together with the code that produced them.

Supplementary Table S1. Pocket versus matched control, AlphaFold models

Within-clade change in each descriptor per +10 °C of optimal growth temperature, in the catalytic neighbourhood (10 Å) and in the size- and burial-matched control patch on the same protein. Three passes are reported: raw descriptor values, values adjusted for local pLDDT, and values standardised by each descriptor's within-clade standard deviation. Intervals are 95% cluster bootstrap over clades; p from within-clade permutation; q by Benjamini-Hochberg.

radius	family	adjustment	descriptor	n	clades	within_sd_pocket	within_sd_control	slope_pocket	slope_pocket_lo	slope_pocket_hi	slope_control	slope_control_lo	slope_control_hi	difference	difference_lo	difference_hi	p_perm	null_sd	q
10	all	raw	n	10674	1485	1.976	1.976	-0.01171	-0.1007	0.0805	-0.01171	-0.1007	0.0805	0	0	0	1	0	
10	all	raw	hydropathy	10674	1485	0.1345	0.1952	-0.003096	-0.00816	0.002217	-0.004802	-0.01245	0.002825	0.001706	-0.007964	0.01133	0.682	0.004384	0.
10	all	raw	resvol	10674	1485	1.875	2.453	0.0116	-0.06659	0.09184	0.243	0.1516	0.3346	-0.2314	-0.3486	-0.1148	0.0005	0.06056	0.00
10	all	raw	f_pos	10674	1485	0.01125	0.01574	0.0004505	5.49e-06	0.0009095	0.00121	0.0005882	0.00184	-0.0007597	-0.001498	-1.39e-05	0.034	0.0003603	0.0
10	all	raw	f_neg	10674	1485	0.0146	0.01871	5.40e-05	-0.0005229	0.0006473	0.001447	0.0006731	0.002257	-0.001393	-0.002404	-0.0004204	0.003	0.0004237	0.
10	all	raw	net_charge	10674	1485	0.7325	0.9147	0.01212	-0.01723	0.04134	-0.005038	-0.04417	0.03413	0.01716	-0.02783	0.06529	0.424	0.02198	0.
10	all	raw	f_aromatic	10674	1485	0.01418	0.0158	-0.0005329	-0.00111	5.30e-05	0.0004548	-0.0002233	0.00117	-0.0009877	-0.001896	-6.21e-05	0.009	0.0003972	0.0
10	all	raw	f_polar	10674	1485	0.02718	0.03044	-0.0002002	-0.001528	0.001036	-0.004099	-0.005354	-0.002804	0.003899	0.001924	0.005628	0	0.0007445	
10	all	raw	f_small	10674	1485	0.02446	0.02735	-0.0004674	-0.001582	0.0006128	-0.002394	-0.003471	-0.001308	0.001926	0.0004819	0.003447	0.005	0.0006962	0.0
10	all	raw	f_gly	10674	1485	0.01481	0.01694	-0.0002649	-0.0008941	0.0003462	-0.0005888	-0.00121	6.35e-05	0.0003239	-0.0005952	0.001203	0.4185	0.000401	0.
10	all	raw	f_pro	10674	1485	0.01037	0.0117	6.35e-06	-0.0004944	0.000506	0.001086	0.0006458	0.001536	-0.00108	-0.001818	-0.0003472	0	0.0002949	
10	all	raw	f_ivywrel	10674	1485	0.02629	0.03154	0.0008708	-0.0004055	0.002152	0.005731	0.004539	0.006942	-0.00486	-0.006439	-0.003299	0	0.0007476	
10	all	raw	sasa_mean	10674	1485	2.408	3.499	0.01613	-0.08628	0.1292	0.08887	-0.05599	0.2392	-0.07274	-0.2074	0.0686	0.321	0.07173	0.
10	all	raw	sasa_total	10674	1485	71.07	136.7	-0.4228	-3.164	2.223	2.251	-2.887	7.821	-2.674	-8.723	3.13	0.3575	2.891	0.
10	all	raw	burial	10674	1485	4.042	5.124	0.01217	-0.2091	0.2031	0.1581	-0.04849	0.36	-0.146	-0.3522	0.06747	0.147	0.1025	0.
10	all	raw	rgyr	10674	1485	0.1603	0.2172	-0.0001643	-0.00671	0.006718	-0.003586	-0.01249	0.005341	0.003421	-0.00483	0.01088	0.3665	0.003773	0.
10	all	raw	plddt	10674	1485	1.853	1.943	0.08042	-0.0008797	0.1553	0.1063	0.02909	0.1843	-0.02585	-0.116	0.06745	0.571	0.04373	0.
10	all	raw	cavity_open_vol	10674	1485	239.2	-	5.055	-5.584	16.25	-	-	-	-	-	-	0.2935	4.884	0.
10	all	raw	cavity_enclosed_vol	10674	1485	37.34	-	0.2978	-1.05	1.762	-	-	-	-	-	-	0.6705	0.7163	0.
10	S1	raw	n	1126	153	1.906	1.906	0.05823	-0.2548	0.3796	0.05823	-0.2548	0.3796	0	0	0	1	0	
10	S1	raw	hydropathy	1126	153	0.1504	0.1795	-0.005934	-0.03208	0.01871	-0.0178	-0.04617	0.01049	0.01187	-0.03067	0.05303	0.474	0.01629	0.
10	S1	raw	resvol	1126	153	2.066	2.511	0.1671	-0.1512	0.4843	0.3659	0.005057	0.733	-0.1988	-0.5969	0.2605	0.396	0.2294	0.
10	S1	raw	f_pos	1126	153	0.01369	0.01803	-0.0005414	-0.002526	0.001892	0.002962	0.0001095	0.005755	-0.003504	-0.006667	-2.44e-05	0.023	0.001537	0.
10	S1	raw	f_neg	1126	153	0.01218	0.0214	0.0005249	-0.001388	0.002838	0.003901	0.0003446	0.00737	-0.003376	-0.007355	0.0005656	0.0605	0.00178	0.
10	S1	raw	net_charge	1126	153	0.7117	1.074	-0.0469	-0.1732	0.08684	-0.03924	-0.244	0.1537	-0.007657	-0.2122	0.2259	0.934	0.09137	
10	S1	raw	f_aromatic	1126	153	0.01765	0.01457	-0.0008174	-0.004333	0.002844	0.0009562	-0.001072	0.002996	-0.001774	-0.005784	0.002465	0.322	0.001772	0.
10	S1	raw	f_polar	1126	153	0.02983	0.0355	0.002899	-0.002814	0.008499	-0.00817	-0.0136	-0.002895	0.01107	0.003804	0.01798	0.0015	0.003323	0.
10	S1	raw	f_small	1126	153	0.02341	0.02901	-0.002592	-0.00579	0.0005765	-0.00367	-0.008265	0.001111	0.001078	-0.003741	0.006003	0.6765	0.00256	0.
10	S1	raw	f_gly	1126	153	0.01628	0.01906	-0.001927	-0.00459	0.0003903	0.001566	-0.001632	0.004619	-0.003493	-0.00725	-8.40e-05	0.055	0.001842	0.
10	S1	raw	f_pro	1126	153	0.01118	0.01307	-0.0002278	-0.001918	0.001523	0.001108	-0.001146	0.003541	-0.001336	-0.004411	0.001476	0.3215	0.001341	0.
10	S1	raw	f_ivywrel	1126	153	0.0286	0.03306	0.001569	-0.003269	0.006652	0.009313	0.00455	0.01393	-0.007743	-0.01443	-0.0005636	0.0095	0.003069	0.0
10	S1	raw	sasa_mean	1126	153	2.229	2.94	0.09126	-0.2505	0.4683	-0.2404	-0.7007	0.2408	0.3317	-0.2124	0.8815	0.149	0.2369	0.
10	S1	raw	sasa_total	1126	153	71.99	112.7	5.811	-5.42	17.44	-6.8	-28.2	13.71	12.61	-9.584	36.34	0.185	9.578	0.
10	S1	raw	burial	1126	153	3.879	4.214	-0.09841	-0.7556	0.5026	0.9236	0.2699	1.558	-1.022	-1.776	-0.227	0.0035	0.3493	0.0
10	S1	raw	rgyr	1126	153	0.1559	0.1868	0.02052	-0.00171	0.04468	0.00155	-0.03227	0.03553	0.01897	-0.01656	0.05888	0.1735	0.01418	0.

Supplementary Table S1 (continued)

radius	family	adjustment	descriptor	n	clades	within_sd_pocket	within_sd_control	slope_pocket	slope_pocket_lo	slope_pocket_hi	slope_control	slope_control_lo	slope_control_hi	difference	difference_lo	difference_hi	p_perm	null_sd	d
10	S1	raw	plddt	1126	153	2.337	2.147	0.01375	-0.3262	0.3221	0.3316	-0.04561	0.7504	-0.3178	-0.8075	0.1351	0.144	0.219	0.
10	S1	raw	cavity_open_vol	1126	153	234	-	0.04331	-45.56	46.33	-	-	-	-	-	-	0.9985	19.28	
10	S1	raw	cavity_enclosed_vol	1126	153	32.52	-	3.11	-2.783	9.481	-	-	-	-	-	-	0.207	2.483	0.
10	S8	raw	n	6216	960	2.307	2.307	-0.03031	-0.1825	0.1151	-0.03031	-0.1825	0.1151	0	0	0	1	0	
10	S8	raw	hydropathy	6216	960	0.1414	0.1883	-0.0007208	-0.008201	0.00682	-0.002199	-0.01179	0.007596	0.001478	-0.01133	0.01374	0.807	0.005991	
10	S8	raw	resvol	6216	960	1.965	2.52	-0.06109	-0.1777	0.05785	0.2638	0.1371	0.388	-0.3249	-0.4962	-0.1399	0	0.08382	
10	S8	raw	f_pos	6216	960	0.01107	0.01538	0.000393	-0.000195	0.0009889	0.001116	0.0003495	0.001888	-0.0007228	-0.001701	0.0002382	0.1175	0.0004687	0.
10	S8	raw	f_neg	6216	960	0.01585	0.01839	-4.34e-05	-0.0009068	0.0008358	0.0008801	-0.0001068	0.002042	-0.0009236	-0.002366	0.0003087	0.1175	0.0005883	0.
10	S8	raw	net_charge	6216	960	0.7887	0.946	0.01454	-0.02735	0.05635	0.01408	-0.04265	0.06922	0.0004557	-0.06374	0.06949	0.9865	0.02962	
10	S8	raw	f_aromatic	6216	960	0.01319	0.01523	-0.000669	-0.00142	3.34e-05	0.0004431	-0.0004947	0.001334	-0.001112	-0.002301	5.06e-05	0.0285	0.000505	0.
10	S8	raw	f_polar	6216	960	0.02896	0.02945	-0.0009967	-0.002904	0.0008236	-0.00281	-0.004347	-0.001235	0.001813	-0.0007058	0.004257	0.0725	0.001008	0.
10	S8	raw	f_small	6216	960	0.02761	0.02828	8.84e-05	-0.001685	0.001881	-0.002662	-0.004103	-0.001189	0.00275	0.0006817	0.004831	0.0055	0.0009877	0.0
10	S8	raw	f_gly	6216	960	0.01514	0.01759	-3.27e-05	-0.000891	0.0008211	-0.001029	-0.001882	-0.0001557	0.000996	-0.0002059	0.002261	0.0905	0.0005709	0.
10	S8	raw	f_pro	6216	960	0.01136	0.01128	-5.56e-05	-0.0007903	0.0006845	0.0007929	0.0002436	0.00134	-0.0008484	-0.001786	0.0001333	0.032	0.0003968	0.
10	S8	raw	f_ivywrel	6216	960	0.02783	0.03136	0.000449	-0.001583	0.002433	0.005472	0.003668	0.007177	-0.005023	-0.007266	-0.002733	0	0.001047	
10	S8	raw	sasa_mean	6216	960	2.77	3.871	0.04206	-0.1287	0.2397	0.2163	-0.0011	0.4489	-0.1742	-0.3749	0.02302	0.0935	0.1038	0.
10	S8	raw	sasa_total	6216	960	78.26	154.8	-0.617	-4.414	3.181	5.96	-2.397	13.76	-6.577	-15.16	2.35	0.1255	4.312	0.
10	S8	raw	burial	6216	960	4.796	5.804	-0.07393	-0.4282	0.2373	-0.08386	-0.4122	0.2216	0.009924	-0.3302	0.3435	0.9475	0.1575	
10	S8	raw	rgyr	6216	960	0.175	0.253	-0.002846	-0.01363	0.007264	-0.005004	-0.0189	0.008591	0.002158	-0.009774	0.01318	0.7075	0.005712	
10	S8	raw	plddt	6216	960	2.062	2.265	0.1222	0.002168	0.242	0.1125	-0.01217	0.24	0.009667	-0.1304	0.1482	0.8835	0.06978	
10	S8	raw	cavity_open_vol	6216	960	284.1	-	13.16	-3.086	31.03	-	-	-	-	-	-	0.091	7.709	0.
10	S8	raw	cavity_enclosed_vol	6216	960	39.91	-	0.03752	-1.945	2.157	-	-	-	-	-	-	0.971	1.007	
10	S9	raw	n	3332	374	1.159	1.159	0.005996	-0.05985	0.07848	0.005996	-0.05985	0.07848	0	0	0	1	0	
10	S9	raw	hydropathy	3332	374	0.114	0.2122	-0.006297	-0.01331	0.000653	-0.006552	-0.02073	0.006963	0.0002553	-0.01543	0.01543	0.9695	0.007025	
10	S9	raw	resvol	3332	374	1.607	2.299	0.1039	0.005858	0.1991	0.1864	0.03242	0.3312	-0.08247	-0.2575	0.1114	0.357	0.08909	0
10	S9	raw	f_pos	3332	374	0.01063	0.01556	0.000725	4.19e-05	0.001459	0.00105	-1.57e-05	0.002141	-0.0003249	-0.001641	0.0009135	0.6245	0.0006197	0.
10	S9	raw	f_neg	3332	374	0.01278	0.0183	0.0001315	-0.0007026	0.0009987	0.001895	0.0007024	0.002985	-0.001763	-0.003176	-0.0003293	0.008	0.000677	0
10	S9	raw	net_charge	3332	374	0.6219	0.7872	0.01877	-0.01817	0.06041	-0.02868	-0.08112	0.02478	0.04745	-0.02232	0.1185	0.124	0.03144	0.
10	S9	raw	f_aromatic	3332	374	0.01431	0.01716	-0.0001963	-0.001039	0.0005949	0.000379	-0.0007656	0.001558	-0.0005753	-0.001966	0.000906	0.406	0.0007005	0
10	S9	raw	f_polar	3332	374	0.02239	0.03038	0.0005096	-0.001081	0.002113	-0.005404	-0.007646	-0.003024	0.005914	0.002896	0.008792	0	0.001144	
10	S9	raw	f_small	3332	374	0.0176	0.02488	-0.0009838	-0.002222	0.0002061	-0.001728	-0.003539	-2.51e-05	0.0007438	-0.001535	0.003033	0.4145	0.0009288	0
10	S9	raw	f_gly	3332	374	0.01362	0.01478	-0.0003331	-0.001218	0.0005608	-0.0002551	-0.001228	0.0007457	-7.81e-05	-0.001463	0.001267	0.905	0.0005959	
10	S9	raw	f_pro	3332	374	0.007836	0.01199	0.0001332	-0.0005972	0.0009007	0.001546	0.0007417	0.002307	-0.001413	-0.002515	-0.0002817	0.002	0.0004671	0.0
10	S9	raw	f_ivywrel	3332	374	0.02217	0.0313	0.001429	-5.88e-05	0.002972	0.005486	0.003748	0.007344	-0.004056	-0.006285	-0.001865	0.0015	0.001163	0.0
10	S9	raw	sasa_mean	3332	374	1.604	2.886	-0.03874	-0.1301	0.04809	-0.05439	-0.2125	0.1286	0.01565	-0.1756	0.1813	0.8585	0.0922	
10	S9	raw	sasa_total	3332	374	54.73	104.3	-1.225	-4.739	2.049	-1.985	-8.814	4.767	0.7602	-6.046	7.707	0.818	3.542	
10	S9	raw	burial	3332	374	2.083	3.897	0.1688	0.06366	0.2835	0.4017	0.1588	0.6392	-0.233	-0.4826	0.02457	0.0575	0.1207	0.
10	S9	raw	rgyr	3332	374	0.1304	0.141	0.000399	-0.007306	0.007442	-0.002168	-0.01087	0.007083	0.002567	-0.006292	0.0113	0.52	0.004057	0.
10	S9	raw	plddt	3332	374	1.095	0.9773	0.02289	-0.06388	0.1031	0.05477	-0.005103	0.1164	-0.03188	-0.116	0.05073	0.397	0.03799	0
10	S9	raw	cavity_open_vol	3332	374	119.6	-	-6.982	-15.61	0.2515	-	-	-	-	-	-	0.0905	4.156	0.
10	S9	raw	cavity_enclosed_vol	3332	374	33.72	-	0.2355	-1.643	2.248	-	-	-	-	-	-	0.807	1.01	
10	all	plddt_adj	n	10674	1485	1.976	1.976	0.0199	-0.0665	0.1125	0.0199	-0.0665	0.1125	0	0	0	1	0	

Supplementary Table S1 (continued)

radius	family	adjustment	descriptor	n	clades	within_sd_pocket	within_sd_control	slope_pocket	slope_pocket_lo	slope_pocket_hi	slope_control	slope_control_lo	slope_control_hi	difference	difference_lo	difference_hi	p_perm	null_sd	d
10	all	plddt_adj	hydropathy	10674	1485	0.1345	0.1952	-0.003075	-0.008315	0.002578	-0.00485	-0.01266	0.003044	0.001775	-0.008104	0.01194	0.689	0.004573	0.
10	all	plddt_adj	resvol	10674	1485	1.875	2.453	0.01949	-0.05927	0.09929	0.2425	0.1485	0.3311	-0.223	-0.3468	-0.09289	0.0005	0.0604	0.0
10	all	plddt_adj	f_pos	10674	1485	0.01125	0.01574	0.0004872	-6.48e-06	0.0009231	0.001211	0.0005845	0.001821	-0.0007237	-0.001519	3.06e-05	0.04	0.0003524	
10	all	plddt_adj	f_neg	10674	1485	0.0146	0.01871	7.78e-05	-0.0005284	0.0006874	0.001453	0.0006663	0.002228	-0.001375	-0.002304	-0.0003928	0.002	0.0004377	0.
10	all	plddt_adj	net_charge	10674	1485	0.7325	0.9147	0.01132	-0.01896	0.04066	-0.005996	-0.04394	0.03129	0.01732	-0.03004	0.06356	0.43	0.02138	0
10	all	plddt_adj	f_aromatic	10674	1485	0.01418	0.0158	-0.0005471	-0.001095	4.26e-05	0.0004475	-0.0002516	0.001131	-0.0009947	-0.001864	-9.57e-05	0.0145	0.0004063	0.0
10	all	plddt_adj	f_polar	10674	1485	0.02718	0.03044	-0.0002436	-0.00158	0.001032	-0.004109	-0.005416	-0.002831	0.003866	0.001995	0.005756	0	0.0007404	
10	all	plddt_adj	f_small	10674	1485	0.02446	0.02735	-0.0005655	-0.001674	0.0005014	-0.002388	-0.003481	-0.001286	0.001823	0.0002828	0.003255	0.009	0.0006774	0
10	all	plddt_adj	f_gly	10674	1485	0.01481	0.01694	-0.0003299	-0.0009157	0.0003068	-0.0005864	-0.001267	4.88e-05	0.0002565	-0.0006132	0.001214	0.5415	0.0004184	0.
10	all	plddt_adj	f_pro	10674	1485	0.01037	0.0117	3.03e-05	-0.0004882	0.0005118	0.00109	0.000628	0.001546	-0.00106	-0.001767	-0.0003908	0	0.0002893	
10	all	plddt_adj	f_ivywrel	10674	1485	0.02629	0.03154	0.0009573	-0.0002876	0.002291	0.00573	0.004553	0.006902	-0.004772	-0.006405	-0.003178	0	0.0007774	
10	all	plddt_adj	sasa_mean	10674	1485	2.408	3.499	0.01491	-0.08877	0.1262	0.08283	-0.06451	0.2321	-0.06792	-0.204	0.06136	0.349	0.0716	0.
10	all	plddt_adj	sasa_total	10674	1485	71.07	136.7	-0.09929	-2.762	2.544	2.997	-2.404	8.311	-3.096	-8.732	2.722	0.284	2.887	0.
10	all	plddt_adj	burial	10674	1485	4.042	5.124	0.02062	-0.1867	0.2104	0.1656	-0.04044	0.365	-0.145	-0.3576	0.06799	0.156	0.1031	0
10	all	plddt_adj	rgyr	10674	1485	0.1603	0.2172	0.0005479	-0.006506	0.007519	-0.001011	-0.009643	0.007932	0.001559	-0.005785	0.008983	0.665	0.003618	0.
10	all	plddt_adj	cavity_open_vol	10674	1485	239.2	-	0.6224	-8.899	11.23	-	-	-	-	-	-	0.888	4.493	0.
10	all	plddt_adj	cavity_enclosed_vol	10674	1485	37.34	-	0.2248	-1.138	1.691	-	-	-	-	-	-	0.7405	0.6953	0.
10	S1	plddt_adj	n	1126	153	1.906	1.906	0.06182	-0.2236	0.3697	0.06182	-0.2236	0.3697	0	0	0	1	0	
10	S1	plddt_adj	hydropathy	1126	153	0.1504	0.1795	-0.005896	-0.03295	0.01819	-0.01781	-0.04584	0.01224	0.01191	-0.03174	0.05103	0.4755	0.01653	0.
10	S1	plddt_adj	resvol	1126	153	2.066	2.511	0.1663	-0.1581	0.5073	0.3659	-0.004249	0.7367	-0.1996	-0.5971	0.2571	0.3805	0.2256	0.
10	S1	plddt_adj	f_pos	1126	153	0.01369	0.01803	-0.0005376	-0.002599	0.001704	0.002964	-7.51e-05	0.005889	-0.003501	-0.006711	8.81e-05	0.0225	0.001559	0.
10	S1	plddt_adj	f_neg	1126	153	0.01218	0.0214	0.0005258	-0.001536	0.002639	0.003906	0.0003091	0.007345	-0.00338	-0.007576	0.0007248	0.0535	0.001766	0.
10	S1	plddt_adj	net_charge	1126	153	0.7117	1.074	-0.04684	-0.1701	0.08671	-0.03938	-0.2229	0.1546	-0.007462	-0.2175	0.209	0.9315	0.09154	
10	S1	plddt_adj	f_aromatic	1126	153	0.01765	0.01457	-0.0008346	-0.004175	0.002876	0.0009468	-0.001209	0.002878	-0.001781	-0.005577	0.002593	0.303	0.001718	0.
10	S1	plddt_adj	f_polar	1126	153	0.02983	0.0355	0.002877	-0.003043	0.008381	-0.008174	-0.01337	-0.002602	0.01105	0.003388	0.0184	0.0015	0.003388	0
10	S1	plddt_adj	f_small	1126	153	0.02341	0.02901	-0.002585	-0.006002	0.0007219	-0.003675	-0.008545	0.0009496	0.001091	-0.003645	0.006049	0.6505	0.002494	0.
10	S1	plddt_adj	f_gly	1126	153	0.01628	0.01906	-0.001926	-0.004559	0.0004824	0.001565	-0.001502	0.004609	-0.003491	-0.007361	-0.0001272	0.0775	0.001956	0.
10	S1	plddt_adj	f_pro	1126	153	0.01118	0.01307	-0.0002224	-0.001863	0.001648	0.001105	-0.001263	0.003576	-0.001327	-0.004341	0.0016	0.345	0.001339	0.
10	S1	plddt_adj	f_ivywrel	1126	153	0.0286	0.03306	0.001569	-0.003022	0.006453	0.009322	0.004774	0.01393	-0.007753	-0.01476	-0.001102	0.0095	0.003051	0
10	S1	plddt_adj	sasa_mean	1126	153	2.229	2.94	0.08945	-0.2385	0.4213	-0.2434	-0.7245	0.2215	0.3328	-0.2131	0.8651	0.161	0.239	0.
10	S1	plddt_adj	sasa_total	1126	153	71.99	112.7	5.806	-5.115	17.79	-6.777	-29.26	13.75	12.58	-11.55	36.7	0.1975	9.917	0
10	S1	plddt_adj	burial	1126	153	3.879	4.214	-0.09312	-0.7027	0.4882	0.9283	0.3162	1.557	-1.021	-1.739	-0.2013	0.004	0.347	0
10	S1	plddt_adj	rgyr	1126	153	0.1559	0.1868	0.0205	-0.001919	0.04492	0.001811	-0.02837	0.03259	0.01869	-0.01548	0.05477	0.1745	0.01383	0.
10	S1	plddt_adj	cavity_open_vol	1126	153	234	-	-0.5647	-39.82	36.2	-	-	-	-	-	-	0.978	16.43	
10	S1	plddt_adj	cavity_enclosed_vol	1126	153	32.52	-	3.105	-2.994	9.108	-	-	-	-	-	-	0.2245	2.53	0.
10	S8	plddt_adj	n	6216	960	2.307	2.307	0.02541	-0.1276	0.1887	0.02541	-0.1276	0.1887	0	0	0	1	0	
10	S8	plddt_adj	hydropathy	6216	960	0.1414	0.1883	-0.0009765	-0.008832	0.006807	-0.002257	-0.01164	0.007414	0.001281	-0.01125	0.01355	0.824	0.005885	
10	S8	plddt_adj	resvol	6216	960	1.965	2.52	-0.04269	-0.1581	0.08289	0.2646	0.1331	0.3918	-0.3073	-0.482	-0.1345	0	0.08028	
10	S8	plddt_adj	f_pos	6216	960	0.01107	0.01538	0.0004556	-0.0001505	0.001068	0.00111	0.0003669	0.001881	-0.0006539	-0.001611	0.000292	0.1585	0.0004572	0.
10	S8	plddt_adj	f_neg	6216	960	0.01585	0.01839	1.82e-05	-0.0009166	0.0009107	0.0008738	-0.00019	0.001967	-0.0008556	-0.002233	0.0003875	0.148	0.0005815	0.
10	S8	plddt_adj	net_charge	6216	960	0.7887	0.946	0.01218	-0.02882	0.05454	0.01261	-0.04026	0.06547	-0.0004266	-0.06476	0.0676	0.9815	0.02912	
10	S8	plddt_adj	f_aromatic	6216	960	0.01319	0.01523	-0.0006524	-0.001409	7.80e-05	0.0004675	-0.0004124	0.001391	-0.00112	-0.002244	-1.96e-05	0.0245	0.0005075	0.

Supplementary Table S1 (continued)

radius	family	adjustment	descriptor	n	clades	within_sd_pocket	within_sd_control	slope_pocket	slope_pocket_lo	slope_pocket_hi	slope_control	slope_control_lo	slope_control_hi	difference	difference_lo	difference_hi	p_perm	null_sd	d
10	S8	plddt_adj	f_polar	6216	960	0.02896	0.02945	-0.001029	-0.002879	0.0008933	-0.002805	-0.004369	-0.001273	0.001776	-0.0007527	0.004293	0.078	0.001012	0.
10	S8	plddt_adj	f_small	6216	960	0.02761	0.02828	-0.0001458	-0.002003	0.001693	-0.00265	-0.00412	-0.001224	0.002504	0.0003379	0.00467	0.0095	0.001007	0
10	S8	plddt_adj	f_gly	6216	960	0.01514	0.01759	-0.0001593	-0.001009	0.0007197	-0.001022	-0.001851	-0.000168	0.0008629	-0.0003953	0.0021	0.12	0.0005646	
10	S8	plddt_adj	f_pro	6216	960	0.01136	0.01128	-1.27e-05	-0.0007108	0.0007283	0.0008024	0.0002558	0.001372	-0.000815	-0.001785	0.000168	0.0445	0.0004048	0.
10	S8	plddt_adj	f_ivywrel	6216	960	0.02783	0.03136	0.0006208	-0.001394	0.002515	0.005446	0.003763	0.007004	-0.004826	-0.007137	-0.002454	0	0.001041	
10	S8	plddt_adj	sasa_mean	6216	960	2.77	3.871	0.03833	-0.1398	0.2254	0.2062	-0.007959	0.4272	-0.1679	-0.3806	0.03391	0.111	0.1055	
10	S8	plddt_adj	sasa_total	6216	960	78.26	154.8	-0.1391	-3.967	3.535	7.287	-1.104	15.27	-7.426	-16.39	1.74	0.081	4.318	0.
10	S8	plddt_adj	burial	6216	960	4.796	5.804	-0.05729	-0.4043	0.253	-0.07353	-0.3959	0.2298	0.01624	-0.3094	0.3437	0.914	0.1581	
10	S8	plddt_adj	rgyr	6216	960	0.175	0.253	-0.00142	-0.01222	0.009573	-0.000506	-0.01514	0.01298	-0.0009142	-0.01154	0.01061	0.865	0.005568	
10	S8	plddt_adj	cavity_open_vol	6216	960	284.1	-	5.77	-9.527	22.16	-	-	-	-	-	-	0.3995	6.949	0.
10	S8	plddt_adj	cavity_enclosed_vol	6216	960	39.91	-	-0.1678	-2.181	1.805	-	-	-	-	-	-	0.8655	0.9924	
10	S9	plddt_adj	n	3332	374	1.159	1.159	0.01025	-0.04935	0.07232	0.01025	-0.04935	0.07232	0	0	0	1	0	
10	S9	plddt_adj	hydropathy	3332	374	0.114	0.2122	-0.006025	-0.01255	0.0006199	-0.006603	-0.02025	0.006518	0.0005781	-0.01478	0.01575	0.935	0.007088	
10	S9	plddt_adj	resvol	3332	374	1.607	2.299	0.1033	0.0103	0.1992	0.1845	0.04674	0.3257	-0.0812	-0.2611	0.09455	0.3685	0.09022	0
10	S9	plddt_adj	f_pos	3332	374	0.01063	0.01556	0.000733	9.02e-05	0.001456	0.001056	-4.37e-05	0.002131	-0.0003231	-0.001583	0.000944	0.6045	0.0006192	0.
10	S9	plddt_adj	f_neg	3332	374	0.01278	0.0183	0.0001143	-0.0007059	0.0009947	0.001908	0.0007916	0.003015	-0.001794	-0.003237	-0.0003707	0.0065	0.0006589	0.0
10	S9	plddt_adj	net_charge	3332	374	0.6219	0.7872	0.01946	-0.01974	0.06169	-0.02903	-0.08168	0.02201	0.04849	-0.01965	0.1181	0.1355	0.03273	0.
10	S9	plddt_adj	f_aromatic	3332	374	0.01431	0.01716	-0.0002166	-0.001064	0.000605	0.0003556	-0.0008795	0.001528	-0.0005721	-0.002014	0.000913	0.392	0.0006877	0
10	S9	plddt_adj	f_polar	3332	374	0.02239	0.03038	0.0004906	-0.001039	0.002066	-0.005428	-0.007775	-0.002983	0.005919	0.002952	0.008743	0	0.001111	
10	S9	plddt_adj	f_small	3332	374	0.0176	0.02488	-0.0009684	-0.002177	0.0002757	-0.001717	-0.003489	3.83e-06	0.0007482	-0.00156	0.00316	0.442	0.0009791	0.
10	S9	plddt_adj	f_gly	3332	374	0.01362	0.01478	-0.0003489	-0.001267	0.0005909	-0.0002572	-0.001247	0.0007211	-9.17e-05	-0.001465	0.001279	0.871	0.0005856	0.
10	S9	plddt_adj	f_pro	3332	374	0.007836	0.01199	0.0001298	-0.0005762	0.0008778	0.001553	0.000766	0.002317	-0.001423	-0.002491	-0.0003668	0.001	0.0004527	0
10	S9	plddt_adj	f_ivywrel	3332	374	0.02217	0.0313	0.001442	-0.000115	0.00298	0.005493	0.003589	0.007274	-0.004051	-0.00626	-0.001846	0	0.001165	
10	S9	plddt_adj	sasa_mean	3332	374	1.604	2.886	-0.03249	-0.1207	0.057	-0.0499	-0.2191	0.1314	0.01741	-0.1659	0.1988	0.85	0.09393	0.
10	S9	plddt_adj	sasa_total	3332	374	54.73	104.3	-0.9536	-4.285	2.01	-1.737	-8.441	5.169	0.7831	-5.927	7.478	0.8165	3.457	0.
10	S9	plddt_adj	burial	3332	374	2.083	3.897	0.1565	0.05421	0.2771	0.3965	0.1659	0.6262	-0.24	-0.4859	0.01886	0.048	0.1209	0
10	S9	plddt_adj	rgyr	3332	374	0.1304	0.141	0.0005432	-0.007338	0.00765	-0.001693	-0.009697	0.006762	0.002236	-0.006589	0.01063	0.5775	0.004055	0.
10	S9	plddt_adj	cavity_open_vol	3332	374	119.6	-	-7.83	-14.58	-1.377	-	-	-	-	-	-	0.0435	3.679	0
10	S9	plddt_adj	cavity_enclosed_vol	3332	374	33.72	-	0.3088	-1.75	2.414	-	-	-	-	-	-	0.7675	1.008	0.
10	all	standardized	n	10674	1485	1.976	1.976	-0.005928	-0.05395	0.03857	-0.005928	-0.05395	0.03857	0	0	0	1	0	
10	all	standardized	hydropathy	10674	1485	0.1345	0.1952	-0.02302	-0.06181	0.01536	-0.0246	-0.06482	0.01431	0.001581	-0.05359	0.06053	0.9465	0.02594	0.
10	all	standardized	resvol	10674	1485	1.875	2.453	0.006185	-0.03506	0.04648	0.09904	0.06065	0.1374	-0.09285	-0.151	-0.03759	0.001	0.02686	0.0
10	all	standardized	f_pos	10674	1485	0.01125	0.01574	0.04005	-0.000422	0.07876	0.0769	0.03734	0.1162	-0.03685	-0.09215	0.01468	0.157	0.02609	0.
10	all	standardized	f_neg	10674	1485	0.0146	0.01871	0.003697	-0.03695	0.04661	0.07735	0.03598	0.1195	-0.07365	-0.1347	-0.01538	0.007	0.02573	0.0
10	all	standardized	net_charge	10674	1485	0.7325	0.9147	0.01655	-0.02267	0.05824	-0.005508	-0.0492	0.03278	0.02206	-0.03151	0.08395	0.3925	0.02554	0.
10	all	standardized	f_aromatic	10674	1485	0.01418	0.0158	-0.03759	-0.07906	0.001992	0.02878	-0.01306	0.07046	-0.06637	-0.1259	-0.005742	0.011	0.02673	0.0
10	all	standardized	f_polar	10674	1485	0.02718	0.03044	-0.007364	-0.05577	0.03752	-0.1347	-0.1801	-0.08895	0.1273	0.05967	0.1928	0	0.02593	
10	all	standardized	f_small	10674	1485	0.02446	0.02735	-0.01911	-0.06701	0.02726	-0.08752	-0.1301	-0.04744	0.06841	0.0107	0.1284	0.005	0.0256	0
10	all	standardized	f_gly	10674	1485	0.01481	0.01694	-0.01788	-0.05908	0.02475	-0.03476	-0.07392	0.002505	0.01688	-0.03861	0.07591	0.5265	0.02583	0.
10	all	standardized	f_pro	10674	1485	0.01037	0.0117	0.000612	-0.04752	0.04916	0.09282	0.0545	0.1316	-0.09221	-0.1554	-0.02959	0	0.02723	
10	all	standardized	f_ivywrel	10674	1485	0.02629	0.03154	0.03312	-0.01522	0.08181	0.1817	0.1439	0.2208	-0.1486	-0.2088	-0.091	0	0.02534	
10	all	standardized	sasa_mean	10674	1485	2.408	3.499	0.0067	-0.03768	0.05406	0.0254	-0.01247	0.06878	-0.0187	-0.06577	0.02675	0.4265	0.02317	0.

Supplementary Table S1 (continued)

radius	family	adjustment	descriptor	n	clades	within_sd_pocket	within_sd_control	slope_pocket	slope_pocket_lo	slope_pocket_hi	slope_control	slope_control_lo	slope_control_hi	difference	difference_lo	difference_hi	p_perm	null_sd	d
10	all	standardized	sasa_total	10674	1485	71.07	136.7	-0.00595	-0.04253	0.03052	0.01646	-0.0225	0.05457	-0.02241	-0.07375	0.03003	0.3815	0.02587	0.
10	all	standardized	burial	10674	1485	4.042	5.124	0.00301	-0.04668	0.04997	0.03087	-0.0103	0.06889	-0.02786	-0.07323	0.02174	0.22	0.02301	0.
10	all	standardized	rgyr	10674	1485	0.1603	0.2172	-0.001025	-0.04491	0.04184	-0.01651	-0.0572	0.02339	0.01548	-0.02427	0.05615	0.4225	0.01942	0.
10	all	standardized	plddt	10674	1485	1.853	1.943	0.0434	0.0003208	0.09007	0.0547	0.01424	0.09233	-0.0113	-0.05914	0.03743	0.62	0.02333	0.
10	all	standardized	cavity_open_vol	10674	1485	239.2	-	0.02113	-0.02383	0.06546	-	-	-	-	-	-	0.302	0.0199	0.
10	all	standardized	cavity_enclosed_vol	10674	1485	37.34	-	0.007976	-0.02739	0.0447	-	-	-	-	-	-	0.6935	0.01902	0.
10	S1	standardized	n	1126	153	1.906	1.906	0.03055	-0.1362	0.1998	0.03055	-0.1362	0.1998	0	0	0	1	0	
10	S1	standardized	hydropathy	1126	153	0.1504	0.1795	-0.03946	-0.2133	0.1118	-0.0992	-0.2571	0.04334	0.05974	-0.181	0.2883	0.538	0.09848	0.
10	S1	standardized	resvol	1126	153	2.066	2.511	0.0809	-0.06707	0.2339	0.1457	0.002545	0.2921	-0.06482	-0.2447	0.13	0.5395	0.1035	0.
10	S1	standardized	f_pos	1126	153	0.01369	0.01803	-0.03956	-0.1922	0.1193	0.1643	0.002529	0.3296	-0.2039	-0.4017	0.001009	0.0365	0.09705	0.
10	S1	standardized	f_neg	1126	153	0.01218	0.0214	0.0431	-0.128	0.2183	0.1823	0.01098	0.3493	-0.1392	-0.3721	0.1157	0.1685	0.1006	0.
10	S1	standardized	net_charge	1126	153	0.7117	1.074	-0.06589	-0.2482	0.1195	-0.03654	-0.2287	0.144	-0.02935	-0.2596	0.2147	0.7615	0.09765	0.
10	S1	standardized	f_aromatic	1126	153	0.01765	0.01457	-0.04631	-0.2497	0.1594	0.06563	-0.07882	0.1957	-0.1119	-0.3493	0.1434	0.3095	0.1085	
10	S1	standardized	f_polar	1126	153	0.02983	0.0355	0.09718	-0.08531	0.2838	-0.2302	-0.3776	-0.0785	0.3273	0.09253	0.5415	0.002	0.1012	0
10	S1	standardized	f_small	1126	153	0.02341	0.02901	-0.1107	-0.2502	0.02675	-0.1265	-0.2969	0.03718	0.0158	-0.1669	0.2038	0.864	0.09477	0.
10	S1	standardized	f_gly	1126	153	0.01628	0.01906	-0.1183	-0.2795	0.02427	0.08217	-0.07604	0.2354	-0.2005	-0.4184	-0.005326	0.066	0.1104	0.
10	S1	standardized	f_pro	1126	153	0.01118	0.01307	-0.02039	-0.166	0.1459	0.08477	-0.1023	0.2715	-0.1052	-0.3438	0.1386	0.353	0.1129	0.
10	S1	standardized	f_ivywrel	1126	153	0.0286	0.03306	0.05488	-0.1155	0.2169	0.2817	0.1362	0.4312	-0.2268	-0.4465	0.003515	0.0245	0.09768	0.
10	S1	standardized	sasa_mean	1126	153	2.229	2.94	0.04093	-0.1117	0.2194	-0.08178	-0.2511	0.09024	0.1227	-0.08466	0.333	0.1595	0.08739	0.
10	S1	standardized	sasa_total	1126	153	71.99	112.7	0.08073	-0.07568	0.2479	-0.06032	-0.2473	0.1367	0.141	-0.09847	0.3875	0.165	0.1035	0.
10	S1	standardized	burial	1126	153	3.879	4.214	-0.02537	-0.1993	0.1224	0.2192	0.069	0.3802	-0.2445	-0.4435	-0.05691	0.007	0.08738	0.
10	S1	standardized	rgyr	1126	153	0.1559	0.1868	0.1316	-0.01394	0.2767	0.008294	-0.1672	0.1947	0.1233	-0.08322	0.3365	0.137	0.08381	0.
10	S1	standardized	plddt	1126	153	2.337	2.147	0.005884	-0.1597	0.145	0.1545	-0.0296	0.3534	-0.1486	-0.3666	0.05344	0.1205	0.09647	0.
10	S1	standardized	cavity_open_vol	1126	153	234	-	0.0001851	-0.186	0.2053	-	-	-	-	-	-	0.9975	0.08129	
10	S1	standardized	cavity_enclosed_vol	1126	153	32.52	-	0.09562	-0.08705	0.2823	-	-	-	-	-	-	0.21	0.07656	0.
10	S8	standardized	n	6216	960	2.307	2.307	-0.01314	-0.07947	0.05439	-0.01314	-0.07947	0.05439	0	0	0	1	0	
10	S8	standardized	hydropathy	6216	960	0.1414	0.1883	-0.005098	-0.06322	0.04851	-0.01168	-0.06793	0.03845	0.00658	-0.06941	0.08827	0.858	0.0359	
10	S8	standardized	resvol	6216	960	1.965	2.52	-0.0311	-0.09217	0.027	0.1047	0.05278	0.1525	-0.1358	-0.2095	-0.05776	0	0.036	
10	S8	standardized	f_pos	6216	960	0.01107	0.01538	0.03549	-0.02103	0.08734	0.07254	0.02134	0.1216	-0.03705	-0.1119	0.03498	0.2775	0.03413	0.
10	S8	standardized	f_neg	6216	960	0.01585	0.01839	-0.00274	-0.05655	0.05088	0.04786	-0.009858	0.1077	-0.0506	-0.1325	0.02207	0.133	0.03369	0.
10	S8	standardized	net_charge	6216	960	0.7887	0.946	0.01843	-0.03267	0.07247	0.01489	-0.04254	0.06983	0.003547	-0.06957	0.07729	0.907	0.03302	
10	S8	standardized	f_aromatic	6216	960	0.01319	0.01523	-0.05071	-0.1071	0.007261	0.02909	-0.03312	0.08633	-0.0798	-0.1605	0.003549	0.027	0.03639	0.
10	S8	standardized	f_polar	6216	960	0.02896	0.02945	-0.03442	-0.09845	0.03054	-0.09542	-0.1498	-0.04408	0.061	-0.02534	0.1493	0.0845	0.03513	0.
10	S8	standardized	f_small	6216	960	0.02761	0.02828	0.0032	-0.0603	0.06809	-0.09411	-0.1479	-0.04248	0.09731	0.0267	0.1755	0.009	0.0362	0
10	S8	standardized	f_gly	6216	960	0.01514	0.01759	-0.002162	-0.05887	0.04906	-0.05848	-0.1097	-0.009622	0.05631	-0.01929	0.1329	0.107	0.03462	0.
10	S8	standardized	f_pro	6216	960	0.01136	0.01128	-0.004892	-0.06592	0.05673	0.0703	0.02117	0.1191	-0.07519	-0.1591	0.01194	0.0345	0.03633	0.
10	S8	standardized	f_ivywrel	6216	960	0.02783	0.03136	0.01613	-0.04927	0.08728	0.1745	0.1179	0.2299	-0.1583	-0.2343	-0.0808	0	0.03711	
10	S8	standardized	sasa_mean	6216	960	2.77	3.871	0.01518	-0.04475	0.08493	0.05587	0.002563	0.1108	-0.04068	-0.1039	0.0175	0.1845	0.03039	0
10	S8	standardized	sasa_total	6216	960	78.26	154.8	-0.007885	-0.05807	0.03882	0.0385	-0.01495	0.08824	-0.04638	-0.1151	0.02369	0.198	0.03618	0
10	S8	standardized	burial	6216	960	4.796	5.804	-0.01542	-0.08806	0.04797	-0.01445	-0.07587	0.03925	-0.0009672	-0.06524	0.06166	0.9725	0.02969	
10	S8	standardized	rgyr	6216	960	0.175	0.253	-0.01627	-0.07739	0.04363	-0.01978	-0.07431	0.03493	0.003515	-0.04949	0.0543	0.882	0.02452	
10	S8	standardized	plddt	6216	960	2.062	2.265	0.05925	0.001924	0.1193	0.04968	-0.00285	0.1033	0.009563	-0.05088	0.07201	0.7495	0.0328	

Supplementary Table S1 (continued)

radius	family	adjustment	descriptor	n	clades	within_sd_pocket	within_sd_control	slope_pocket	slope_pocket_lo	slope_pocket_hi	slope_control	slope_control_lo	slope_control_hi	difference	difference_lo	difference_hi	p_perm	null_sd	d
10	S8	standardized	cavity_open_vol	6216	960	284.1	-	0.04631	-0.01573	0.1086	-	-	-	-	-	-	0.092	0.02736	0.
10	S8	standardized	cavity_enclosed_vol	6216	960	39.91	-	0.00094	-0.04819	0.04977	-	-	-	-	-	-	0.967	0.02423	
10	S9	standardized	n	3332	374	1.159	1.159	0.005175	-0.05275	0.06056	0.005175	-0.05275	0.06056	0	0	0	1	0	
10	S9	standardized	hydropathy	3332	374	0.114	0.2122	-0.05522	-0.1119	0.002662	-0.03088	-0.09633	0.03144	-0.02434	-0.1094	0.07103	0.5605	0.04191	
10	S9	standardized	resvol	3332	374	1.607	2.299	0.06467	0.007785	0.1277	0.08105	0.01617	0.1428	-0.01638	-0.103	0.07481	0.6895	0.04275	
10	S9	standardized	f_pos	3332	374	0.01063	0.01556	0.06823	0.007257	0.1372	0.06749	-0.004559	0.1354	0.0007384	-0.08787	0.09027	0.9865	0.0457	
10	S9	standardized	f_neg	3332	374	0.01278	0.0183	0.01028	-0.05172	0.07312	0.1036	0.041	0.162	-0.09327	-0.1812	-0.0001219	0.0375	0.04414	0.
10	S9	standardized	net_charge	3332	374	0.6219	0.7872	0.03019	-0.03088	0.09319	-0.03643	-0.1001	0.02819	0.06662	-0.02995	0.1627	0.1515	0.0459	0.
10	S9	standardized	f_aromatic	3332	374	0.01431	0.01716	-0.01372	-0.07512	0.0461	0.02208	-0.04563	0.08876	-0.0358	-0.1252	0.05168	0.4155	0.04416	0.
10	S9	standardized	f_polar	3332	374	0.02239	0.03038	0.02276	-0.04611	0.09048	-0.1779	-0.2512	-0.1007	0.2006	0.08678	0.3038	0	0.04317	
10	S9	standardized	f_small	3332	374	0.0176	0.02488	-0.0559	-0.1298	0.01073	-0.06944	-0.1413	-0.003143	0.01354	-0.09326	0.1225	0.754	0.04427	
10	S9	standardized	f_gly	3332	374	0.01362	0.01478	-0.02445	-0.08949	0.04259	-0.01726	-0.08347	0.04684	-0.007196	-0.1036	0.08954	0.864	0.04234	
10	S9	standardized	f_pro	3332	374	0.007836	0.01199	0.017	-0.07116	0.1106	0.129	0.06472	0.1934	-0.112	-0.223	0.0001479	0.011	0.04457	0.0
10	S9	standardized	f_ivywrel	3332	374	0.02217	0.0313	0.06445	-0.005843	0.133	0.1753	0.1196	0.233	-0.1108	-0.1947	-0.02453	0.011	0.04307	0.0
10	S9	standardized	sasa_mean	3332	374	1.604	2.886	-0.02415	-0.07807	0.02665	-0.01885	-0.07601	0.04433	-0.005303	-0.07968	0.06399	0.897	0.03909	
10	S9	standardized	sasa_total	3332	374	54.73	104.3	-0.02238	-0.08307	0.03707	-0.01903	-0.08496	0.04184	-0.00335	-0.08193	0.07259	0.927	0.04168	
10	S9	standardized	burial	3332	374	2.083	3.897	0.08102	0.02913	0.1357	0.1031	0.04076	0.1619	-0.02207	-0.09632	0.05967	0.5815	0.03946	
10	S9	standardized	rgyr	3332	374	0.1304	0.141	0.003059	-0.05937	0.0582	-0.01538	-0.07854	0.04996	0.01844	-0.05236	0.08214	0.5605	0.03165	
10	S9	standardized	plddt	3332	374	1.095	0.9773	0.0209	-0.06012	0.09458	0.05604	-0.008009	0.1176	-0.03514	-0.117	0.04198	0.315	0.03472	0
10	S9	standardized	cavity_open_vol	3332	374	119.6	-	-0.05837	-0.1287	0.001951	-	-	-	-	-	-	0.0975	0.03512	0.
10	S9	standardized	cavity_enclosed_vol	3332	374	33.72	-	0.006986	-0.05227	0.0676	-	-	-	-	-	-	0.8115	0.02965	

Supplementary Table S2. Pocket versus matched control, ESMFold models

As Supplementary Table S1, computed on structures predicted with ESMFold for sequences that AlphaFold DB does not cover. The two structure sets barely overlap.

radius	family	adjustment	descriptor	n	clades	within_sd_pocket	within_sd_control	slope_pocket	slope_pocket_lo	slope_pocket_hi	slope_control	slope_control_lo	slope_control_hi	difference	difference_lo	difference_hi	p_perm	null_sd	q
10	all	raw	n	5445	390	2.031	2.031	-0.004254	-0.1213	0.114	-0.004254	-0.1213	0.114	0	0	0	1	0	
10	all	raw	hydropathy	5445	390	0.1447	0.1944	-0.0011	-0.009377	0.008322	-0.01434	-0.02518	-0.003117	0.01324	-0.0005979	0.02764	0.0195	0.005615	0.05
10	all	raw	resvol	5445	390	2.065	2.608	-0.03046	-0.1509	0.1019	0.179	0.04605	0.3123	-0.2094	-0.3987	-0.0159	0.015	0.08156	0.05
10	all	raw	f_pos	5445	390	0.01195	0.01704	-0.0001768	-0.0008888	0.0005855	0.002437	0.001549	0.003321	-0.002614	-0.003811	-0.001315	0	0.0004878	
10	all	raw	f_neg	5445	390	0.01465	0.01992	-0.0005468	-0.001595	0.0004297	0.001289	0.0002612	0.002337	-0.001836	-0.003257	-0.0004909	0.0005	0.0005611	0.002
10	all	raw	net_charge	5445	390	0.7633	1.019	0.01411	-0.0348	0.06536	0.05083	-0.002646	0.1035	-0.03671	-0.1104	0.04569	0.2225	0.0306	0.3
10	all	raw	f_aromatic	5445	390	0.01632	0.01579	0.0002572	-0.0007685	0.001276	0.0007636	2.22e-05	0.001528	-0.0005064	-0.001937	0.0007947	0.341	0.0005412	0.4
10	all	raw	f_polar	5445	390	0.02976	0.03216	0.001273	-0.001028	0.003265	-0.004354	-0.005943	-0.002678	0.005627	0.003214	0.007957	0	0.0009597	
10	all	raw	f_small	5445	390	0.02601	0.02948	0.0003387	-0.001266	0.002005	-0.001356	-0.003075	0.0002795	0.001694	-0.0004487	0.003877	0.0685	0.0009056	0.1
10	all	raw	f_gly	5445	390	0.01566	0.0185	-0.000312	-0.001165	0.0005533	-3.21e-05	-0.0009717	0.0008737	-0.0002799	-0.001543	0.0009836	0.5875	0.0005278	0.7
10	all	raw	f_pro	5445	390	0.01124	0.01225	4.12e-05	-0.0006499	0.0007306	0.0009437	0.0003769	0.001469	-0.0009024	-0.001763	-3.21e-05	0.0165	0.0003637	0.05
10	all	raw	f_ivywrel	5445	390	0.02819	0.03325	0.001009	-0.0005519	0.002819	0.004942	0.003163	0.006657	-0.003933	-0.006167	-0.001466	0.0005	0.001018	0.002
10	all	raw	sasa_mean	5445	390	2.308	3.587	0.05949	-0.0511	0.1828	0.03288	-0.1638	0.2316	0.02661	-0.1837	0.2335	0.769	0.08967	0.8
10	all	raw	sasa_total	5445	390	78.81	142.4	1.654	-2.069	5.696	0.7684	-7.076	8.208	0.8856	-7.884	9.437	0.8095	3.746	0.8
10	all	raw	burial	5445	390	3.877	5.145	-0.007462	-0.2485	0.2307	0.1996	-0.0939	0.4774	-0.2071	-0.5118	0.09639	0.1235	0.1334	0.2
10	all	raw	rgyr	5445	390	0.1759	0.2041	-0.0006855	-0.009159	0.00767	-0.001632	-0.01223	0.008378	0.0009462	-0.007474	0.00941	0.829	0.004143	0.8
10	all	raw	plddt	5445	390	1.647	2.24	0.03558	-0.06337	0.1253	0.1285	0.02351	0.2324	-0.09291	-0.2164	0.02748	0.1255	0.06012	0.2
10	all	raw	cavity_open_vol	5445	390	250.9	-	10.19	-5.233	25.49	-	-	-	-	-	-	0.1445	6.94	0.2
10	all	raw	cavity_enclosed_vol	5445	390	45.6	-	-0.5794	-2.605	1.552	-	-	-	-	-	-	0.598	1.058	0.7
10	S1	raw	n	1343	74	1.542	1.542	0.1243	-0.04718	0.3051	0.1243	-0.04718	0.3051	0	0	0	1	0	
10	S1	raw	hydropathy	1343	74	0.1477	0.1861	-0.00649	-0.02795	0.01468	-0.01881	-0.04017	0.005143	0.01232	-0.01957	0.04211	0.391	0.01418	0.5
10	S1	raw	resvol	1343	74	1.916	2.64	0.04683	-0.1622	0.2564	0.395	0.03185	0.7315	-0.3481	-0.7146	0.03875	0.0685	0.1943	0.1
10	S1	raw	f_pos	1343	74	0.01302	0.0192	-0.0005738	-0.002554	0.001488	0.005037	0.002808	0.007481	-0.00561	-0.008808	-0.002544	0	0.001369	
10	S1	raw	f_neg	1343	74	0.01047	0.02288	0.0004849	-0.001305	0.002475	0.002765	-0.0001976	0.005583	-0.00228	-0.005689	0.001099	0.1405	0.001563	0.
10	S1	raw	net_charge	1343	74	0.6239	1.129	-0.03812	-0.1364	0.06664	0.08345	-0.06446	0.2365	-0.1216	-0.2985	0.05971	0.1215	0.08108	0.2
10	S1	raw	f_aromatic	1343	74	0.01743	0.0155	-0.0002626	-0.002922	0.002455	-0.0004657	-0.00197	0.0009976	0.0002031	-0.002482	0.003034	0.887	0.001447	0.9
10	S1	raw	f_polar	1343	74	0.02929	0.03779	0.001502	-0.003274	0.006468	-0.007158	-0.01192	-0.002237	0.008659	0.00303	0.01396	0.001	0.002793	0.006
10	S1	raw	f_small	1343	74	0.02276	0.03065	-0.0001245	-0.003111	0.002928	-0.004338	-0.008899	0.0003355	0.004214	-0.001561	0.009985	0.0595	0.002273	0.1
10	S1	raw	f_gly	1343	74	0.0169	0.02056	-0.001039	-0.003203	0.0009434	0.000831	-0.00205	0.003787	-0.00187	-0.005189	0.001469	0.209	0.001491	0.
10	S1	raw	f_pro	1343	74	0.01055	0.01413	2.24e-05	-0.001672	0.001755	-0.0007292	-0.002323	0.0007388	0.0007515	-0.001513	0.003142	0.483	0.001067	0.6
10	S1	raw	f_ivywrel	1343	74	0.02772	0.03474	-0.000105	-0.003834	0.004153	0.009719	0.004688	0.01458	-0.009824	-0.01576	-0.003137	0.0005	0.002666	0.00
10	S1	raw	sasa_mean	1343	74	2.288	2.963	0.04567	-0.2459	0.3066	-0.002682	-0.3716	0.346	0.04835	-0.388	0.4645	0.8195	0.2141	0.9
10	S1	raw	sasa_total	1343	74	73.79	119.2	6.427	-2.682	15.5	4.991	-8.917	19.62	1.436	-15.56	17.3	0.85	8.169	0.9
10	S1	raw	burial	1343	74	3.466	4.205	0.0648	-0.4375	0.5695	0.6682	0.132	1.176	-0.6034	-1.094	-0.03329	0.0615	0.3253	0.1
10	S1	raw	rgyr	1343	74	0.1536	0.1577	0.01293	-0.003754	0.0314	0.003241	-0.01334	0.02346	0.009688	-0.01027	0.02874	0.2765	0.008858	0.4
10	S1	raw	plddt	1343	74	1.039	1.976	0.1241	-0.04549	0.2893	0.4138	0.08913	0.7619	-0.2897	-0.6102	0.01676	0.033	0.1352	0.1

Supplementary Table S2 (continued)

radius	family	adjustment	descriptor	n	clades	within_sd_pocket	within_sd_control	slope_pocket	slope_pocket_lo	slope_pocket_hi	slope_control	slope_control_lo	slope_control_hi	difference	difference_lo	difference_hi	p_perm	null_sd	q
10	S1	raw	cavity_open_vol	1343	74	147.6	-	6.365	-14.06	26.26	-	-	-	-	-	-	0.55	10.35	0.6
10	S1	raw	cavity_enclosed_vol	1343	74	29.87	-	-3.429	-7.715	0.4882	-	-	-	-	-	-	0.073	1.896	0.1
10	S8	raw	n	3147	256	2.36	2.36	-0.007466	-0.1827	0.1632	-0.007466	-0.1827	0.1632	0	0	0	1	0	
10	S8	raw	hydropathy	3147	256	0.1441	0.1918	0.002886	-0.007538	0.0139	-0.0114	-0.02617	0.003142	0.01429	-0.005158	0.03342	0.0465	0.00715	0.2
10	S8	raw	resvol	3147	256	2.165	2.639	-0.004591	-0.1712	0.1793	0.1493	-0.03838	0.3149	-0.1539	-0.4087	0.1237	0.135	0.103	0.3
10	S8	raw	f_pos	3147	256	0.01121	0.01639	6.56e-05	-0.0006479	0.0008088	0.002329	0.001273	0.003399	-0.002263	-0.003543	-0.0009441	0	0.0005938	
10	S8	raw	f_neg	3147	256	0.01536	0.01898	-0.000347	-0.001392	0.0008027	0.0004076	-0.0008874	0.001801	-0.0007546	-0.002269	0.0007024	0.2755	0.0006966	0.6
10	S8	raw	net_charge	3147	256	0.7977	1.014	0.01333	-0.03756	0.06476	0.08308	0.01512	0.1448	-0.06975	-0.142	0.009807	0.0635	0.03782	0.2
10	S8	raw	f_aromatic	3147	256	0.01512	0.01482	0.0005887	-0.0004704	0.001711	0.0005638	-0.0003879	0.001382	2.50e-05	-0.001318	0.001466	0.9715	0.0006363	
10	S8	raw	f_polar	3147	256	0.03056	0.02971	-0.0009562	-0.003652	0.001627	-0.003674	-0.00576	-0.001458	0.002718	-0.0003744	0.005889	0.035	0.001249	0.2
10	S8	raw	f_small	3147	256	0.02887	0.02984	0.0003303	-0.002088	0.002686	-0.0007716	-0.002932	0.001437	0.001102	-0.002086	0.003813	0.3725	0.001239	0.7
10	S8	raw	f_gly	3147	256	0.01568	0.01838	-2.94e-05	-0.001261	0.001162	-0.0003751	-0.001641	0.0008172	0.0003457	-0.001302	0.002084	0.6045	0.0006699	0.8
10	S8	raw	f_pro	3147	256	0.01249	0.01149	0.0001539	-0.0008127	0.001036	0.001193	0.0005341	0.001816	-0.001039	-0.00221	8.71e-05	0.0385	0.000484	0.2
10	S8	raw	f_ivywrel	3147	256	0.02853	0.03291	0.002009	1.06e-05	0.004014	0.004414	0.002239	0.006509	-0.002405	-0.005168	0.0004429	0.071	0.001299	0.2
10	S8	raw	sasa_mean	3147	256	2.451	3.978	0.1304	-0.03008	0.3072	0.08424	-0.1919	0.3914	0.04619	-0.2657	0.3303	0.7025	0.1247	0.9
10	S8	raw	sasa_total	3147	256	82.68	157.8	3.519	-1.859	8.893	2.031	-8.618	13.7	1.487	-12.33	13.95	0.7675	5.305	0.9
10	S8	raw	burial	3147	256	4.397	5.783	-0.06376	-0.4257	0.2644	-0.01404	-0.4426	0.4037	-0.04971	-0.5048	0.4639	0.79	0.182	0.9
10	S8	raw	rgyr	3147	256	0.1859	0.234	-0.004074	-0.01585	0.007165	-0.0002782	-0.01568	0.01473	-0.003796	-0.01598	0.00732	0.504	0.005874	0.8
10	S8	raw	plddt	3147	256	1.834	2.325	-0.02832	-0.1629	0.1022	0.02591	-0.1037	0.1615	-0.05423	-0.2344	0.1124	0.5085	0.08315	0.8
10	S8	raw	cavity_open_vol	3147	256	306.6	-	10.4	-12.49	34.29	-	-	-	-	-	-	0.32	10.39	0.6
10	S8	raw	cavity_enclosed_vol	3147	256	49.63	-	-0.2793	-2.778	2.358	-	-	-	-	-	-	0.857	1.472	0.9
10	S9	raw	n	955	61	1.343	1.343	-0.07217	-0.3088	0.08737	-0.07217	-0.3088	0.08737	0	0	0	1	0	
10	S9	raw	hydropathy	955	61	0.1421	0.2131	-0.008229	-0.02698	0.01551	-0.01885	-0.04082	0.004271	0.01062	-0.01307	0.04078	0.365	0.01186	0.4
10	S9	raw	resvol	955	61	1.893	2.451	-0.1341	-0.3387	0.1997	0.1268	-0.1071	0.3639	-0.2609	-0.6197	0.2002	0.082	0.1514	0.1
10	S9	raw	f_pos	955	61	0.0126	0.01587	-0.0005074	-0.002187	0.001794	0.001121	-0.001514	0.003026	-0.001628	-0.004934	0.002573	0.1235	0.001055	0.2
10	S9	raw	f_neg	955	61	0.0171	0.01841	-0.001684	-0.005046	0.0007103	0.00261	0.0009399	0.004528	-0.004295	-0.008981	-0.0007811	0.001	0.00121	0.6
10	S9	raw	net_charge	955	61	0.8219	0.8598	0.0496	-0.07479	0.2281	-0.05043	-0.1764	0.04317	0.1	-0.09877	0.3869	0.0975	0.06038	0.2
10	S9	raw	f_aromatic	955	61	0.01777	0.01894	-0.0001332	-0.002464	0.002843	0.002054	0.0001105	0.004512	-0.002187	-0.006181	0.001686	0.081	0.001226	0.1
10	S9	raw	f_polar	955	61	0.02767	0.03128	0.006872	0.003577	0.01	-0.004373	-0.007663	0.0001481	0.01125	0.005859	0.01551	0	0.001909	
10	S9	raw	f_small	955	61	0.01954	0.02636	0.0006407	-0.001172	0.00288	-0.001074	-0.003455	0.001498	0.001715	-0.001853	0.005366	0.297	0.001633	0.4
10	S9	raw	f_gly	955	61	0.01369	0.01562	-0.0005951	-0.001967	0.001104	0.0003236	-0.001327	0.001963	-0.0009187	-0.002914	0.001512	0.3515	0.000978	0.4
10	S9	raw	f_pro	955	61	0.007055	0.01179	-0.0002629	-0.001064	0.0004347	0.001356	4.74e-05	0.00258	-0.001619	-0.003069	-0.0001796	0.0065	0.0006056	0.04
10	S9	raw	f_ivywrel	955	61	0.02764	0.03218	-0.0008288	-0.003694	0.00339	0.003364	-0.0001197	0.007314	-0.004193	-0.008251	0.001308	0.03	0.001884	0.1
10	S9	raw	sasa_mean	955	61	1.794	2.972	-0.1129	-0.249	0.09228	-0.07217	-0.3897	0.2003	-0.04075	-0.341	0.3412	0.7895	0.1525	0.8
10	S9	raw	sasa_total	955	61	72.22	116	-5.958	-12.45	-0.2138	-4.779	-19.78	5.694	-1.179	-11.86	14.35	0.835	5.813	0.8
10	S9	raw	burial	955	61	2.255	3.974	0.09056	-0.1081	0.2756	0.4509	0.1123	0.7804	-0.3603	-0.6976	-0.03934	0.0665	0.1956	0.1
10	S9	raw	rgyr	955	61	0.1713	0.1483	-0.0002901	-0.02391	0.01597	-0.007783	-0.02701	0.006805	0.007492	-0.006448	0.02072	0.269	0.006698	0.4
10	S9	raw	plddt	955	61	1.68	2.286	0.1386	-0.08848	0.2954	0.2074	-0.02652	0.3803	-0.06882	-0.2639	0.121	0.504	0.09974	0.6
10	S9	raw	cavity_open_vol	955	61	136	-	11.77	-4.649	24.74	-	-	-	-	-	-	0.049	5.953	0.1
10	S9	raw	cavity_enclosed_vol	955	61	49.81	-	0.4787	-5.189	5.209	-	-	-	-	-	-	0.8455	2.339	0.8
10	all	plddt_adj	n	5445	390	2.031	2.031	0.009879	-0.103	0.1245	0.009879	-0.103	0.1245	0	0	0	1	0	
10	all	plddt_adj	hydropathy	5445	390	0.1447	0.1944	-0.0009489	-0.009219	0.007848	-0.01434	-0.02505	-0.003788	0.01339	4.56e-05	0.0274	0.016	0.005543	0.04

Supplementary Table S2 (continued)

radius	family	adjustment	descriptor	n	clades	within_sd_pocket	within_sd_control	slope_pocket	slope_pocket_lo	slope_pocket_hi	slope_control	slope_control_lo	slope_control_hi	difference	difference_lo	difference_hi	p_perm	null_sd	q
10	all	plddt_adj	resvol	5445	390	2.065	2.608	-0.02362	-0.143	0.1167	0.1792	0.04549	0.3104	-0.2028	-0.3974	-0.00148	0.0115	0.0779	0.04
10	all	plddt_adj	f_pos	5445	390	0.01195	0.01704	-0.0001692	-0.0008696	0.0005456	0.002435	0.001548	0.003283	-0.002604	-0.003798	-0.001321	0	0.0005029	
10	all	plddt_adj	f_neg	5445	390	0.01465	0.01992	-0.000536	-0.001511	0.0003881	0.001297	0.0002539	0.002254	-0.001833	-0.003286	-0.0004894	0.0015	0.0005642	0.00
10	all	plddt_adj	net_charge	5445	390	0.7633	1.019	0.01324	-0.03813	0.06137	0.05042	-0.001521	0.09971	-0.03718	-0.1134	0.04044	0.2095	0.02956	0.3
10	all	plddt_adj	f_aromatic	5445	390	0.01632	0.01579	0.00029	-0.0006688	0.001253	0.0007612	1.69e-05	0.001475	-0.0004712	-0.001799	0.000857	0.383	0.0005425	0.5
10	all	plddt_adj	f_polar	5445	390	0.02976	0.03216	0.001225	-0.00102	0.003292	-0.00437	-0.005957	-0.002578	0.005595	0.00307	0.008151	0	0.0009883	
10	all	plddt_adj	f_small	5445	390	0.02601	0.02948	0.0002814	-0.001328	0.001846	-0.001357	-0.002997	0.0003011	0.001638	-0.0005276	0.003695	0.08	0.0009215	0
10	all	plddt_adj	f_gly	5445	390	0.01566	0.0185	-0.0003513	-0.001271	0.0005306	-2.98e-05	-0.0009732	0.0009269	-0.0003215	-0.001571	0.0008951	0.54	0.0005252	0.7
10	all	plddt_adj	f_pro	5445	390	0.01124	0.01225	5.44e-05	-0.0006517	0.0006879	0.000949	0.0004091	0.001468	-0.0008946	-0.001745	-2.76e-05	0.0145	0.0003648	0.04
10	all	plddt_adj	f_ivywrel	5445	390	0.02819	0.03325	0.001058	-0.0005236	0.002757	0.00495	0.003182	0.006739	-0.003893	-0.00612	-0.001575	0.0005	0.001012	0.
10	all	plddt_adj	sasa_mean	5445	390	2.308	3.587	0.06738	-0.04921	0.192	0.03375	-0.1603	0.2286	0.03363	-0.1773	0.2397	0.7335	0.09104	0.8
10	all	plddt_adj	sasa_total	5445	390	78.81	142.4	2.114	-1.437	5.911	1.188	-6.477	8.928	0.9261	-7.928	9.931	0.8095	3.833	0.9
10	all	plddt_adj	burial	5445	390	3.877	5.145	-0.01274	-0.263	0.2149	0.1992	-0.07896	0.4657	-0.2119	-0.5237	0.09884	0.11	0.1306	0
10	all	plddt_adj	rgyr	5445	390	0.1759	0.2041	-7.35e-05	-0.009119	0.008663	-0.0004656	-0.009827	0.009291	0.0003921	-0.007754	0.008549	0.927	0.00399	0.9
10	all	plddt_adj	cavity_open_vol	5445	390	250.9	-	8.181	-4.759	22.73	-	-	-	-	-	-	0.1885	6.302	0.3
10	all	plddt_adj	cavity_enclosed_vol	5445	390	45.6	-	-0.5188	-2.587	1.583	-	-	-	-	-	-	0.655	1.119	0.8
10	S1	plddt_adj	n	1343	74	1.542	1.542	0.1445	-0.02477	0.3071	0.1445	-0.02477	0.3071	0	0	0	1	0	
10	S1	plddt_adj	hydropathy	1343	74	0.1477	0.1861	-0.007797	-0.02984	0.01271	-0.01929	-0.04029	0.003686	0.01149	-0.01885	0.04089	0.439	0.01426	0.6
10	S1	plddt_adj	resvol	1343	74	1.916	2.64	0.05903	-0.1379	0.2605	0.4042	0.04688	0.7606	-0.3451	-0.7238	0.05097	0.084	0.1971	0.
10	S1	plddt_adj	f_pos	1343	74	0.01302	0.0192	-0.0004462	-0.002315	0.001627	0.005141	0.002782	0.007483	-0.005588	-0.008618	-0.002375	0	0.001357	
10	S1	plddt_adj	f_neg	1343	74	0.01047	0.02288	0.0005836	-0.001255	0.002473	0.002833	-0.0001423	0.005638	-0.00225	-0.005645	0.001084	0.145	0.001568	0
10	S1	plddt_adj	net_charge	1343	74	0.6239	1.129	-0.03805	-0.1341	0.07174	0.08481	-0.06287	0.2458	-0.1229	-0.2944	0.05105	0.1265	0.07981	0.2
10	S1	plddt_adj	f_aromatic	1343	74	0.01743	0.0155	-0.0002431	-0.003056	0.002326	-0.0004299	-0.001923	0.001117	0.0001868	-0.002617	0.002854	0.9095	0.001437	0.
10	S1	plddt_adj	f_polar	1343	74	0.02929	0.03779	0.001197	-0.003593	0.005858	-0.007461	-0.01215	-0.002844	0.008658	0.002833	0.01392	0.001	0.002729	0.
10	S1	plddt_adj	f_small	1343	74	0.02276	0.03065	-0.0002057	-0.003147	0.00286	-0.004395	-0.008808	0.0004666	0.004189	-0.001702	0.009857	0.0635	0.002257	0.2
10	S1	plddt_adj	f_gly	1343	74	0.0169	0.02056	-0.001195	-0.003142	0.0007473	0.0008432	-0.001967	0.003741	-0.002038	-0.005339	0.001249	0.185	0.001554	0.
10	S1	plddt_adj	f_pro	1343	74	0.01055	0.01413	0.0001874	-0.001416	0.001859	-0.0007503	-0.00231	0.0006788	0.0009377	-0.001128	0.003296	0.38	0.001071	0
10	S1	plddt_adj	f_ivywrel	1343	74	0.02772	0.03474	-7.70e-05	-0.004293	0.004126	0.009737	0.004788	0.01477	-0.009814	-0.01588	-0.003456	0	0.002671	
10	S1	plddt_adj	sasa_mean	1343	74	2.288	2.963	0.1206	-0.1286	0.351	0.02155	-0.3455	0.3665	0.09901	-0.3183	0.5093	0.6375	0.2062	0.8
10	S1	plddt_adj	sasa_total	1343	74	73.79	119.2	9.134	1.011	17.1	6.375	-7.725	20.74	2.759	-13.67	17.96	0.73	7.969	0.
10	S1	plddt_adj	burial	1343	74	3.466	4.205	-0.01349	-0.4978	0.4557	0.6583	0.159	1.152	-0.6718	-1.165	-0.137	0.0305	0.3117	0.1
10	S1	plddt_adj	rgyr	1343	74	0.1536	0.1577	0.01619	0.001586	0.03192	0.005812	-0.01042	0.02469	0.01037	-0.00876	0.0295	0.228	0.008925	0.3
10	S1	plddt_adj	cavity_open_vol	1343	74	147.6	-	2.724	-15.57	21.32	-	-	-	-	-	-	0.792	10.31	0.
10	S1	plddt_adj	cavity_enclosed_vol	1343	74	29.87	-	-3.11	-7.206	0.7549	-	-	-	-	-	-	0.098	1.917	0.
10	S8	plddt_adj	n	3147	256	2.36	2.36	-0.02016	-0.1828	0.1377	-0.02016	-0.1828	0.1377	0	0	0	1	0	
10	S8	plddt_adj	hydropathy	3147	256	0.1441	0.1918	0.00272	-0.007108	0.01345	-0.01148	-0.02631	0.00325	0.0142	-0.004894	0.03272	0.038	0.00717	0.
10	S8	plddt_adj	resvol	3147	256	2.165	2.639	-0.009979	-0.1749	0.163	0.1491	-0.03805	0.3205	-0.1591	-0.4041	0.1264	0.1315	0.1047	0.3
10	S8	plddt_adj	f_pos	3147	256	0.01121	0.01639	6.04e-05	-0.0006481	0.0007753	0.002332	0.001264	0.003422	-0.002272	-0.003613	-0.0009687	0	0.0005899	
10	S8	plddt_adj	f_neg	3147	256	0.01536	0.01898	-0.0003524	-0.001418	0.0007266	0.0004054	-0.0008839	0.001795	-0.0007577	-0.002216	0.0006908	0.29	0.0007079	0
10	S8	plddt_adj	net_charge	3147	256	0.7977	1.014	0.01403	-0.03458	0.06718	0.08335	0.01321	0.1479	-0.06932	-0.1466	0.01617	0.0645	0.038	0.1
10	S8	plddt_adj	f_aromatic	3147	256	0.01512	0.01482	0.0005646	-0.000466	0.001653	0.0005659	-0.0003549	0.001342	-1.24e-06	-0.001344	0.001356	0.9975	0.00063	
10	S8	plddt_adj	f_polar	3147	256	0.03056	0.02971	-0.0009174	-0.003584	0.001628	-0.003664	-0.005638	-0.001459	0.002746	-0.000573	0.00573	0.0255	0.001244	0.

Supplementary Table S2 (continued)

radius	family	adjustment	descriptor	n	clades	within_sd_pocket	within_sd_control	slope_pocket	slope_pocket_lo	slope_pocket_hi	slope_control	slope_control_lo	slope_control_hi	difference	difference_lo	difference_hi	p_perm	null_sd	q
10	S8	plddt_adj	f_small	3147	256	0.02887	0.02984	0.0003859	-0.002096	0.002653	-0.0007724	-0.003024	0.001497	0.001158	-0.001639	0.003842	0.348	0.00123	0.6
10	S8	plddt_adj	f_gly	3147	256	0.01568	0.01838	-3.53e-06	-0.001233	0.001127	-0.0003754	-0.001601	0.0007415	0.0003719	-0.001272	0.002034	0.572	0.0006683	0.8
10	S8	plddt_adj	f_pro	3147	256	0.01249	0.01149	0.0001403	-0.0008592	0.001043	0.001191	0.0005302	0.001824	-0.001051	-0.00223	3.66e-05	0.029	0.0004801	0.
10	S8	plddt_adj	f_ivywrel	3147	256	0.02853	0.03291	0.001971	-0.0001755	0.004018	0.004403	0.002238	0.006457	-0.002432	-0.005383	0.0004641	0.0555	0.00127	0.1
10	S8	plddt_adj	sasa_mean	3147	256	2.451	3.978	0.1254	-0.04151	0.2906	0.08474	-0.1906	0.3831	0.04068	-0.2806	0.3251	0.747	0.1245	0.9
10	S8	plddt_adj	sasa_total	3147	256	82.68	157.8	3.193	-1.581	8.663	1.707	-8.806	13.69	1.487	-12.47	13.91	0.787	5.278	0.9
10	S8	plddt_adj	burial	3147	256	4.397	5.783	-0.06225	-0.4125	0.2847	-0.01453	-0.4433	0.4269	-0.04771	-0.5169	0.4536	0.7915	0.1824	0.9
10	S8	plddt_adj	rgyr	3147	256	0.1859	0.234	-0.004349	-0.01465	0.006461	-0.001257	-0.01491	0.01185	-0.003091	-0.0139	0.007587	0.585	0.005439	0.8
10	S8	plddt_adj	cavity_open_vol	3147	256	306.6	-	12.3	-7.786	32.47	-	-	-	-	-	-	0.1935	9.441	0.4
10	S8	plddt_adj	cavity_enclosed_vol	3147	256	49.63	-	-0.3037	-2.759	2.576	-	-	-	-	-	-	0.8325	1.484	0.9
10	S9	plddt_adj	n	955	61	1.343	1.343	-0.02742	-0.2662	0.1626	-0.02742	-0.2662	0.1626	0	0	0	1	0	
10	S9	plddt_adj	hydropathy	955	61	0.1421	0.2131	-0.007374	-0.02592	0.01578	-0.02012	-0.04167	0.003651	0.01275	-0.01203	0.04381	0.277	0.0116	0.3
10	S9	plddt_adj	resvol	955	61	1.893	2.451	-0.1027	-0.3083	0.1906	0.1207	-0.1136	0.3617	-0.2234	-0.5724	0.2309	0.1445	0.1555	0.2
10	S9	plddt_adj	f_pos	955	61	0.0126	0.01587	-0.000542	-0.002132	0.00169	0.001085	-0.001354	0.002932	-0.001627	-0.004579	0.002587	0.117	0.001018	0.2
10	S9	plddt_adj	f_neg	955	61	0.0171	0.01841	-0.001623	-0.005076	0.0008154	0.002704	0.0009716	0.00456	-0.004327	-0.008868	-0.0008432	0.0005	0.001214	0.0
10	S9	plddt_adj	net_charge	955	61	0.8219	0.8598	0.04399	-0.08457	0.2224	-0.05565	-0.1818	0.0418	0.09965	-0.1127	0.3849	0.1015	0.06096	0.2
10	S9	plddt_adj	f_aromatic	955	61	0.01777	0.01894	4.00e-05	-0.002182	0.002794	0.002017	0.0001074	0.004537	-0.001977	-0.006216	0.0017	0.117	0.00125	0.2
10	S9	plddt_adj	f_polar	955	61	0.02767	0.03128	0.006796	0.003289	0.009925	-0.004339	-0.007716	-2.68e-05	0.01114	0.005641	0.01531	0	0.001865	
10	S9	plddt_adj	f_small	955	61	0.01954	0.02636	0.0005372	-0.001211	0.002695	-0.001067	-0.003443	0.001444	0.001604	-0.001605	0.005094	0.317	0.00161	0.4
10	S9	plddt_adj	f_gly	955	61	0.01369	0.01562	-0.0008445	-0.002184	0.0008845	0.0003626	-0.001399	0.002128	-0.001207	-0.003273	0.001372	0.1995	0.0009295	0.2
10	S9	plddt_adj	f_pro	955	61	0.007055	0.01179	-0.0003313	-0.001115	0.0003586	0.001426	5.94e-05	0.002597	-0.001758	-0.003138	-0.0002408	0.0025	0.0006005	0.
10	S9	plddt_adj	f_ivywrel	955	61	0.02764	0.03218	-0.0005588	-0.00319	0.003392	0.003322	-0.000108	0.00753	-0.003881	-0.007917	0.001088	0.049	0.001935	0.1
10	S9	plddt_adj	sasa_mean	955	61	1.794	2.972	-0.08701	-0.2266	0.08815	-0.06095	-0.3509	0.2044	-0.02606	-0.3101	0.3334	0.8635	0.1553	0.9
10	S9	plddt_adj	sasa_total	955	61	72.22	116	-4.06	-10.74	1.407	-2.998	-18.01	8.794	-1.062	-12.73	14.04	0.8545	5.848	0.9
10	S9	plddt_adj	burial	955	61	2.255	3.974	0.05496	-0.1305	0.2605	0.443	0.08444	0.7795	-0.388	-0.7033	-0.04195	0.047	0.2003	0.1
10	S9	plddt_adj	rgyr	955	61	0.1713	0.1483	0.005554	-0.01797	0.02374	-0.00336	-0.02501	0.01246	0.008914	-0.005068	0.0228	0.1735	0.006595	0.2
10	S9	plddt_adj	cavity_open_vol	955	61	136	-	7.431	-6.676	21.11	-	-	-	-	-	-	0.1635	5.231	0.2
10	S9	plddt_adj	cavity_enclosed_vol	955	61	49.81	-	1.086	-5.232	6.247	-	-	-	-	-	-	0.64	2.275	0.
10	all	standardized	n	5445	390	2.031	2.031	-0.002095	-0.05974	0.05595	-0.002095	-0.05974	0.05595	0	0	0	1	0	
10	all	standardized	hydropathy	5445	390	0.1447	0.1944	-0.007602	-0.066	0.05451	-0.07378	-0.133	-0.0182	0.06618	-0.01433	0.1568	0.048	0.03268	0.1
10	all	standardized	resvol	5445	390	2.065	2.608	-0.01475	-0.07475	0.05118	0.06863	0.01839	0.1197	-0.08338	-0.1617	0.002575	0.0085	0.03192	0.6
10	all	standardized	f_pos	5445	390	0.01195	0.01704	-0.0148	-0.07224	0.04614	0.143	0.09214	0.1928	-0.1578	-0.2429	-0.07325	0	0.03275	
10	all	standardized	f_neg	5445	390	0.01465	0.01992	-0.03732	-0.1046	0.0247	0.06472	0.0128	0.1185	-0.102	-0.1872	-0.02559	0.0005	0.03267	0.002
10	all	standardized	net_charge	5445	390	0.7633	1.019	0.01849	-0.04375	0.08346	0.04986	-0.003636	0.1008	-0.03137	-0.1181	0.06222	0.3165	0.03217	0.4
10	all	standardized	f_aromatic	5445	390	0.01632	0.01579	0.01576	-0.0422	0.07731	0.04837	0.0005499	0.09644	-0.03261	-0.1199	0.05228	0.3345	0.03438	0.4
10	all	standardized	f_polar	5445	390	0.02976	0.03216	0.04276	-0.03213	0.1151	-0.1354	-0.1876	-0.07981	0.1782	0.09286	0.262	0	0.03155	
10	all	standardized	f_small	5445	390	0.02601	0.02948	0.01302	-0.04572	0.07441	-0.04599	-0.102	0.008354	0.05901	-0.01981	0.1349	0.0645	0.03227	0.1
10	all	standardized	f_gly	5445	390	0.01566	0.0185	-0.01992	-0.07335	0.03735	-0.001736	-0.05291	0.04639	-0.01818	-0.09084	0.05668	0.563	0.03103	0.6
10	all	standardized	f_pro	5445	390	0.01124	0.01225	0.003666	-0.05856	0.05863	0.07705	0.03057	0.1181	-0.07339	-0.1445	-0.002239	0.018	0.03078	0.
10	all	standardized	f_ivywrel	5445	390	0.02819	0.03325	0.03578	-0.02266	0.09583	0.1486	0.09628	0.2011	-0.1129	-0.1846	-0.03635	0.0005	0.03214	0.002
10	all	standardized	sasa_mean	5445	390	2.308	3.587	0.02577	-0.02301	0.08146	0.009169	-0.04555	0.06572	0.01661	-0.05341	0.08573	0.555	0.02837	0.6
10	all	standardized	sasa_total	5445	390	78.81	142.4	0.02099	-0.02932	0.06857	0.005397	-0.04819	0.06192	0.01559	-0.06003	0.09287	0.605	0.03111	0.6

Supplementary Table S2 (continued)

radius	family	adjustment	descriptor	n	clades	within_sd_pocket	within_sd_control	slope_pocket	slope_pocket_lo	slope_pocket_hi	slope_control	slope_control_lo	slope_control_hi	difference	difference_lo	difference_hi	p_perm	null_sd	q
10	all	standardized	burial	5445	390	3.877	5.145	-0.001925	-0.06531	0.06028	0.0388	-0.01942	0.09188	-0.04073	-0.111	0.03143	0.158	0.02866	0.3
10	all	standardized	rgyr	5445	390	0.1759	0.2041	-0.003897	-0.05548	0.04466	-0.007994	-0.05757	0.04175	0.004097	-0.03827	0.04652	0.851	0.02065	0.8
10	all	standardized	plddt	5445	390	1.647	2.24	0.02161	-0.0429	0.08082	0.05736	0.008538	0.1065	-0.03576	-0.1038	0.02852	0.231	0.02981	0.
10	all	standardized	cavity_open_vol	5445	390	250.9	-	0.04063	-0.01834	0.1008	-	-	-	-	-	-	0.136	0.02649	0.2
10	all	standardized	cavity_enclosed_vol	5445	390	45.6	-	-0.0127	-0.0599	0.03509	-	-	-	-	-	-	0.598	0.02408	0.6
10	S1	standardized	n	1343	74	1.542	1.542	0.0806	-0.02974	0.199	0.0806	-0.02974	0.199	0	0	0	1	0	
10	S1	standardized	hydropathy	1343	74	0.1477	0.1861	-0.04394	-0.1955	0.1036	-0.1011	-0.2152	0.02129	0.05717	-0.1267	0.2361	0.5045	0.08625	0.6
10	S1	standardized	resvol	1343	74	1.916	2.64	0.02444	-0.07987	0.1304	0.1496	0.008442	0.2832	-0.1252	-0.2749	0.04017	0.1355	0.08333	0.3
10	S1	standardized	f_pos	1343	74	0.01302	0.0192	-0.04408	-0.2011	0.1233	0.2623	0.1436	0.3839	-0.3063	-0.5112	-0.1002	0.0005	0.08334	0.0
10	S1	standardized	f_neg	1343	74	0.01047	0.02288	0.0463	-0.1293	0.2319	0.1208	-0.0119	0.2469	-0.07454	-0.2905	0.1336	0.4085	0.09278	0.6
10	S1	standardized	net_charge	1343	74	0.6239	1.129	-0.0611	-0.2224	0.1097	0.07392	-0.05687	0.208	-0.135	-0.3391	0.06741	0.1295	0.08864	0.3
10	S1	standardized	f_aromatic	1343	74	0.01743	0.0155	-0.01506	-0.1749	0.1398	-0.03003	-0.1264	0.06289	0.01497	-0.1479	0.174	0.8535	0.08467	0.9
10	S1	standardized	f_polar	1343	74	0.02929	0.03779	0.05127	-0.1168	0.2166	-0.1894	-0.3117	-0.06301	0.2407	0.07307	0.3996	0.0035	0.08449	0.02
10	S1	standardized	f_small	1343	74	0.02276	0.03065	-0.005471	-0.1379	0.1312	-0.1416	-0.2936	0.009388	0.1361	-0.07989	0.3446	0.1105	0.08519	0.3
10	S1	standardized	f_gly	1343	74	0.0169	0.02056	-0.06152	-0.1857	0.05493	0.04042	-0.08915	0.1758	-0.1019	-0.2711	0.05601	0.2215	0.08287	0.4
10	S1	standardized	f_pro	1343	74	0.01055	0.01413	0.002118	-0.163	0.1753	-0.05162	-0.167	0.05328	0.05374	-0.1387	0.2669	0.5195	0.08386	0.6
10	S1	standardized	f_ivywrel	1343	74	0.02772	0.03474	-0.003787	-0.1486	0.148	0.2798	0.1281	0.4196	-0.2836	-0.4761	-0.07021	0.001	0.08401	0.0
10	S1	standardized	sasa_mean	1343	74	2.288	2.963	0.01996	-0.1126	0.1321	-0.0009053	-0.1172	0.1244	0.02086	-0.1598	0.179	0.8035	0.08159	0.
10	S1	standardized	sasa_total	1343	74	73.79	119.2	0.0871	-0.05178	0.2168	0.04187	-0.07275	0.1723	0.04522	-0.1305	0.211	0.5685	0.07932	0.6
10	S1	standardized	burial	1343	74	3.466	4.205	0.01869	-0.1292	0.1702	0.1589	0.03648	0.2801	-0.1402	-0.2714	0.0103	0.1005	0.08461	0.3
10	S1	standardized	rgyr	1343	74	0.1536	0.1577	0.08418	-0.02992	0.1996	0.02055	-0.09289	0.1453	0.06363	-0.06646	0.1921	0.2635	0.05731	0.
10	S1	standardized	plddt	1343	74	1.039	1.976	0.1194	-0.03589	0.2785	0.2094	0.04091	0.3764	-0.08995	-0.2831	0.1005	0.2825	0.08223	0.
10	S1	standardized	cavity_open_vol	1343	74	147.6	-	0.04314	-0.09404	0.1867	-	-	-	-	-	-	0.562	0.07201	0.6
10	S1	standardized	cavity_enclosed_vol	1343	74	29.87	-	-0.1148	-0.2646	0.01703	-	-	-	-	-	-	0.0675	0.06349	0.3
10	S8	standardized	n	3147	256	2.36	2.36	-0.003164	-0.07653	0.0726	-0.003164	-0.07653	0.0726	0	0	0	1	0	
10	S8	standardized	hydropathy	3147	256	0.1441	0.1918	0.02003	-0.0473	0.09439	-0.05947	-0.1404	0.01412	0.0795	-0.03049	0.1954	0.057	0.04204	0.2
10	S8	standardized	resvol	3147	256	2.165	2.639	-0.00212	-0.07971	0.0811	0.05659	-0.01108	0.1206	-0.05871	-0.1717	0.06124	0.1705	0.04306	0.4
10	S8	standardized	f_pos	3147	256	0.01121	0.01639	0.005856	-0.06081	0.06877	0.1421	0.07746	0.2082	-0.1363	-0.2324	-0.04561	0.0015	0.04249	0.0
10	S8	standardized	f_neg	3147	256	0.01536	0.01898	-0.02259	-0.08992	0.04723	0.02148	-0.0442	0.09404	-0.04407	-0.1325	0.03679	0.2825	0.04075	0.6
10	S8	standardized	net_charge	3147	256	0.7977	1.014	0.01671	-0.04734	0.08005	0.0819	0.01675	0.1474	-0.06519	-0.1501	0.02295	0.111	0.04123	0.4
10	S8	standardized	f_aromatic	3147	256	0.01512	0.01482	0.03895	-0.03231	0.1092	0.03805	-0.02084	0.09129	0.000901	-0.08971	0.09861	0.986	0.04174	
10	S8	standardized	f_polar	3147	256	0.03056	0.02971	-0.03129	-0.1191	0.05678	-0.1236	-0.1941	-0.05434	0.09235	-0.01309	0.1919	0.0255	0.04124	0.1
10	S8	standardized	f_small	3147	256	0.02887	0.02984	0.01144	-0.07283	0.09037	-0.02586	-0.1043	0.0473	0.0373	-0.05739	0.1342	0.3775	0.04182	0.6
10	S8	standardized	f_gly	3147	256	0.01568	0.01838	-0.001875	-0.0769	0.07268	-0.02041	-0.08923	0.04069	0.01853	-0.07593	0.1144	0.634	0.03877	0.8
10	S8	standardized	f_pro	3147	256	0.01249	0.01149	0.01232	-0.06538	0.08757	0.1039	0.04433	0.1555	-0.09153	-0.1843	0.0005053	0.025	0.03974	0.1
10	S8	standardized	f_ivywrel	3147	256	0.02853	0.03291	0.07043	7.45e-05	0.1445	0.1341	0.06659	0.1975	-0.06369	-0.1563	0.03005	0.135	0.04359	0.4
10	S8	standardized	sasa_mean	3147	256	2.451	3.978	0.05322	-0.01041	0.1216	0.02118	-0.04591	0.0941	0.03204	-0.05929	0.116	0.3915	0.03667	0.6
10	S8	standardized	sasa_total	3147	256	82.68	157.8	0.04256	-0.01607	0.1063	0.01288	-0.05359	0.08305	0.02969	-0.06953	0.1211	0.4505	0.03959	0.6
10	S8	standardized	burial	3147	256	4.397	5.783	-0.0145	-0.1011	0.06749	-0.002429	-0.08461	0.06837	-0.01207	-0.1042	0.09409	0.7395	0.03549	0.8
10	S8	standardized	rgyr	3147	256	0.1859	0.234	-0.02192	-0.08408	0.03782	-0.001189	-0.0655	0.06179	-0.02073	-0.07707	0.03259	0.441	0.02676	0.6
10	S8	standardized	plddt	3147	256	1.834	2.325	-0.01545	-0.09043	0.05316	0.01114	-0.05224	0.06921	-0.02659	-0.1162	0.0576	0.5	0.04103	0.6
10	S8	standardized	cavity_open_vol	3147	256	306.6	-	0.03393	-0.04212	0.1094	-	-	-	-	-	-	0.3225	0.03411	0.6

Supplementary Table S2 (continued)

radius	family	adjustment	descriptor	n	clades	within_sd_pocket	within_sd_control	slope_pocket	slope_pocket_lo	slope_pocket_hi	slope_control	slope_control_lo	slope_control_hi	difference	difference_lo	difference_hi	p_perm	null_sd	q
10	S8	standardized	cavity_enclosed_vol	3147	256	49.63	-	-0.005629	-0.05726	0.0502	-	-	-	-	-	-	0.8435	0.02946	0.9
10	S9	standardized	n	955	61	1.343	1.343	-0.05373	-0.2211	0.06709	-0.05373	-0.2211	0.06709	0	0	0	1	0	
10	S9	standardized	hydropathy	955	61	0.1421	0.2131	-0.05792	-0.1838	0.1101	-0.08846	-0.1981	0.0221	0.03054	-0.1017	0.2086	0.636	0.063	0.7
10	S9	standardized	resvol	955	61	1.893	2.451	-0.07083	-0.1805	0.1064	0.05176	-0.04122	0.1466	-0.1226	-0.2796	0.1126	0.0695	0.06811	0.2
10	S9	standardized	f_pos	955	61	0.0126	0.01587	-0.04027	-0.1697	0.1347	0.07065	-0.08619	0.1918	-0.1109	-0.3352	0.1857	0.139	0.07513	0.2
10	S9	standardized	f_neg	955	61	0.0171	0.01841	-0.09853	-0.2987	0.04081	0.1418	0.05034	0.2412	-0.2403	-0.4873	-0.05365	0.0005	0.07012	0.00
10	S9	standardized	net_charge	955	61	0.8219	0.8598	0.06035	-0.094	0.2774	-0.05865	-0.211	0.05239	0.119	-0.1244	0.4708	0.1025	0.07399	0.2
10	S9	standardized	f_aromatic	955	61	0.01777	0.01894	-0.007495	-0.1355	0.1557	0.1084	0.01069	0.2469	-0.1159	-0.3473	0.08701	0.0885	0.06685	0.2
10	S9	standardized	f_polar	955	61	0.02767	0.03128	0.2483	0.1243	0.3574	-0.1398	-0.2416	0.007793	0.3881	0.2016	0.5242	0	0.06519	
10	S9	standardized	f_small	955	61	0.01954	0.02636	0.03279	-0.0632	0.1386	-0.04074	-0.1314	0.0602	0.07353	-0.07975	0.2301	0.286	0.06761	0.4
10	S9	standardized	f_gly	955	61	0.01369	0.01562	-0.04348	-0.136	0.08924	0.02072	-0.0905	0.1253	-0.0642	-0.2012	0.1144	0.338	0.06675	0.
10	S9	standardized	f_pro	955	61	0.007055	0.01179	-0.03727	-0.1546	0.05688	0.115	0.005275	0.216	-0.1523	-0.3126	-0.009321	0.0145	0.06046	0.09
10	S9	standardized	f_ivywrel	955	61	0.02764	0.03218	-0.02998	-0.1387	0.12	0.1045	-0.01271	0.2257	-0.1345	-0.2662	0.06135	0.0355	0.06514	0.1
10	S9	standardized	sasa_mean	955	61	1.794	2.972	-0.06296	-0.1422	0.04461	-0.02428	-0.1298	0.07306	-0.03868	-0.1594	0.1095	0.5175	0.06067	0.6
10	S9	standardized	sasa_total	955	61	72.22	116	-0.08249	-0.1725	0.001346	-0.04121	-0.165	0.04958	-0.04128	-0.1539	0.1035	0.495	0.0602	0.6
10	S9	standardized	burial	955	61	2.255	3.974	0.04015	-0.04966	0.1248	0.1135	0.02245	0.1972	-0.07332	-0.1785	0.02851	0.2175	0.06026	0.3
10	S9	standardized	rgyr	955	61	0.1713	0.1483	-0.001694	-0.1344	0.09264	-0.05249	-0.1927	0.0449	0.05079	-0.03112	0.1469	0.2305	0.04171	0.3
10	S9	standardized	plddt	955	61	1.68	2.286	0.08248	-0.05851	0.1827	0.09074	-0.005731	0.1629	-0.00826	-0.1221	0.08192	0.874	0.04673	0.9
10	S9	standardized	cavity_open_vol	955	61	136	-	0.08656	-0.0373	0.1832	-	-	-	-	-	-	0.038	0.04289	0.1
10	S9	standardized	cavity_enclosed_vol	955	61	49.81	-	0.009611	-0.1087	0.1152	-	-	-	-	-	-	0.819	0.04635	0.9

Supplementary Table S3. Pocket versus matched control at a 6 Å radius

As Supplementary Table S1 with the catalytic neighbourhood defined by a 6 Å radius, testing whether the choice of radius carries the result.

radius	family	adjustment	descriptor	n	clades	within_sd_pocket	within_sd_control	slope_pocket	slope_pocket_lo	slope_pocket_hi	slope_control	slope_control_lo	slope_control_hi	difference	difference_lo	difference_hi	p_perm	null_sd	c
6	all	raw	n	10635	1481	0.8892	1.97	0.001715	-0.03505	0.04105	-0.01486	-0.1048	0.07653	0.01657	-0.06235	0.09377	0.6175	0.03323	0.
6	all	raw	hydropathy	10635	1481	0.2123	0.1954	0.006641	-0.001639	0.01499	-0.004822	-0.01258	0.002774	0.01146	0.0005736	0.02309	0.033	0.005417	0.
6	all	raw	resvol	10635	1481	2.905	2.454	-0.05567	-0.1745	0.06891	0.2437	0.1544	0.3351	-0.2993	-0.446	-0.1468	0	0.0719	
6	all	raw	f_pos	10635	1481	0.01642	0.01574	-4.07e-05	-0.000674	0.0006011	0.001224	0.0006153	0.001872	-0.001265	-0.002142	-0.0004147	0.003	0.000436	0.
6	all	raw	f_neg	10635	1481	0.02356	0.01872	-1.33e-05	-0.0009833	0.0009074	0.001467	0.0007057	0.002242	-0.00148	-0.002622	-0.0003348	0.006	0.0005399	0.
6	all	raw	net_charge	10635	1481	0.3674	0.915	-0.002563	-0.01732	0.01207	-0.005229	-0.04313	0.0338	0.002666	-0.0384	0.04233	0.883	0.0179	0.
6	all	raw	f_aromatic	10635	1481	0.02191	0.01579	-0.001076	-0.002039	-0.0001369	0.000449	-0.00023	0.001089	-0.001525	-0.002614	-0.0003606	0.0035	0.000516	0.
6	all	raw	f_polar	10635	1481	0.04344	0.03044	-0.00109	-0.002692	0.0005278	-0.004145	-0.005446	-0.002857	0.003055	0.0009936	0.005175	0.0005	0.0009657	0.
6	all	raw	f_small	10635	1481	0.03526	0.02737	-0.000226	-0.001568	0.001113	-0.002393	-0.003544	-0.00132	0.002167	0.0004584	0.003968	0.0095	0.0008375	0.
6	all	raw	f_gly	10635	1481	0.02049	0.01694	0.0002362	-0.0005575	0.001027	-0.0005783	-0.001252	4.05e-05	0.0008145	-0.0002376	0.001875	0.106	0.0005039	0.
6	all	raw	f_pro	10635	1481	0.01377	0.01171	-0.0003843	-0.001184	0.0003069	0.001083	0.0006165	0.001534	-0.001467	-0.002438	-0.0006007	0	0.0003661	
6	all	raw	f_ivywrel	10635	1481	0.04074	0.03156	0.0006472	-0.001041	0.002382	0.00578	0.004595	0.007036	-0.005133	-0.007166	-0.00302	0	0.0009414	
6	all	raw	sasa_mean	10635	1481	3.114	3.498	-0.04382	-0.1918	0.09884	0.08972	-0.04485	0.2352	-0.1335	-0.3021	0.02833	0.0765	0.07674	0.
6	all	raw	sasa_total	10635	1481	32.86	136.6	-0.839	-2.293	0.6005	2.184	-3.161	7.502	-3.023	-8.84	2.794	0.265	2.706	0.
6	all	raw	burial	10635	1481	4.907	5.119	-0.003399	-0.2362	0.2219	0.1568	-0.06133	0.3586	-0.1602	-0.395	0.07391	0.1695	0.1165	0.
6	all	raw	rgyr	10635	1481	0.1788	0.2166	-0.003987	-0.01253	0.003538	-0.003756	-0.01289	0.005534	-0.0002312	-0.0104	0.009524	0.956	0.004664	0
6	all	raw	plddt	10635	1481	2.373	1.93	0.06849	-0.03357	0.1695	0.1068	0.02627	0.1861	-0.03834	-0.1457	0.06346	0.459	0.05223	0.
6	all	raw	cavity_open_vol	10635	1481	238.7	-	5.651	-5.175	16.97	-	-	-	-	-	-	0.241	4.765	0.
6	all	raw	cavity_enclosed_vol	10635	1481	37.32	-	0.4544	-1.045	1.932	-	-	-	-	-	-	0.511	0.7132	0.
6	S1	raw	n	1125	153	0.7488	1.907	-0.07089	-0.2018	0.05954	0.05796	-0.2548	0.3796	-0.1288	-0.4186	0.1363	0.341	0.132	0.
6	S1	raw	hydropathy	1125	153	0.1985	0.1795	-0.02177	-0.05941	0.01528	-0.01787	-0.04625	0.01046	-0.003894	-0.0543	0.04506	0.8355	0.01897	0
6	S1	raw	resvol	1125	153	3.438	2.511	-0.1381	-0.7897	0.496	0.367	0.006601	0.733	-0.5051	-1.224	0.2448	0.101	0.3106	0.
6	S1	raw	f_pos	1125	153	0.01986	0.01798	0.0003538	-0.003452	0.004554	0.002981	0.0001279	0.00578	-0.002627	-0.007337	0.002279	0.2075	0.00203	0
6	S1	raw	f_neg	1125	153	0.01828	0.02141	-0.0011	-0.003112	0.0009514	0.003906	0.0003459	0.007399	-0.005006	-0.008795	-0.00114	0.009	0.001932	0.
6	S1	raw	net_charge	1125	153	0.3719	1.073	0.02387	-0.03679	0.08774	-0.03863	-0.2432	0.154	0.0625	-0.1364	0.2797	0.4415	0.08252	0
6	S1	raw	f_aromatic	1125	153	0.0259	0.01458	-0.0004903	-0.00451	0.003655	0.0009535	-0.001077	0.002995	-0.001444	-0.006093	0.003413	0.509	0.002094	0.
6	S1	raw	f_polar	1125	153	0.04345	0.0355	0.005096	-0.003834	0.0146	-0.008182	-0.01361	-0.002925	0.01328	0.003889	0.02296	0.0025	0.004507	0.
6	S1	raw	f_small	1125	153	0.03549	0.029	0.0008226	-0.004826	0.006772	-0.003683	-0.008308	0.0011	0.004506	-0.002074	0.01108	0.1585	0.003274	0.
6	S1	raw	f_gly	1125	153	0.02119	0.01906	0.001137	-0.00304	0.005196	0.001559	-0.001653	0.004611	-0.0004224	-0.005367	0.004827	0.8365	0.002172	0
6	S1	raw	f_pro	1125	153	0.01526	0.01308	-0.0005296	-0.00413	0.003247	0.001105	-0.001146	0.003537	-0.001634	-0.005969	0.002897	0.346	0.001713	0.
6	S1	raw	f_ivywrel	1125	153	0.04007	0.03307	0.002155	-0.005822	0.01079	0.00932	0.004566	0.01394	-0.007165	-0.0162	0.002595	0.0655	0.003869	0.
6	S1	raw	sasa_mean	1125	153	3.606	2.941	-0.1442	-0.7757	0.4723	-0.2399	-0.7007	0.2413	0.0957	-0.6357	0.8464	0.7495	0.3006	0
6	S1	raw	sasa_total	1125	153	47.89	112.8	-2.937	-12.03	5.416	-6.786	-28.2	13.71	3.849	-20.17	27.57	0.6935	9.776	0.
6	S1	raw	burial	1125	153	5.395	4.216	0.181	-0.8028	1.101	0.9235	0.2698	1.558	-0.7426	-1.757	0.312	0.09	0.4366	0.
6	S1	raw	rgyr	1125	153	0.1301	0.1869	-0.006899	-0.03355	0.02153	0.001545	-0.03227	0.03553	-0.008444	-0.04976	0.03547	0.6285	0.01709	0
6	S1	raw	plddt	1125	153	2.918	2.122	0.07906	-0.353	0.4924	0.3272	-0.05598	0.7461	-0.2482	-0.8188	0.3115	0.3465	0.259	0.

Supplementary Table S3 (continued)

radius	family	adjustment	descriptor	n	clades	within_sd_pocket	within_sd_control	slope_pocket	slope_pocket_lo	slope_pocket_hi	slope_control	slope_control_lo	slope_control_hi	difference	difference_lo	difference_hi	p_perm	null_sd	
6	S1	raw	cavity_open_vol	1125	153	234.1	-	0.08809	-45.47	46.33	-	-	-	-	-	-	0.998	18.62	0
6	S1	raw	cavity_enclosed_vol	1125	153	32.51	-	3.127	-2.783	9.482	-	-	-	-	-	-	0.224	2.514	0
6	S8	raw	n	6178	956	1.067	2.301	0.02921	-0.03167	0.09032	-0.03586	-0.1953	0.1125	0.06507	-0.06802	0.1966	0.2125	0.05233	0.
6	S8	raw	hydropathy	6178	956	0.2352	0.1886	0.009327	-0.002952	0.02225	-0.002213	-0.01207	0.007362	0.01154	-0.003749	0.02758	0.129	0.007604	0.
6	S8	raw	resvol	6178	956	2.824	2.521	0.002331	-0.1706	0.1712	0.265	0.1318	0.3963	-0.2627	-0.4686	-0.0467	0.007	0.09402	0
6	S8	raw	f_pos	6178	956	0.01261	0.0154	2.94e-05	-0.0005098	0.0005674	0.001138	0.0003796	0.001914	-0.001109	-0.00206	-0.0001345	0.0225	0.0004805	0.
6	S8	raw	f_neg	6178	956	0.02886	0.01841	0.0005436	-0.0008964	0.002027	0.0009111	-0.0001007	0.002039	-0.0003675	-0.002102	0.001303	0.6415	0.0008241	0.
6	S8	raw	net_charge	6178	956	0.4119	0.9469	-0.008368	-0.02931	0.01217	0.01378	-0.04064	0.06631	-0.02215	-0.07473	0.03566	0.3725	0.02537	0.
6	S8	raw	f_aromatic	6178	956	0.02104	0.0152	-0.001415	-0.002827	-0.0001572	0.0004333	-0.0005068	0.001305	-0.001849	-0.003361	-0.0003965	0.0045	0.0006736	0
6	S8	raw	f_polar	6178	956	0.0499	0.02945	-0.002637	-0.005261	-0.0002806	-0.002883	-0.004403	-0.001257	0.0002457	-0.002825	0.00314	0.864	0.001412	0
6	S8	raw	f_small	6178	956	0.03993	0.02831	-0.001091	-0.003148	0.001064	-0.00266	-0.004081	-0.001249	0.001569	-0.0009031	0.004219	0.2045	0.001214	0.
6	S8	raw	f_gly	6178	956	0.01879	0.0176	-0.0001397	-0.001126	0.0009289	-0.001012	-0.001931	-0.0001256	0.0008723	-0.0004401	0.002206	0.174	0.0006317	0.
6	S8	raw	f_pro	6178	956	0.01553	0.01128	-0.0006422	-0.001896	0.0004065	0.0007863	0.0002348	0.001319	-0.001429	-0.002931	-0.0002184	0.005	0.0005124	0
6	S8	raw	f_ivywrel	6178	956	0.04568	0.03141	0.001842	-0.0007662	0.004626	0.005556	0.003852	0.007292	-0.003714	-0.006683	-0.0006196	0.008	0.001387	0
6	S8	raw	sasa_mean	6178	956	3.415	3.873	0.01663	-0.2043	0.245	0.2184	-0.003554	0.4519	-0.2017	-0.4379	0.04774	0.0765	0.1139	0.
6	S8	raw	sasa_total	6178	956	33.41	154.6	0.0837	-1.969	1.903	5.86	-2.08	13.96	-5.776	-14.82	2.963	0.1775	4.273	0.
6	S8	raw	burial	6178	956	5.647	5.801	-0.1476	-0.5472	0.2117	-0.0875	-0.4061	0.21	-0.06011	-0.4061	0.2818	0.722	0.1775	0.
6	S8	raw	rgyr	6178	956	0.1869	0.2524	0.0002626	-0.01337	0.01088	-0.005308	-0.01896	0.008815	0.00557	-0.008363	0.01929	0.408	0.006701	0.
6	S8	raw	plddt	6178	956	2.748	2.253	0.1005	-0.07162	0.269	0.114	-0.01149	0.2353	-0.01355	-0.1926	0.1686	0.859	0.08564	0
6	S8	raw	cavity_open_vol	6178	956	283.6	-	14.23	-3.624	32.08	-	-	-	-	-	-	0.0765	8.011	0.
6	S8	raw	cavity_enclosed_vol	6178	956	39.89	-	0.3075	-1.719	2.368	-	-	-	-	-	-	0.7445	1.032	0.
6	S9	raw	n	3332	374	0.4644	1.159	-0.02699	-0.05342	-0.0003799	0.005996	-0.05985	0.07848	-0.03298	-0.1044	0.0361	0.3465	0.03578	0.
6	S9	raw	hydropathy	3332	374	0.1673	0.2122	0.007471	-0.002633	0.01862	-0.006552	-0.02073	0.006963	0.01402	-0.002197	0.03209	0.076	0.00779	0.
6	S9	raw	resvol	3332	374	2.855	2.299	-0.1332	-0.3008	0.03336	0.1864	0.03242	0.3312	-0.3195	-0.5256	-0.103	0.0045	0.1112	0.
6	S9	raw	f_pos	3332	374	0.02078	0.01556	-0.0002207	-0.001611	0.001177	0.00105	-1.57e-05	0.002141	-0.001271	-0.003052	0.0004058	0.119	0.0008222	0.
6	S9	raw	f_neg	3332	374	0.01066	0.0183	-0.000674	-0.001544	2.78e-05	0.001895	0.0007024	0.002985	-0.002569	-0.003954	-0.001195	0	0.0006336	
6	S9	raw	net_charge	3332	374	0.2633	0.7872	0.001546	-0.01565	0.01849	-0.02868	-0.08112	0.02478	0.03023	-0.02734	0.08598	0.2195	0.02505	0.
6	S9	raw	f_aromatic	3332	374	0.02199	0.01716	-0.0006197	-0.001956	0.0007683	0.000379	-0.0007656	0.001558	-0.0009987	-0.002615	0.0008206	0.2565	0.0008802	0.
6	S9	raw	f_polar	3332	374	0.02765	0.03038	0.0002096	-0.001372	0.001791	-0.005404	-0.007646	-0.003024	0.005614	0.00294	0.008259	0	0.001247	
6	S9	raw	f_small	3332	374	0.02421	0.02488	0.0009563	-0.0004976	0.002374	-0.001728	-0.003539	-2.51e-05	0.002684	0.0004565	0.00497	0.0105	0.001042	0.0
6	S9	raw	f_gly	3332	374	0.0231	0.01478	0.0006845	-0.001009	0.002187	-0.0002551	-0.001228	0.0007457	0.0009396	-0.000968	0.00281	0.257	0.0008282	0.
6	S9	raw	f_pro	3332	374	0.008844	0.01199	7.42e-05	-0.00058	0.0008334	0.001546	0.0007417	0.002307	-0.001472	-0.0024	-0.0005793	0.0015	0.0004792	0.00
6	S9	raw	f_ivywrel	3332	374	0.02976	0.0313	-0.001494	-0.003268	0.0002326	0.005486	0.003748	0.007344	-0.00698	-0.009431	-0.004524	0	0.001293	
6	S9	raw	sasa_mean	3332	374	2.223	2.886	-0.1205	-0.2644	0.02105	-0.05439	-0.2125	0.1286	-0.06606	-0.2963	0.1503	0.501	0.09696	0.
6	S9	raw	sasa_total	3332	374	24.49	104.3	-1.881	-3.699	-0.1805	-1.985	-8.814	4.767	0.1043	-7.101	7.291	0.971	3.394	0
6	S9	raw	burial	3332	374	2.81	3.897	0.1912	0.0099	0.376	0.4017	0.1588	0.6392	-0.2105	-0.4988	0.1105	0.111	0.132	0.
6	S9	raw	rgyr	3332	374	0.1776	0.141	-0.01008	-0.02118	6.05e-05	-0.002168	-0.01087	0.007083	-0.007908	-0.02326	0.00533	0.234	0.006722	0.
6	S9	raw	plddt	3332	374	1.038	0.9773	0.01339	-0.04775	0.07148	0.05477	-0.005103	0.1164	-0.04137	-0.1139	0.02767	0.2385	0.03525	0.
6	S9	raw	cavity_open_vol	3332	374	119.6	-	-6.982	-15.61	0.2515	-	-	-	-	-	-	0.0905	4.156	0.
6	S9	raw	cavity_enclosed_vol	3332	374	33.72	-	0.2355	-1.643	2.248	-	-	-	-	-	-	0.807	1.01	0.
6	all	plddt_adj	n	10635	1481	0.8892	1.97	0.01179	-0.02333	0.04687	0.004886	-0.08679	0.1004	0.006906	-0.07389	0.08969	0.844	0.03319	0.
6	all	plddt_adj	hydropathy	10635	1481	0.2123	0.1954	0.006616	-0.002072	0.01497	-0.00486	-0.01229	0.002672	0.01148	0.000519	0.02309	0.028	0.005263	0

Supplementary Table S3 (continued)

radius	family	adjustment	descriptor	n	clades	within_sd_pocket	within_sd_control	slope_pocket	slope_pocket_lo	slope_pocket_hi	slope_control	slope_control_lo	slope_control_hi	difference	difference_lo	difference_hi	p_perm	null_sd	
6	all	plddt_adj	resvol	10635	1481	2.905	2.454	-0.04483	-0.1652	0.07488	0.2425	0.1519	0.3347	-0.2873	-0.4313	-0.1417	0	0.06919	
6	all	plddt_adj	f_pos	10635	1481	0.01642	0.01574	-4.61e-06	-0.0006722	0.000621	0.001226	0.0005805	0.001843	-0.00123	-0.002148	-0.000366	0.005	0.000423	0
6	all	plddt_adj	f_neg	10635	1481	0.02356	0.01872	7.51e-05	-0.000844	0.001037	0.001466	0.0006714	0.002264	-0.00139	-0.002525	-0.0002168	0.012	0.000533	0
6	all	plddt_adj	net_charge	10635	1481	0.3674	0.915	-0.003471	-0.01735	0.01184	-0.00561	-0.04536	0.0317	0.002139	-0.0375	0.04291	0.9165	0.01851	0.
6	all	plddt_adj	f_aromatic	10635	1481	0.02191	0.01579	-0.001044	-0.001955	-0.000144	0.0004466	-0.0002421	0.001129	-0.001491	-0.002629	-0.0003163	0.004	0.0005121	0.
6	all	plddt_adj	f_polar	10635	1481	0.04344	0.03044	-0.001286	-0.003003	0.0003799	-0.00415	-0.005504	-0.002837	0.002864	0.0006891	0.00502	0.0015	0.0009502	0.0
6	all	plddt_adj	f_small	10635	1481	0.03526	0.02737	-0.0003271	-0.00166	0.001057	-0.002377	-0.003491	-0.001284	0.00205	0.0003107	0.003801	0.01	0.0008184	0.0
6	all	plddt_adj	f_gly	10635	1481	0.02049	0.01694	0.0001912	-0.0006399	0.001079	-0.0005706	-0.001227	9.32e-05	0.0007617	-0.0003448	0.001834	0.106	0.0004742	0.
6	all	plddt_adj	f_pro	10635	1481	0.01377	0.01171	-0.0003491	-0.001127	0.0003501	0.001086	0.0006322	0.001547	-0.001435	-0.00237	-0.0006135	0	0.0003503	
6	all	plddt_adj	f_ivywrel	10635	1481	0.04074	0.03156	0.000744	-0.0009545	0.00248	0.005771	0.004563	0.006963	-0.005027	-0.007015	-0.002976	0	0.0009577	
6	all	plddt_adj	sasa_mean	10635	1481	3.114	3.498	-0.05103	-0.1863	0.08946	0.0832	-0.05868	0.2274	-0.1342	-0.309	0.04088	0.0785	0.07567	0.
6	all	plddt_adj	sasa_total	10635	1481	32.86	136.6	-0.7828	-2.236	0.6182	2.541	-2.888	7.986	-3.324	-9.263	2.522	0.234	2.782	0
6	all	plddt_adj	burial	10635	1481	4.907	5.119	0.0189	-0.2398	0.2434	0.1648	-0.05666	0.3711	-0.1459	-0.3751	0.08575	0.1925	0.1131	0.
6	all	plddt_adj	rgyr	10635	1481	0.1788	0.2166	-0.003335	-0.01139	0.004298	-0.002205	-0.01053	0.006222	-0.00113	-0.01062	0.008888	0.801	0.004634	0.
6	all	plddt_adj	cavity_open_vol	10635	1481	238.7	-	2.794	-6.724	12.54	-	-	-	-	-	-	0.5275	4.45	0.
6	all	plddt_adj	cavity_enclosed_vol	10635	1481	37.32	-	0.2494	-1.112	1.7	-	-	-	-	-	-	0.723	0.7135	0.
6	S1	plddt_adj	n	1125	153	0.7488	1.907	-0.06567	-0.1901	0.05423	0.07459	-0.1919	0.3654	-0.1403	-0.4103	0.1062	0.2825	0.1292	0
6	S1	plddt_adj	hydropathy	1125	153	0.1985	0.1795	-0.02112	-0.05865	0.01664	-0.0179	-0.04573	0.01237	-0.003219	-0.05329	0.04458	0.868	0.01921	0
6	S1	plddt_adj	resvol	1125	153	3.438	2.511	-0.1203	-0.711	0.5255	0.366	0.002344	0.7363	-0.4864	-1.145	0.2553	0.126	0.3104	0.
6	S1	plddt_adj	f_pos	1125	153	0.01986	0.01798	0.0003864	-0.003399	0.00447	0.002984	-6.19e-05	0.005918	-0.002598	-0.007465	0.002642	0.205	0.002033	0.
6	S1	plddt_adj	f_neg	1125	153	0.01828	0.02141	-0.001086	-0.003118	0.0009905	0.00393	0.00031	0.007371	-0.005016	-0.008862	-0.001146	0.009	0.001948	0
6	S1	plddt_adj	net_charge	1125	153	0.3719	1.073	0.02348	-0.03367	0.08571	-0.03951	-0.222	0.1557	0.06299	-0.1373	0.2519	0.4405	0.08164	0.
6	S1	plddt_adj	f_aromatic	1125	153	0.0259	0.01458	-0.0004816	-0.004352	0.003762	0.0009061	-0.001236	0.002872	-0.001388	-0.005709	0.003644	0.5105	0.00208	0.
6	S1	plddt_adj	f_polar	1125	153	0.04345	0.0355	0.004891	-0.00371	0.01361	-0.008199	-0.01345	-0.002629	0.01309	0.00366	0.02258	0.002	0.004419	0
6	S1	plddt_adj	f_small	1125	153	0.03549	0.029	0.0007368	-0.006037	0.006282	-0.003701	-0.008562	0.0009418	0.004438	-0.002577	0.01094	0.171	0.003243	0.
6	S1	plddt_adj	f_gly	1125	153	0.02119	0.01906	0.001037	-0.003099	0.005187	0.001563	-0.001446	0.004612	-0.0005267	-0.006	0.004661	0.799	0.002136	0
6	S1	plddt_adj	f_pro	1125	153	0.01526	0.01308	-0.0005679	-0.003964	0.003073	0.001087	-0.001296	0.003536	-0.001655	-0.005567	0.002743	0.3515	0.001737	0.
6	S1	plddt_adj	f_ivywrel	1125	153	0.04007	0.03307	0.002289	-0.005342	0.01039	0.00936	0.004814	0.01396	-0.007071	-0.01623	0.001897	0.0595	0.003742	0
6	S1	plddt_adj	sasa_mean	1125	153	3.606	2.941	-0.1688	-0.7506	0.364	-0.2557	-0.7308	0.2103	0.08689	-0.653	0.7879	0.7905	0.3026	0
6	S1	plddt_adj	sasa_total	1125	153	47.89	112.8	-3.191	-11.52	4.192	-6.778	-29.34	14.27	3.586	-20.41	28.04	0.731	10.08	0
6	S1	plddt_adj	burial	1125	153	5.395	4.216	0.2308	-0.5293	1.07	0.9483	0.3492	1.582	-0.7174	-1.672	0.3037	0.087	0.4082	0.
6	S1	plddt_adj	rgyr	1125	153	0.1301	0.1869	-0.006879	-0.03351	0.0214	0.002714	-0.02636	0.0333	-0.009593	-0.05241	0.0287	0.55	0.01682	0.
6	S1	plddt_adj	cavity_open_vol	1125	153	234.1	-	-2.916	-40.09	31.21	-	-	-	-	-	-	0.851	15.73	0
6	S1	plddt_adj	cavity_enclosed_vol	1125	153	32.51	-	3.011	-2.934	8.934	-	-	-	-	-	-	0.226	2.495	0.
6	S8	plddt_adj	n	6178	956	1.067	2.301	0.04631	-0.01198	0.1034	-0.003166	-0.1532	0.1549	0.04947	-0.09149	0.1757	0.334	0.05054	0
6	S8	plddt_adj	hydropathy	6178	956	0.2352	0.1886	0.009115	-0.004142	0.02193	-0.00226	-0.01236	0.007655	0.01138	-0.004733	0.02764	0.126	0.007481	0
6	S8	plddt_adj	resvol	6178	956	2.824	2.521	0.01659	-0.1559	0.1795	0.264	0.1339	0.394	-0.2474	-0.4594	-0.03517	0.0085	0.09606	0
6	S8	plddt_adj	f_pos	6178	956	0.01261	0.0154	9.15e-05	-0.0004453	0.0006321	0.001137	0.0003807	0.001924	-0.001046	-0.001999	-0.0001101	0.031	0.000472	0.
6	S8	plddt_adj	f_neg	6178	956	0.02886	0.01841	0.0007035	-0.0008645	0.00223	0.0009007	-0.000114	0.002023	-0.0001972	-0.002073	0.00154	0.8085	0.0008214	0.
6	S8	plddt_adj	net_charge	6178	956	0.4119	0.9469	-0.009954	-0.03214	0.01263	0.01329	-0.04323	0.06934	-0.02325	-0.07993	0.03252	0.366	0.02539	0.
6	S8	plddt_adj	f_aromatic	6178	956	0.02104	0.0152	-0.001372	-0.002725	-0.0001708	0.0004482	-0.0004238	0.001399	-0.00182	-0.003403	-0.0003577	0.0085	0.0006587	0
6	S8	plddt_adj	f_polar	6178	956	0.0499	0.02945	-0.002949	-0.005332	-0.000561	-0.002881	-0.004375	-0.00137	-6.72e-05	-0.002885	0.002832	0.969	0.001383	0

Supplementary Table S3 (continued)

radius	family	adjustment	descriptor	n	clades	within_sd_pocket	within_sd_control	slope_pocket	slope_pocket_lo	slope_pocket_hi	slope_control	slope_control_lo	slope_control_hi	difference	difference_lo	difference_hi	p_perm	null_sd	d
6	S8	plddt_adj	f_small	6178	956	0.03993	0.02831	-0.001247	-0.003377	0.0009545	-0.002631	-0.004094	-0.001207	0.001384	-0.001248	0.004031	0.267	0.001244	0.
6	S8	plddt_adj	f_gly	6178	956	0.01879	0.0176	-0.0001864	-0.001146	0.0008148	-0.0009997	-0.001885	-0.0001688	0.0008133	-0.0004913	0.002086	0.1735	0.0006054	0.
6	S8	plddt_adj	f_pro	6178	956	0.01553	0.01128	-0.0005693	-0.001797	0.0004794	0.0007931	0.0002373	0.001337	-0.001362	-0.002839	-0.0001287	0.008	0.0005249	0
6	S8	plddt_adj	f_ivywrel	6178	956	0.04568	0.03141	0.001997	-0.0005146	0.004737	0.005533	0.003789	0.007248	-0.003536	-0.006362	-0.000292	0.0135	0.001404	0.0
6	S8	plddt_adj	sasa_mean	6178	956	3.415	3.873	0.00716	-0.2097	0.2194	0.2075	-0.01025	0.4298	-0.2004	-0.4304	0.03667	0.082	0.1159	0
6	S8	plddt_adj	sasa_total	6178	956	33.41	154.6	0.2297	-1.677	2.007	6.459	-1.532	15.03	-6.229	-15.4	1.939	0.136	4.118	0
6	S8	plddt_adj	burial	6178	956	5.647	5.801	-0.1136	-0.4902	0.2466	-0.07546	-0.3981	0.245	-0.03813	-0.3685	0.3216	0.815	0.1718	0.
6	S8	plddt_adj	rgyr	6178	956	0.1869	0.2524	0.001279	-0.0109	0.01243	-0.002791	-0.017	0.01109	0.00407	-0.00997	0.0185	0.5355	0.006684	0.
6	S8	plddt_adj	cavity_open_vol	6178	956	283.6	-	9.744	-5.929	26.05	-	-	-	-	-	-	0.163	7.031	0.
6	S8	plddt_adj	cavity_enclosed_vol	6178	956	39.89	-	-0.04974	-2.135	1.882	-	-	-	-	-	-	0.9625	0.9694	0
6	S9	plddt_adj	n	3332	374	0.4644	1.159	-0.02615	-0.05272	0.0004061	0.006246	-0.05846	0.06912	-0.0324	-0.1002	0.03543	0.3325	0.03389	0
6	S9	plddt_adj	hydropathy	3332	374	0.1673	0.2122	0.007463	-0.00281	0.01838	-0.00659	-0.02032	0.006518	0.01405	-0.00395	0.03119	0.074	0.007761	0.
6	S9	plddt_adj	resvol	3332	374	2.855	2.299	-0.1304	-0.3056	0.04645	0.1851	0.04673	0.3254	-0.3155	-0.5396	-0.1057	0.0065	0.112	0.
6	S9	plddt_adj	f_pos	3332	374	0.02078	0.01556	-0.000225	-0.001583	0.001141	0.001055	-4.28e-05	0.00212	-0.00128	-0.002948	0.000408	0.1395	0.0008447	0.
6	S9	plddt_adj	f_neg	3332	374	0.01066	0.0183	-0.0006698	-0.001499	1.63e-05	0.0019	0.0007874	0.002999	-0.002569	-0.003897	-0.001265	0	0.0006165	
6	S9	plddt_adj	net_charge	3332	374	0.2633	0.7872	0.001523	-0.01559	0.01908	-0.02869	-0.08132	0.02337	0.03022	-0.02468	0.08566	0.242	0.02567	0.
6	S9	plddt_adj	f_aromatic	3332	374	0.02199	0.01716	-0.0005974	-0.001979	0.000724	0.0003681	-0.0008704	0.001529	-0.0009655	-0.002791	0.0008579	0.2535	0.000851	0.
6	S9	plddt_adj	f_polar	3332	374	0.02765	0.03038	0.000204	-0.001395	0.001792	-0.005417	-0.007774	-0.002958	0.005621	0.002783	0.008223	0	0.001249	
6	S9	plddt_adj	f_small	3332	374	0.02421	0.02488	0.0009351	-0.0004451	0.0023	-0.001721	-0.003489	-1.60e-05	0.002656	0.0003472	0.004976	0.0175	0.001071	0.
6	S9	plddt_adj	f_gly	3332	374	0.0231	0.01478	0.0006626	-0.0008495	0.00219	-0.0002541	-0.001252	0.000722	0.0009168	-0.0008693	0.002765	0.269	0.0008316	0.
6	S9	plddt_adj	f_pro	3332	374	0.008844	0.01199	7.87e-05	-0.0005411	0.0008696	0.001552	0.0007635	0.002315	-0.001473	-0.00235	-0.0004772	0.0005	0.0004828	0.0
6	S9	plddt_adj	f_ivywrel	3332	374	0.02976	0.0313	-0.001507	-0.0032	0.0002681	0.005479	0.003556	0.007249	-0.006986	-0.009567	-0.004279	0	0.001325	
6	S9	plddt_adj	sasa_mean	3332	374	2.223	2.886	-0.1165	-0.2676	0.02917	-0.04971	-0.2229	0.1304	-0.06676	-0.3081	0.1562	0.485	0.09854	0.
6	S9	plddt_adj	sasa_total	3332	374	24.49	104.3	-1.837	-3.553	-0.245	-1.86	-8.665	5.055	0.02271	-7.54	7.339	0.997	3.348	0
6	S9	plddt_adj	burial	3332	374	2.81	3.897	0.1825	-0.00129	0.3698	0.396	0.1652	0.6267	-0.2135	-0.5142	0.09826	0.11	0.1331	
6	S9	plddt_adj	rgyr	3332	374	0.1776	0.141	-0.009723	-0.02103	0.0006783	-0.001997	-0.0104	0.006918	-0.007726	-0.02202	0.005677	0.2325	0.006516	0.
6	S9	plddt_adj	cavity_open_vol	3332	374	119.6	-	-7.162	-15.38	0.09558	-	-	-	-	-	-	0.078	3.985	0.
6	S9	plddt_adj	cavity_enclosed_vol	3332	374	33.72	-	0.2372	-1.785	2.293	-	-	-	-	-	-	0.8155	1.005	0.
6	all	standardized	n	10635	1481	0.8892	1.97	0.001929	-0.03809	0.04331	-0.007541	-0.05553	0.0389	0.009469	-0.03549	0.05416	0.624	0.01938	0.
6	all	standardized	hydropathy	10635	1481	0.2123	0.1954	0.03128	-0.006041	0.07324	-0.02468	-0.06344	0.01152	0.05596	0.001163	0.1146	0.031	0.0257	0.0
6	all	standardized	resvol	10635	1481	2.905	2.454	-0.01917	-0.06242	0.02158	0.0993	0.06053	0.1374	-0.1185	-0.1742	-0.06386	0	0.02738	
6	all	standardized	f_pos	10635	1481	0.01642	0.01574	-0.002482	-0.04374	0.03444	0.07777	0.03796	0.1192	-0.08025	-0.1371	-0.02679	0.002	0.02621	0.00
6	all	standardized	f_neg	10635	1481	0.02356	0.01872	-0.0005649	-0.04001	0.03716	0.07835	0.03556	0.1203	-0.07891	-0.1355	-0.02375	0.0025	0.02553	0.00
6	all	standardized	net_charge	10635	1481	0.3674	0.915	-0.006976	-0.04649	0.03264	-0.005715	-0.04789	0.03489	-0.001262	-0.05341	0.05245	0.964	0.02627	0
6	all	standardized	f_aromatic	10635	1481	0.02191	0.01579	-0.04913	-0.09079	-0.008569	0.02844	-0.01484	0.07128	-0.07757	-0.1336	-0.0196	0.0055	0.02755	0.0
6	all	standardized	f_polar	10635	1481	0.04344	0.03044	-0.02509	-0.06481	0.01297	-0.1362	-0.1801	-0.09224	0.1111	0.05196	0.172	0	0.02591	
6	all	standardized	f_small	10635	1481	0.03526	0.02737	-0.00641	-0.04366	0.03265	-0.08745	-0.1258	-0.04643	0.08104	0.02517	0.1388	0.002	0.02712	0.00
6	all	standardized	f_gly	10635	1481	0.02049	0.01694	0.01153	-0.03022	0.0514	-0.03414	-0.07378	0.004775	0.04567	-0.01207	0.1027	0.0735	0.02548	0.
6	all	standardized	f_pro	10635	1481	0.01377	0.01171	-0.02791	-0.08419	0.02445	0.09252	0.05476	0.1313	-0.1204	-0.1929	-0.0592	0.0005	0.02796	0.00
6	all	standardized	f_ivywrel	10635	1481	0.04074	0.03156	0.01589	-0.02708	0.05623	0.1831	0.1462	0.2222	-0.1672	-0.223	-0.1141	0	0.02562	
6	all	standardized	sasa_mean	10635	1481	3.114	3.498	-0.01407	-0.05923	0.03205	0.02565	-0.01522	0.06754	-0.03972	-0.09088	0.0106	0.0815	0.02262	0.
6	all	standardized	sasa_total	10635	1481	32.86	136.6	-0.02553	-0.06741	0.01542	0.01599	-0.02346	0.05673	-0.04152	-0.1074	0.01842	0.1435	0.02751	0.

Supplementary Table S3 (continued)

radius	family	adjustment	descriptor	n	clades	within_sd_pocket	within_sd_control	slope_pocket	slope_pocket_lo	slope_pocket_hi	slope_control	slope_control_lo	slope_control_hi	difference	difference_lo	difference_hi	p_perm	null_sd	d
6	all	standardized	burial	10635	1481	4.907	5.119	-0.0006927	-0.04735	0.04506	0.03062	-0.01186	0.0692	-0.03132	-0.07719	0.01799	0.1665	0.02283	0.
6	all	standardized	rgyr	10635	1481	0.1788	0.2166	-0.02229	-0.07265	0.01813	-0.01733	-0.05922	0.02179	-0.004958	-0.05723	0.044	0.833	0.02289	0.
6	all	standardized	plddt	10635	1481	2.373	1.93	0.02886	-0.01029	0.07092	0.05534	0.01655	0.09494	-0.02648	-0.07233	0.02207	0.277	0.02426	0.
6	all	standardized	cavity_open_vol	10635	1481	238.7	-	0.02367	-0.02219	0.06835	-	-	-	-	-	-	0.245	0.02052	0.
6	all	standardized	cavity_enclosed_vol	10635	1481	37.32	-	0.01218	-0.02477	0.05142	-	-	-	-	-	-	0.5175	0.01899	0.
6	S1	standardized	n	1125	153	0.7488	1.907	-0.09468	-0.2683	0.07886	0.03039	-0.1365	0.1996	-0.1251	-0.3008	0.04178	0.135	0.08322	0.
6	S1	standardized	hydropathy	1125	153	0.1985	0.1795	-0.1097	-0.2915	0.07805	-0.09961	-0.2575	0.04334	-0.01006	-0.2582	0.243	0.9275	0.102	0
6	S1	standardized	resvol	1125	153	3.438	2.511	-0.04016	-0.2185	0.1527	0.1462	0.003088	0.2922	-0.1863	-0.4345	0.04942	0.0665	0.103	0.
6	S1	standardized	f_pos	1125	153	0.01986	0.01798	0.01781	-0.1797	0.2368	0.1658	0.003685	0.3313	-0.148	-0.3999	0.1161	0.164	0.1057	0.
6	S1	standardized	f_neg	1125	153	0.01828	0.02141	-0.06017	-0.1729	0.04678	0.1824	0.01116	0.3495	-0.2426	-0.4318	-0.05539	0.0105	0.09526	0.0
6	S1	standardized	net_charge	1125	153	0.3719	1.073	0.06419	-0.1025	0.2341	-0.03599	-0.2281	0.1446	0.1002	-0.1396	0.3436	0.3065	0.1007	0.
6	S1	standardized	f_aromatic	1125	153	0.0259	0.01458	-0.01893	-0.1724	0.1395	0.06542	-0.07922	0.1955	-0.08434	-0.2892	0.1378	0.4055	0.09769	0.
6	S1	standardized	f_polar	1125	153	0.04345	0.0355	0.1173	-0.08945	0.3307	-0.2304	-0.3778	-0.07849	0.3477	0.1095	0.5786	0.002	0.1123	0
6	S1	standardized	f_small	1125	153	0.03549	0.029	0.02318	-0.1471	0.1947	-0.127	-0.2989	0.03631	0.1502	-0.05811	0.3482	0.136	0.1001	0.
6	S1	standardized	f_gly	1125	153	0.02119	0.01906	0.05363	-0.139	0.2327	0.08179	-0.0764	0.235	-0.02816	-0.2793	0.2126	0.7875	0.1096	0.
6	S1	standardized	f_pro	1125	153	0.01526	0.01308	-0.0347	-0.259	0.2117	0.08448	-0.1023	0.2712	-0.1192	-0.394	0.1833	0.3395	0.1219	0.
6	S1	standardized	f_ivywrel	1125	153	0.04007	0.03307	0.05378	-0.141	0.2722	0.2819	0.1362	0.4314	-0.2281	-0.4765	0.01568	0.024	0.1028	0
6	S1	standardized	sasa_mean	1125	153	3.606	2.941	-0.04	-0.2159	0.1335	-0.08159	-0.2508	0.09073	0.04159	-0.1912	0.2635	0.6535	0.09124	0.
6	S1	standardized	sasa_total	1125	153	47.89	112.8	-0.06132	-0.2499	0.109	-0.06018	-0.2471	0.1366	-0.001141	-0.2983	0.2688	0.992	0.1159	0
6	S1	standardized	burial	1125	153	5.395	4.216	0.03354	-0.1586	0.1911	0.2191	0.06897	0.38	-0.1855	-0.4066	0.02199	0.0365	0.0883	0.
6	S1	standardized	rgyr	1125	153	0.1301	0.1869	-0.05302	-0.2572	0.1731	0.008265	-0.1672	0.1946	-0.06129	-0.355	0.2204	0.5755	0.1072	0
6	S1	standardized	plddt	1125	153	2.918	2.122	0.02709	-0.1273	0.1659	0.1542	-0.03395	0.3563	-0.1271	-0.3591	0.1016	0.225	0.1032	0.
6	S1	standardized	cavity_open_vol	1125	153	234.1	-	0.0003763	-0.1859	0.2057	-	-	-	-	-	-	0.997	0.08312	0
6	S1	standardized	cavity_enclosed_vol	1125	153	32.51	-	0.09621	-0.08583	0.2829	-	-	-	-	-	-	0.2165	0.07772	0.
6	S8	standardized	n	6178	956	1.067	2.301	0.02737	-0.03178	0.08493	-0.01559	-0.08103	0.0491	0.04296	-0.01843	0.1017	0.073	0.02413	0.
6	S8	standardized	hydropathy	6178	956	0.2352	0.1886	0.03966	-0.01547	0.0924	-0.01173	-0.06489	0.03699	0.05139	-0.0201	0.128	0.143	0.03505	0.
6	S8	standardized	resvol	6178	956	2.824	2.521	0.0008254	-0.05915	0.05915	0.1051	0.05328	0.1562	-0.1043	-0.18	-0.02553	0.0045	0.03548	0.
6	S8	standardized	f_pos	6178	956	0.01261	0.0154	0.002332	-0.04142	0.04428	0.07389	0.02579	0.1244	-0.07156	-0.1381	-0.006461	0.034	0.03306	0.
6	S8	standardized	f_neg	6178	956	0.02886	0.01841	0.01884	-0.02972	0.07133	0.0495	-0.01055	0.1129	-0.03066	-0.1067	0.04214	0.3495	0.0344	0.
6	S8	standardized	net_charge	6178	956	0.4119	0.9469	-0.02032	-0.07501	0.03243	0.01455	-0.04323	0.06968	-0.03487	-0.1037	0.03807	0.3115	0.03484	0.
6	S8	standardized	f_aromatic	6178	956	0.02104	0.0152	-0.06728	-0.1323	-0.008342	0.0285	-0.02987	0.08669	-0.09578	-0.1747	-0.01437	0.012	0.03623	0
6	S8	standardized	f_polar	6178	956	0.0499	0.02945	-0.05285	-0.1042	-0.004583	-0.0979	-0.1481	-0.04513	0.04505	-0.03138	0.1176	0.197	0.03464	0.
6	S8	standardized	f_small	6178	956	0.03993	0.02831	-0.02733	-0.08029	0.03166	-0.09397	-0.148	-0.04333	0.06665	-0.008918	0.1451	0.07	0.03578	0.
6	S8	standardized	f_gly	6178	956	0.01879	0.0176	-0.007433	-0.06247	0.04608	-0.0575	-0.109	-0.006314	0.05007	-0.0242	0.1257	0.1475	0.03427	0.
6	S8	standardized	f_pro	6178	956	0.01553	0.01128	-0.04136	-0.1261	0.02776	0.06969	0.0222	0.1198	-0.111	-0.2131	-0.02613	0.0035	0.03686	0.
6	S8	standardized	f_ivywrel	6178	956	0.04568	0.03141	0.04032	-0.01643	0.09605	0.1769	0.1212	0.2268	-0.1366	-0.2072	-0.05766	0	0.03532	
6	S8	standardized	sasa_mean	6178	956	3.415	3.873	0.004871	-0.05565	0.06722	0.05638	0.0006302	0.1162	-0.05151	-0.119	0.01363	0.099	0.03134	0
6	S8	standardized	sasa_total	6178	956	33.41	154.6	0.002505	-0.05503	0.05892	0.0379	-0.01649	0.0898	-0.03539	-0.1217	0.04664	0.3435	0.0368	0.
6	S8	standardized	burial	6178	956	5.647	5.801	-0.02614	-0.09482	0.03961	-0.01508	-0.0734	0.04225	-0.01105	-0.07249	0.05309	0.7185	0.03041	0.
6	S8	standardized	rgyr	6178	956	0.1869	0.2524	0.001405	-0.06777	0.0581	-0.02103	-0.07526	0.03269	0.02243	-0.04574	0.08368	0.4375	0.02924	0.
6	S8	standardized	plddt	6178	956	2.748	2.253	0.03657	-0.01831	0.09902	0.05062	-0.006452	0.1074	-0.01405	-0.07823	0.05211	0.654	0.03251	0.
6	S8	standardized	cavity_open_vol	6178	956	283.6	-	0.05019	-0.01338	0.111	-	-	-	-	-	-	0.067	0.02718	0.

Supplementary Table S3 (continued)

radius	family	adjustment	descriptor	n	clades	within_sd_pocket	within_sd_control	slope_pocket	slope_pocket_lo	slope_pocket_hi	slope_control	slope_control_lo	slope_control_hi	difference	difference_lo	difference_hi	p_perm	null_sd	
6	S8	standardized	cavity_enclosed_vol	6178	956	39.89	-	0.007708	-0.04284	0.05989	-	-	-	-	-	-	0.7715	0.02523	0.
6	S9	standardized	n	3332	374	0.4644	1.159	-0.05811	-0.12	-0.002311	0.005175	-0.05275	0.06056	-0.06329	-0.1375	0.01147	0.096	0.03804	0.
6	S9	standardized	hydropathy	3332	374	0.1673	0.2122	0.04464	-0.01349	0.1113	-0.03088	-0.09633	0.03144	0.07552	-0.0105	0.1666	0.0685	0.0414	0.
6	S9	standardized	resvol	3332	374	2.855	2.299	-0.04665	-0.1043	0.01112	0.08105	0.01617	0.1428	-0.1277	-0.2085	-0.04689	0.0015	0.04204	0.06
6	S9	standardized	f_pos	3332	374	0.02078	0.01556	-0.01062	-0.07591	0.05583	0.06749	-0.004559	0.1354	-0.07811	-0.1742	0.01385	0.0785	0.04504	0.
6	S9	standardized	f_neg	3332	374	0.01066	0.0183	-0.06322	-0.1437	-0.0009437	0.1036	0.041	0.162	-0.1668	-0.2657	-0.07995	0	0.04303	
6	S9	standardized	net_charge	3332	374	0.2633	0.7872	0.005872	-0.06244	0.072	-0.03643	-0.1001	0.02819	0.0423	-0.05442	0.1327	0.3555	0.04384	0.
6	S9	standardized	f_aromatic	3332	374	0.02199	0.01716	-0.02818	-0.08984	0.03248	0.02208	-0.04563	0.08876	-0.05026	-0.1408	0.04319	0.256	0.04437	0.
6	S9	standardized	f_polar	3332	374	0.02765	0.03038	0.007581	-0.05071	0.06581	-0.1779	-0.2512	-0.1007	0.1854	0.09526	0.2728	0	0.04463	
6	S9	standardized	f_small	3332	374	0.02421	0.02488	0.0395	-0.01856	0.09695	-0.06944	-0.1413	-0.003143	0.1089	0.01744	0.2062	0.01	0.0436	0.0
6	S9	standardized	f_gly	3332	374	0.0231	0.01478	0.02963	-0.0386	0.09545	-0.01726	-0.08347	0.04684	0.04689	-0.04563	0.1407	0.27	0.04313	0.
6	S9	standardized	f_pro	3332	374	0.008844	0.01199	0.008389	-0.06241	0.09609	0.129	0.06472	0.1934	-0.1206	-0.2056	-0.0316	0.006	0.04471	0.
6	S9	standardized	f_ivywrel	3332	374	0.02976	0.0313	-0.05021	-0.1102	0.01359	0.1753	0.1196	0.233	-0.2255	-0.3089	-0.1372	0	0.0427	
6	S9	standardized	sasa_mean	3332	374	2.223	2.886	-0.05418	-0.119	0.006491	-0.01885	-0.07601	0.04433	-0.03533	-0.1315	0.0486	0.351	0.038	0.
6	S9	standardized	sasa_total	3332	374	24.49	104.3	-0.07681	-0.1478	-0.005752	-0.01903	-0.08496	0.04184	-0.05778	-0.1584	0.03985	0.183	0.0437	0.
6	S9	standardized	burial	3332	374	2.81	3.897	0.06805	0.002334	0.1364	0.1031	0.04076	0.1619	-0.03505	-0.1215	0.06301	0.3805	0.04057	0.
6	S9	standardized	rgyr	3332	374	0.1776	0.141	-0.05672	-0.1183	0.002458	-0.01538	-0.07854	0.04996	-0.04135	-0.133	0.04452	0.3105	0.0409	0.
6	S9	standardized	plddt	3332	374	1.038	0.9773	0.0129	-0.04726	0.0681	0.05604	-0.008009	0.1176	-0.04314	-0.1101	0.02957	0.1995	0.03336	0.
6	S9	standardized	cavity_open_vol	3332	374	119.6	-	-0.05837	-0.1287	0.001951	-	-	-	-	-	-	0.0975	0.03512	0.
6	S9	standardized	cavity_enclosed_vol	3332	374	33.72	-	0.006986	-0.05227	0.0676	-	-	-	-	-	-	0.8115	0.02965	0.

Supplementary Table S4. Pocket versus matched control at an 8 Å radius

As Supplementary Table S1 with an 8 Å radius.

radius	family	adjustment	descriptor	n	clades	within_sd_pocket	within_sd_control	slope_pocket	slope_pocket_lo	slope_pocket_hi	slope_control	slope_control_lo	slope_control_hi	difference	difference_lo	difference_hi	p_perm	null_sd	d
8	all	raw	n	10669	1484	1.289	1.974	-0.05032	-0.1152	0.01237	-0.01385	-0.1042	0.07815	-0.03647	-0.1218	0.03492	0.22	0.03003	0.
8	all	raw	hydropathy	10669	1484	0.1619	0.1953	0.001055	-0.005443	0.007398	-0.00483	-0.01257	0.002594	0.005885	-0.00392	0.01605	0.235	0.004815	0.
8	all	raw	resvol	10669	1484	2.457	2.453	-0.07003	-0.1851	0.04333	0.2437	0.143	0.3364	-0.3138	-0.4749	-0.1582	0	0.0674	
8	all	raw	f_pos	10669	1484	0.01305	0.01574	0.0001971	-0.000299	0.0006812	0.001209	0.0006066	0.001864	-0.001012	-0.001739	-0.0002747	0.008	0.0003802	0
8	all	raw	f_neg	10669	1484	0.01538	0.0187	-0.0001743	-0.0007881	0.0004811	0.001445	0.000692	0.002259	-0.001619	-0.002599	-0.0006705	0	0.0004458	
8	all	raw	net_charge	10669	1484	0.4817	0.9147	0.01125	-0.008986	0.03016	-0.004951	-0.04389	0.03185	0.0162	-0.02286	0.05736	0.392	0.01877	0.
8	all	raw	f_aromatic	10669	1484	0.01953	0.0158	-0.001101	-0.001967	-0.0003095	0.0004594	-0.0002436	0.001147	-0.001561	-0.002695	-0.00047	0.0005	0.0004729	0.00
8	all	raw	f_polar	10669	1484	0.03152	0.03043	-0.0004494	-0.001796	0.0007962	-0.004095	-0.005408	-0.002815	0.003646	0.001726	0.005568	0	0.0008117	
8	all	raw	f_small	10669	1484	0.02906	0.02735	0.0001185	-0.001199	0.001441	-0.002403	-0.003531	-0.001272	0.002522	0.0008005	0.004353	0.0005	0.0007493	0.00
8	all	raw	f_gly	10669	1484	0.01712	0.01693	-0.0001513	-0.0008189	0.0005171	-0.0005929	-0.001229	4.81e-05	0.0004416	-0.0004018	0.001343	0.3155	0.0004401	0.
8	all	raw	f_pro	10669	1484	0.009662	0.0117	0.0001457	-0.0002811	0.0005379	0.001089	0.0006345	0.001551	-0.0009433	-0.001603	-0.0003407	0.0005	0.0002831	0.00
8	all	raw	f_ivywrel	10669	1484	0.0319	0.03154	0.0002467	-0.001074	0.00162	0.005722	0.00449	0.006849	-0.005475	-0.007331	-0.003615	0	0.0008609	
8	all	raw	sasa_mean	10669	1484	2.598	3.499	-0.01997	-0.1387	0.1038	0.08826	-0.06016	0.2362	-0.1082	-0.2556	0.04673	0.1375	0.07209	0.
8	all	raw	sasa_total	10669	1484	48.69	136.7	-1.661	-4.064	0.5768	2.168	-3.047	7.536	-3.83	-9.933	2.055	0.2015	2.924	0.
8	all	raw	burial	10669	1484	4.358	5.124	0.0445	-0.1738	0.2626	0.1589	-0.05749	0.3653	-0.1144	-0.3248	0.1175	0.292	0.1083	0.
8	all	raw	rgyr	10669	1484	0.1888	0.217	-0.0051	-0.01475	0.003775	-0.003822	-0.0125	0.004619	-0.001278	-0.01273	0.009561	0.803	0.005019	0
8	all	raw	plddt	10669	1484	2.046	1.942	0.08368	-0.007293	0.1777	0.1079	0.02435	0.1829	-0.02418	-0.1318	0.07565	0.6175	0.04803	0.
8	all	raw	cavity_open_vol	10669	1484	239.3	-	5.222	-5.474	16.55	-	-	-	-	-	-	0.2815	4.821	0.
8	all	raw	cavity_enclosed_vol	10669	1484	37.33	-	0.2507	-1.185	1.654	-	-	-	-	-	-	0.7255	0.7016	0.
8	S1	raw	n	1126	153	1.554	1.906	-0.07765	-0.4309	0.2498	0.05823	-0.2548	0.3796	-0.1359	-0.319	0.045	0.205	0.1084	0.
8	S1	raw	hydropathy	1126	153	0.179	0.1795	-0.02505	-0.05689	0.00785	-0.0178	-0.04617	0.01049	-0.007244	-0.05418	0.03487	0.6745	0.01797	0.
8	S1	raw	resvol	1126	153	2.601	2.511	-0.07262	-0.5778	0.4266	0.3659	0.005057	0.733	-0.4385	-1.065	0.1196	0.118	0.2766	0.
8	S1	raw	f_pos	1126	153	0.01399	0.01803	-0.0008201	-0.003711	0.001963	0.002962	0.0001095	0.005755	-0.003782	-0.007488	2.19e-05	0.022	0.001653	0.
8	S1	raw	f_neg	1126	153	0.01398	0.0214	0.001412	-0.001157	0.004203	0.003901	0.0003446	0.00737	-0.00249	-0.006644	0.001502	0.181	0.001877	0.
8	S1	raw	net_charge	1126	153	0.445	1.074	-0.04759	-0.152	0.04376	-0.03924	-0.244	0.1537	-0.008348	-0.2115	0.1951	0.9105	0.08603	0.
8	S1	raw	f_aromatic	1126	153	0.01917	0.01457	-0.0008257	-0.004495	0.003466	0.0009562	-0.001072	0.002996	-0.001782	-0.00601	0.003038	0.3795	0.001989	0.
8	S1	raw	f_polar	1126	153	0.03531	0.0355	0.003863	-0.002514	0.01038	-0.00817	-0.0136	-0.002895	0.01203	0.003517	0.01999	0.001	0.003778	0.
8	S1	raw	f_small	1126	153	0.02944	0.02901	0.000763	-0.002699	0.004768	-0.00367	-0.008265	0.001111	0.004433	-0.001495	0.01079	0.1305	0.002915	0.
8	S1	raw	f_gly	1126	153	0.01882	0.01906	-0.001421	-0.004163	0.001166	0.001566	-0.001632	0.004619	-0.002987	-0.006743	0.001034	0.143	0.002039	0.
8	S1	raw	f_pro	1126	153	0.01252	0.01307	0.0008764	-0.001343	0.003144	0.001108	-0.001146	0.003541	-0.0002319	-0.003238	0.002659	0.8755	0.001407	0.
8	S1	raw	f_ivywrel	1126	153	0.03604	0.03306	-0.001938	-0.008391	0.005417	0.009313	0.00455	0.01393	-0.01125	-0.01861	-0.003155	0.001	0.003554	0.
8	S1	raw	sasa_mean	1126	153	2.72	2.94	0.1051	-0.3058	0.5149	-0.2404	-0.7007	0.2408	0.3455	-0.2393	0.9237	0.1835	0.2607	0.
8	S1	raw	sasa_total	1126	153	47.68	112.7	1.429	-4.975	7.408	-6.8	-28.2	13.71	8.228	-14.02	30.39	0.402	9.771	0.
8	S1	raw	burial	1126	153	4.484	4.214	-0.0125	-0.6961	0.6683	0.9236	0.2699	1.558	-0.9361	-1.756	-0.07934	0.0145	0.383	0.0
8	S1	raw	rgyr	1126	153	0.1669	0.1868	0.001679	-0.02512	0.02907	0.00155	-0.03227	0.03553	0.0001298	-0.0339	0.03511	0.993	0.01544	0.
8	S1	raw	plddt	1126	153	2.659	2.147	0.04444	-0.3655	0.4205	0.3316	-0.04561	0.7504	-0.2871	-0.8204	0.2282	0.2415	0.2433	0.

Supplementary Table S4 (continued)

radius	family	adjustment	descriptor	n	clades	within_sd_pocket	within_sd_control	slope_pocket	slope_pocket_lo	slope_pocket_hi	slope_control	slope_control_lo	slope_control_hi	difference	difference_lo	difference_hi	p_perm	null_sd	c
8	S1	raw	cavity_open_vol	1126	153	234	-	0.04331	-45.56	46.33	-	-	-	-	-	-	0.9985	19.28	0.
8	S1	raw	cavity_enclosed_vol	1126	153	32.52	-	3.11	-2.783	9.481	-	-	-	-	-	-	0.207	2.483	0.
8	S8	raw	n	6211	959	1.467	2.305	-0.08341	-0.1861	0.01866	-0.03406	-0.1845	0.1153	-0.04936	-0.1834	0.07437	0.288	0.04672	0.
8	S8	raw	hydropathy	6211	959	0.1734	0.1884	0.005583	-0.003345	0.01569	-0.002246	-0.01279	0.007405	0.00783	-0.005291	0.02203	0.2295	0.006388	0.
8	S8	raw	resvol	6211	959	2.618	2.52	-0.1043	-0.2808	0.07052	0.2652	0.1368	0.3901	-0.3694	-0.598	-0.1313	0	0.09769	
8	S8	raw	f_pos	6211	959	0.01273	0.01538	0.0002017	-0.0003999	0.0008004	0.001114	0.0003092	0.001928	-0.0009126	-0.001826	5.65e-05	0.0685	0.0004864	0.
8	S8	raw	f_neg	6211	959	0.01737	0.01838	-6.60e-05	-0.001102	0.0009661	0.0008764	-0.0001567	0.002019	-0.0009424	-0.002378	0.0003779	0.1175	0.0006009	0
8	S8	raw	net_charge	6211	959	0.5383	0.9461	0.01122	-0.01767	0.04048	0.01425	-0.04303	0.06551	-0.003027	-0.05558	0.05583	0.914	0.02731	0.
8	S8	raw	f_aromatic	6211	959	0.02104	0.01524	-0.001507	-0.002719	-0.0002859	0.0004512	-0.000498	0.001361	-0.001958	-0.003667	-0.0003605	0.0045	0.0006615	0.
8	S8	raw	f_polar	6211	959	0.03509	0.02943	-0.001965	-0.003759	-0.0001719	-0.002803	-0.004332	-0.001266	0.0008377	-0.001487	0.003233	0.4265	0.001066	0.
8	S8	raw	f_small	6211	959	0.0333	0.02828	0.0003393	-0.00183	0.002336	-0.002678	-0.00418	-0.00122	0.003017	0.0003177	0.005605	0.007	0.001123	0.0
8	S8	raw	f_gly	6211	959	0.01712	0.01758	-4.90e-05	-0.0009722	0.0008803	-0.001036	-0.001936	-0.0001662	0.0009873	-0.0002302	0.002177	0.0965	0.000598	0.
8	S8	raw	f_pro	6211	959	0.01027	0.01128	4.79e-05	-0.0005924	0.0006624	0.0007974	0.0002399	0.001369	-0.0007495	-0.001619	0.0001005	0.046	0.0003638	0.
8	S8	raw	f_ivywrel	6211	959	0.03442	0.03136	0.0004538	-0.001531	0.002411	0.005457	0.003657	0.007123	-0.005003	-0.007725	-0.00223	0	0.001252	
8	S8	raw	sasa_mean	6211	959	2.859	3.873	-0.03113	-0.2279	0.1586	0.2153	-0.002942	0.439	-0.2465	-0.4703	-0.0292	0.023	0.1057	0.
8	S8	raw	sasa_total	6211	959	51.21	154.8	-2.792	-6.496	0.6067	5.819	-2.2	13.14	-8.61	-17.92	0.8123	0.0455	4.318	0.
8	S8	raw	burial	6211	959	5.087	5.805	-0.002394	-0.3633	0.3294	-0.08268	-0.3999	0.2186	0.08029	-0.2531	0.4334	0.6155	0.1589	0.
8	S8	raw	rgyr	6211	959	0.1985	0.2528	-0.01042	-0.02552	0.003534	-0.005418	-0.01988	0.00828	-0.005	-0.02392	0.01173	0.493	0.007249	0.
8	S8	raw	plddt	6211	959	2.306	2.264	0.1196	-0.02059	0.2695	0.1153	-0.01058	0.2428	0.004293	-0.1569	0.1685	0.9585	0.07329	0.
8	S8	raw	cavity_open_vol	6211	959	284.1	-	13.45	-3.713	31.7	-	-	-	-	-	-	0.0795	7.761	0.
8	S8	raw	cavity_enclosed_vol	6211	959	39.9	-	-0.04475	-2.126	2.011	-	-	-	-	-	-	0.9645	0.9911	0.
8	S9	raw	n	3332	374	0.7047	1.159	0.006503	-0.03512	0.04731	0.005996	-0.05985	0.07848	0.000507	-0.06692	0.06574	0.986	0.03447	0
8	S9	raw	hydropathy	3332	374	0.1305	0.2122	-0.001467	-0.009705	0.006444	-0.006552	-0.02073	0.006963	0.005086	-0.01161	0.02137	0.493	0.007242	0.
8	S9	raw	resvol	3332	374	2.064	2.299	-0.01332	-0.126	0.1052	0.1864	0.03242	0.3312	-0.1997	-0.3796	-0.009723	0.0315	0.09533	0.0
8	S9	raw	f_pos	3332	374	0.01329	0.01556	0.0003672	-0.000509	0.001293	0.00105	-1.57e-05	0.002141	-0.0006827	-0.002025	0.0006202	0.284	0.000635	0.
8	S9	raw	f_neg	3332	374	0.01135	0.0183	-0.0006058	-0.001374	0.0002596	0.001895	0.0007024	0.002985	-0.002501	-0.003943	-0.0009643	0	0.0006609	
8	S9	raw	net_charge	3332	374	0.368	0.7872	0.02121	-0.004154	0.04583	-0.02868	-0.08112	0.02478	0.04989	-0.009286	0.1094	0.057	0.02573	0.
8	S9	raw	f_aromatic	3332	374	0.01645	0.01716	-0.0004797	-0.001403	0.0004487	0.000379	-0.0007656	0.001558	-0.0008587	-0.0023	0.0005829	0.2245	0.0007304	0.
8	S9	raw	f_polar	3332	374	0.02155	0.03038	0.0012	-0.0004371	0.002808	-0.005404	-0.007646	-0.003024	0.006604	0.003715	0.009463	0	0.001153	
8	S9	raw	f_small	3332	374	0.01853	0.02488	-0.0003699	-0.001485	0.0006576	-0.001728	-0.003539	-2.51e-05	0.001358	-0.0005804	0.003398	0.1545	0.0009505	0.
8	S9	raw	f_gly	3332	374	0.01652	0.01478	-8.58e-05	-0.001071	0.0009204	-0.0002551	-0.001228	0.0007457	0.0001693	-0.001165	0.001554	0.8045	0.0006729	0.
8	S9	raw	f_pro	3332	374	0.007029	0.01199	0.0001727	-0.0002245	0.0006354	0.001546	0.0007417	0.002307	-0.001374	-0.002211	-0.0005236	0.002	0.0004334	0.
8	S9	raw	f_ivywrel	3332	374	0.02473	0.0313	0.0003162	-0.0012	0.001783	0.005486	0.003748	0.007344	-0.005169	-0.007541	-0.00282	0	0.001185	
8	S9	raw	sasa_mean	3332	374	1.968	2.886	-0.02388	-0.1421	0.09952	-0.05439	-0.2125	0.1286	0.03051	-0.1678	0.2153	0.748	0.09669	0.
8	S9	raw	sasa_total	3332	374	43.97	104.3	-0.4114	-3.304	2.629	-1.985	-8.814	4.767	1.574	-5.181	8.524	0.632	3.484	0.
8	S9	raw	burial	3332	374	2.405	3.897	0.1298	-0.008939	0.2575	0.4017	0.1588	0.6392	-0.2719	-0.5122	-0.02583	0.0315	0.1274	0.0
8	S9	raw	rgyr	3332	374	0.1768	0.141	0.001928	-0.009473	0.01428	-0.002168	-0.01087	0.007083	0.004096	-0.009435	0.01684	0.4725	0.006145	0.
8	S9	raw	plddt	3332	374	1.043	0.9773	0.03112	-0.0317	0.09761	0.05477	-0.005103	0.1164	-0.02365	-0.0988	0.05196	0.508	0.03646	0.
8	S9	raw	cavity_open_vol	3332	374	119.6	-	-6.982	-15.61	0.2515	-	-	-	-	-	-	0.0905	4.156	0.
8	S9	raw	cavity_enclosed_vol	3332	374	33.72	-	0.2355	-1.643	2.248	-	-	-	-	-	-	0.807	1.01	0.
8	all	plddt_adj	n	10669	1484	1.289	1.974	-0.02692	-0.08259	0.02551	0.01689	-0.07389	0.1088	-0.04381	-0.1282	0.03575	0.147	0.03047	0.
8	all	plddt_adj	hydropathy	10669	1484	0.1619	0.1953	0.001182	-0.005115	0.007651	-0.004877	-0.01245	0.002733	0.006059	-0.004053	0.01599	0.1915	0.004598	0.

Supplementary Table S4 (continued)

radius	family	adjustment	descriptor	n	clades	within_sd_pocket	within_sd_control	slope_pocket	slope_pocket_lo	slope_pocket_hi	slope_control	slope_control_lo	slope_control_hi	difference	difference_lo	difference_hi	p_perm	null_sd	c
8	all	plddt_adj	resvol	10669	1484	2.457	2.453	-0.05663	-0.161	0.0496	0.2431	0.1444	0.3345	-0.2997	-0.4508	-0.151	0	0.06675	
8	all	plddt_adj	f_pos	10669	1484	0.01305	0.01574	0.0002558	-0.0002482	0.0007496	0.001213	0.000625	0.001841	-0.0009574	-0.001749	-0.0001948	0.008	0.0003667	0
8	all	plddt_adj	f_neg	10669	1484	0.01538	0.0187	-0.0001489	-0.0007846	0.0005357	0.001449	0.0006449	0.002247	-0.001597	-0.002599	-0.000625	0	0.0004357	
8	all	plddt_adj	net_charge	10669	1484	0.4817	0.9147	0.01082	-0.008733	0.03044	-0.005645	-0.04232	0.03253	0.01647	-0.0251	0.05789	0.3835	0.01887	0
8	all	plddt_adj	f_aromatic	10669	1484	0.01953	0.0158	-0.001091	-0.001899	-0.0002681	0.0004533	-0.0002457	0.001112	-0.001545	-0.002585	-0.0004676	0.002	0.0004822	0.00
8	all	plddt_adj	f_polar	10669	1484	0.03152	0.03043	-0.0005425	-0.001861	0.0007253	-0.004106	-0.005402	-0.002784	0.003564	0.0016	0.005555	0	0.0007947	
8	all	plddt_adj	f_small	10669	1484	0.02906	0.02735	-3.20e-05	-0.001208	0.001174	-0.002393	-0.003503	-0.001241	0.002361	0.0006743	0.00401	0.0015	0.000755	0.
8	all	plddt_adj	f_gly	10669	1484	0.01712	0.01693	-0.0002072	-0.0008636	0.0004398	-0.0005893	-0.001243	6.71e-05	0.000382	-0.0005698	0.001255	0.3925	0.0004375	0
8	all	plddt_adj	f_pro	10669	1484	0.009662	0.0117	0.0001561	-0.0002593	0.0005776	0.00109	0.0006704	0.001556	-0.0009336	-0.001557	-0.0003523	0.0005	0.0002731	0.
8	all	plddt_adj	f_ivywrel	10669	1484	0.0319	0.03154	0.0004079	-0.0008739	0.00173	0.005718	0.004577	0.006959	-0.00531	-0.007159	-0.003474	0	0.0008468	
8	all	plddt_adj	sasa_mean	10669	1484	2.598	3.499	-0.02856	-0.1503	0.09761	0.07955	-0.06418	0.2239	-0.1081	-0.2602	0.03938	0.141	0.07403	0.
8	all	plddt_adj	sasa_total	10669	1484	48.69	136.7	-1.533	-3.936	0.6626	2.804	-2.252	8.303	-4.336	-10.79	1.374	0.1295	2.835	0.
8	all	plddt_adj	burial	10669	1484	4.358	5.124	0.06742	-0.1761	0.2755	0.17	-0.04032	0.3757	-0.1025	-0.3325	0.128	0.326	0.1034	0.
8	all	plddt_adj	rgyr	10669	1484	0.1888	0.217	-0.003773	-0.01322	0.005225	-0.00137	-0.009882	0.007629	-0.002402	-0.0146	0.009419	0.6165	0.004844	0.
8	all	plddt_adj	cavity_open_vol	10669	1484	239.3	-	0.7156	-8.854	10.65	-	-	-	-	-	-	0.877	4.363	0.
8	all	plddt_adj	cavity_enclosed_vol	10669	1484	37.33	-	0.04636	-1.273	1.409	-	-	-	-	-	-	0.942	0.6981	0
8	S1	plddt_adj	n	1126	153	1.554	1.906	-0.06649	-0.3734	0.2194	0.06856	-0.2001	0.3679	-0.135	-0.3224	0.03986	0.2005	0.107	0.
8	S1	plddt_adj	hydropathy	1126	153	0.179	0.1795	-0.02466	-0.05383	0.006662	-0.01785	-0.0458	0.01241	-0.006813	-0.04885	0.03559	0.7015	0.01826	0.
8	S1	plddt_adj	resvol	1126	153	2.601	2.511	-0.06348	-0.4974	0.3825	0.3662	-0.001478	0.7374	-0.4297	-0.9801	0.1146	0.108	0.2688	0.
8	S1	plddt_adj	f_pos	1126	153	0.01399	0.01803	-0.0007889	-0.003797	0.001809	0.002969	-9.10e-05	0.005904	-0.003758	-0.007649	0.0002133	0.0285	0.001689	0.
8	S1	plddt_adj	f_neg	1126	153	0.01398	0.0214	0.00138	-0.0009617	0.0039	0.003915	0.0003015	0.007348	-0.002535	-0.00639	0.001468	0.1745	0.001859	0.
8	S1	plddt_adj	net_charge	1126	153	0.445	1.074	-0.04673	-0.1472	0.03922	-0.03955	-0.2228	0.1555	-0.007184	-0.2054	0.1781	0.929	0.0832	0.
8	S1	plddt_adj	f_aromatic	1126	153	0.01917	0.01457	-0.0007955	-0.004284	0.003304	0.000927	-0.001217	0.002863	-0.001723	-0.00591	0.003011	0.3595	0.001928	0.
8	S1	plddt_adj	f_polar	1126	153	0.03531	0.0355	0.00375	-0.002746	0.009686	-0.008178	-0.01344	-0.002585	0.01193	0.003526	0.01979	0.001	0.003818	0
8	S1	plddt_adj	f_small	1126	153	0.02944	0.02901	0.0007276	-0.002797	0.00451	-0.003688	-0.008552	0.0009453	0.004416	-0.00116	0.01067	0.133	0.002933	0.
8	S1	plddt_adj	f_gly	1126	153	0.01882	0.01906	-0.001442	-0.004231	0.001364	0.00156	-0.001513	0.004606	-0.003003	-0.007003	0.001011	0.148	0.002068	0.
8	S1	plddt_adj	f_pro	1126	153	0.01252	0.01307	0.0008936	-0.001396	0.003324	0.001096	-0.001271	0.003574	-0.0002028	-0.003205	0.002719	0.8895	0.001408	0.
8	S1	plddt_adj	f_ivywrel	1126	153	0.03604	0.03306	-0.001848	-0.008016	0.004836	0.009341	0.004841	0.01393	-0.01119	-0.01889	-0.00353	0.0005	0.003432	0
8	S1	plddt_adj	sasa_mean	1126	153	2.72	2.94	0.09597	-0.2818	0.4627	-0.2492	-0.7245	0.2106	0.3452	-0.2569	0.9132	0.1895	0.2613	0.
8	S1	plddt_adj	sasa_total	1126	153	47.68	112.7	1.419	-4.816	7.421	-6.749	-29.28	14.06	8.168	-14.82	31.51	0.4025	10.01	0.
8	S1	plddt_adj	burial	1126	153	4.484	4.214	0.004756	-0.5818	0.6871	0.9375	0.3389	1.558	-0.9327	-1.753	-0.03131	0.0125	0.3789	0
8	S1	plddt_adj	rgyr	1126	153	0.1669	0.1868	0.002357	-0.02314	0.02638	0.002294	-0.02746	0.0326	6.30e-05	-0.0365	0.03166	0.995	0.01536	0
8	S1	plddt_adj	cavity_open_vol	1126	153	234	-	-1.749	-40.05	32.33	-	-	-	-	-	-	0.916	16.05	0.
8	S1	plddt_adj	cavity_enclosed_vol	1126	153	32.52	-	3.071	-2.976	9.073	-	-	-	-	-	-	0.227	2.527	0.
8	S8	plddt_adj	n	6211	959	1.467	2.305	-0.04685	-0.1324	0.03467	0.01717	-0.1373	0.1865	-0.06402	-0.2118	0.06004	0.179	0.04745	0.
8	S8	plddt_adj	hydropathy	6211	959	0.1734	0.1884	0.005458	-0.003934	0.01531	-0.002252	-0.01187	0.006818	0.007711	-0.005255	0.02184	0.2245	0.00646	0.
8	S8	plddt_adj	resvol	6211	959	2.618	2.52	-0.0842	-0.247	0.08215	0.265	0.1398	0.3906	-0.3492	-0.5658	-0.1284	0	0.0984	
8	S8	plddt_adj	f_pos	6211	959	0.01273	0.01538	0.0002886	-0.0002856	0.0008645	0.00111	0.0003468	0.001851	-0.0008219	-0.001752	0.0001065	0.099	0.000499	0.
8	S8	plddt_adj	f_neg	6211	959	0.01737	0.01838	1.63e-05	-0.00106	0.001031	0.0008663	-0.0001654	0.001947	-0.0008501	-0.002341	0.0004455	0.143	0.0005974	0.
8	S8	plddt_adj	net_charge	6211	959	0.5383	0.9461	0.009456	-0.02129	0.03914	0.0132	-0.03937	0.06825	-0.003745	-0.06003	0.05291	0.897	0.02677	0
8	S8	plddt_adj	f_aromatic	6211	959	0.02104	0.01524	-0.001496	-0.002719	-0.00027	0.0004714	-0.0004249	0.001417	-0.001967	-0.003597	-0.0003897	0.0025	0.0006703	0
8	S8	plddt_adj	f_polar	6211	959	0.03509	0.02943	-0.002076	-0.003824	-0.0001871	-0.002802	-0.004383	-0.001218	0.0007259	-0.001621	0.003315	0.511	0.001085	0

Supplementary Table S4 (continued)

radius	family	adjustment	descriptor	n	clades	within_sd_pocket	within_sd_control	slope_pocket	slope_pocket_lo	slope_pocket_hi	slope_control	slope_control_lo	slope_control_hi	difference	difference_lo	difference_hi	p_perm	null_sd	d
8	S8	plddt_adj	f_small	6211	959	0.0333	0.02828	6.80e-05	-0.001888	0.001992	-0.002655	-0.004191	-0.001255	0.002723	0.00027	0.005175	0.014	0.001111	0
8	S8	plddt_adj	f_gly	6211	959	0.01712	0.01758	-0.0001251	-0.00103	0.0007495	-0.001026	-0.001928	-0.0001746	0.0009014	-0.0003076	0.002149	0.1405	0.0006153	0.
8	S8	plddt_adj	f_pro	6211	959	0.01027	0.01128	6.10e-05	-0.0005716	0.0006382	0.0008017	0.0002767	0.001378	-0.0007406	-0.001685	8.50e-05	0.053	0.0003706	0.
8	S8	plddt_adj	f_ivywrel	6211	959	0.03442	0.03136	0.0007204	-0.00111	0.002636	0.005433	0.003632	0.007079	-0.004713	-0.007278	-0.001884	0	0.001188	
8	S8	plddt_adj	sasa_mean	6211	959	2.859	3.873	-0.04626	-0.2398	0.1453	0.2006	-0.01403	0.4113	-0.2469	-0.4765	-0.02604	0.019	0.1072	0.
8	S8	plddt_adj	sasa_total	6211	959	51.21	154.8	-2.64	-6.376	0.5893	6.873	-1.14	14.58	-9.513	-19.06	-0.2955	0.0255	4.279	0.
8	S8	plddt_adj	burial	6211	959	5.087	5.805	0.03966	-0.3588	0.3931	-0.06629	-0.4059	0.2459	0.106	-0.2464	0.4587	0.512	0.1603	0
8	S8	plddt_adj	rgyr	6211	959	0.1985	0.2528	-0.008593	-0.02262	0.004033	-0.001382	-0.01579	0.01224	-0.007211	-0.02602	0.01076	0.332	0.007437	0.
8	S8	plddt_adj	cavity_open_vol	6211	959	284.1	-	6.279	-9.741	22.72	-	-	-	-	-	-	0.3565	6.995	0.
8	S8	plddt_adj	cavity_enclosed_vol	6211	959	39.9	-	-0.4352	-2.405	1.562	-	-	-	-	-	-	0.661	0.9886	0.
8	S9	plddt_adj	n	3332	374	0.7047	1.159	0.009982	-0.02909	0.05001	0.00958	-0.05301	0.07378	0.0004023	-0.06798	0.0659	0.994	0.03275	0
8	S9	plddt_adj	hydropathy	3332	374	0.1305	0.2122	-0.001173	-0.009554	0.006727	-0.00669	-0.02039	0.006438	0.005517	-0.01091	0.02209	0.4505	0.007286	0.
8	S9	plddt_adj	resvol	3332	374	2.064	2.299	-0.0137	-0.1299	0.1072	0.1837	0.04591	0.3247	-0.1974	-0.3705	-0.01746	0.0405	0.09665	0.
8	S9	plddt_adj	f_pos	3332	374	0.01329	0.01556	0.0003832	-0.0004608	0.001266	0.001066	-3.99e-05	0.002143	-0.0006831	-0.001998	0.0005922	0.289	0.0006454	0.
8	S9	plddt_adj	f_neg	3332	374	0.01135	0.0183	-0.0006381	-0.001449	0.0001961	0.001915	0.0007984	0.003016	-0.002553	-0.003972	-0.001145	0	0.0006413	
8	S9	plddt_adj	net_charge	3332	374	0.368	0.7872	0.02219	-0.00404	0.04858	-0.02893	-0.08182	0.02267	0.05112	-0.008547	0.1074	0.0485	0.02644	0.
8	S9	plddt_adj	f_aromatic	3332	374	0.01645	0.01716	-0.0005095	-0.001492	0.0004536	0.0003505	-0.0008874	0.001513	-0.00086	-0.002322	0.0006511	0.221	0.0007105	0.
8	S9	plddt_adj	f_polar	3332	374	0.02155	0.03038	0.001206	-0.0003882	0.002742	-0.005444	-0.00779	-0.00299	0.00665	0.003729	0.009513	0	0.001122	
8	S9	plddt_adj	f_small	3332	374	0.01853	0.02488	-0.0003592	-0.001409	0.0007007	-0.001714	-0.003481	-1.32e-05	0.001355	-0.0006177	0.003373	0.1735	0.0009816	0
8	S9	plddt_adj	f_gly	3332	374	0.01652	0.01478	-0.0001297	-0.001077	0.0009479	-0.0002554	-0.001255	0.0007226	0.0001257	-0.001187	0.001513	0.8515	0.000668	0.
8	S9	plddt_adj	f_pro	3332	374	0.007029	0.01199	0.0001625	-0.0002387	0.0006052	0.001558	0.0007741	0.002321	-0.001395	-0.002264	-0.0005354	0.0005	0.0004237	0.0
8	S9	plddt_adj	f_ivywrel	3332	374	0.02473	0.0313	0.0002842	-0.001216	0.001822	0.005483	0.003593	0.007257	-0.005199	-0.007527	-0.002938	0	0.001202	
8	S9	plddt_adj	sasa_mean	3332	374	1.968	2.886	-0.01327	-0.131	0.112	-0.0457	-0.2157	0.1331	0.03243	-0.1674	0.232	0.729	0.09777	0.
8	S9	plddt_adj	sasa_total	3332	374	43.97	104.3	-0.1679	-2.789	2.907	-1.647	-8.333	5.235	1.479	-5.565	8.299	0.667	3.434	0.
8	S9	plddt_adj	burial	3332	374	2.405	3.897	0.1085	-0.02259	0.2459	0.3922	0.1588	0.6198	-0.2837	-0.5214	-0.04848	0.0205	0.1269	0.
8	S9	plddt_adj	rgyr	3332	374	0.1768	0.141	0.002675	-0.008334	0.01431	-0.001641	-0.01005	0.007203	0.004316	-0.008753	0.01683	0.5005	0.006348	0
8	S9	plddt_adj	cavity_open_vol	3332	374	119.6	-	-7.856	-15.59	-0.7471	-	-	-	-	-	-	0.0455	3.84	0.
8	S9	plddt_adj	cavity_enclosed_vol	3332	374	33.72	-	0.2817	-1.737	2.381	-	-	-	-	-	-	0.7875	1.005	0.
8	all	standardized	n	10669	1484	1.289	1.974	-0.03903	-0.09037	0.009495	-0.007018	-0.0584	0.04102	-0.03201	-0.07997	0.01161	0.0545	0.01702	0.
8	all	standardized	hydropathy	10669	1484	0.1619	0.1953	0.006515	-0.03343	0.04894	-0.02474	-0.06474	0.01492	0.03125	-0.02834	0.09143	0.2285	0.02585	0.
8	all	standardized	resvol	10669	1484	2.457	2.453	-0.0285	-0.07419	0.01832	0.09937	0.06048	0.1354	-0.1279	-0.1873	-0.06147	0	0.02744	
8	all	standardized	f_pos	10669	1484	0.01305	0.01574	0.0151	-0.02417	0.05422	0.07685	0.04012	0.1159	-0.06175	-0.1133	-0.009986	0.0195	0.02676	0.0
8	all	standardized	f_neg	10669	1484	0.01538	0.0187	-0.01133	-0.05433	0.03084	0.07727	0.03744	0.1199	-0.0886	-0.1465	-0.03438	0	0.02491	
8	all	standardized	net_charge	10669	1484	0.4817	0.9147	0.02336	-0.01909	0.06408	-0.005413	-0.04805	0.03627	0.02877	-0.02476	0.08615	0.259	0.02564	0.
8	all	standardized	f_aromatic	10669	1484	0.01953	0.0158	-0.05639	-0.09848	-0.01602	0.02907	-0.01468	0.07165	-0.08546	-0.1461	-0.02281	0.0005	0.02747	0.
8	all	standardized	f_polar	10669	1484	0.03152	0.03043	-0.01426	-0.05595	0.02783	-0.1346	-0.1791	-0.09047	0.1203	0.05741	0.1848	0	0.02594	
8	all	standardized	f_small	10669	1484	0.02906	0.02735	0.004076	-0.03936	0.04684	-0.08786	-0.1285	-0.04807	0.09194	0.03385	0.1523	0.002	0.02723	0.00
8	all	standardized	f_gly	10669	1484	0.01712	0.01693	-0.008834	-0.04724	0.03141	-0.03502	-0.07284	0.001793	0.02618	-0.02762	0.08122	0.3245	0.02638	0
8	all	standardized	f_pro	10669	1484	0.009662	0.0117	0.01508	-0.02979	0.0583	0.09304	0.05454	0.1304	-0.07796	-0.1393	-0.021	0.0025	0.02555	0.00
8	all	standardized	f_ivywrel	10669	1484	0.0319	0.03154	0.007735	-0.03392	0.04873	0.1814	0.1436	0.2207	-0.1737	-0.2303	-0.1142	0	0.02687	
8	all	standardized	sasa_mean	10669	1484	2.598	3.499	-0.007685	-0.0534	0.03929	0.02522	-0.01536	0.06732	-0.03291	-0.08203	0.01414	0.147	0.02288	0.
8	all	standardized	sasa_total	10669	1484	48.69	136.7	-0.03411	-0.0874	0.009809	0.01586	-0.02194	0.05565	-0.04998	-0.1243	0.01196	0.0865	0.02905	0.

Supplementary Table S4 (continued)

radius	family	adjustment	descriptor	n	clades	within_sd_pocket	within_sd_control	slope_pocket	slope_pocket_lo	slope_pocket_hi	slope_control	slope_control_lo	slope_control_hi	difference	difference_lo	difference_hi	p_perm	null_sd	d
8	all	standardized	burial	10669	1484	4.358	5.124	0.01021	-0.03989	0.0605	0.03102	-0.01031	0.07069	-0.0208	-0.06735	0.03166	0.3555	0.02245	0.
8	all	standardized	rgyr	10669	1484	0.1888	0.217	-0.02701	-0.07981	0.02114	-0.01761	-0.06033	0.02281	-0.009398	-0.06834	0.04566	0.7025	0.02477	0.
8	all	standardized	plddt	10669	1484	2.046	1.942	0.0409	-0.002947	0.0869	0.05554	0.01515	0.09586	-0.01464	-0.06489	0.03819	0.543	0.02303	0.
8	all	standardized	cavity_open_vol	10669	1484	239.3	-	0.02183	-0.02637	0.06698	-	-	-	-	-	-	0.289	0.02023	0.
8	all	standardized	cavity_enclosed_vol	10669	1484	37.33	-	0.006717	-0.02814	0.04435	-	-	-	-	-	-	0.7445	0.01932	0.
8	S1	standardized	n	1126	153	1.554	1.906	-0.04997	-0.2705	0.1692	0.03055	-0.1362	0.1998	-0.08052	-0.1979	0.0389	0.204	0.06338	0.
8	S1	standardized	hydropathy	1126	153	0.179	0.1795	-0.1399	-0.3077	0.03589	-0.0992	-0.2571	0.04334	-0.04074	-0.2664	0.1904	0.678	0.09959	0.
8	S1	standardized	resvol	1126	153	2.601	2.511	-0.02792	-0.2127	0.1535	0.1457	0.002545	0.2921	-0.1736	-0.4073	0.04999	0.118	0.1114	0.
8	S1	standardized	f_pos	1126	153	0.01399	0.01803	-0.05861	-0.2685	0.1223	0.1643	0.002529	0.3296	-0.2229	-0.4742	0.01157	0.033	0.1038	0.
8	S1	standardized	f_neg	1126	153	0.01398	0.0214	0.101	-0.07382	0.2936	0.1823	0.01098	0.3493	-0.0813	-0.294	0.1612	0.437	0.1013	0.
8	S1	standardized	net_charge	1126	153	0.445	1.074	-0.1069	-0.3409	0.09145	-0.03654	-0.2287	0.144	-0.07039	-0.3258	0.1793	0.4845	0.1018	0.
8	S1	standardized	f_aromatic	1126	153	0.01917	0.01457	-0.04307	-0.2237	0.1785	0.06563	-0.07882	0.1957	-0.1087	-0.3366	0.1518	0.348	0.1145	0
8	S1	standardized	f_polar	1126	153	0.03531	0.0355	0.1094	-0.07321	0.2842	-0.2302	-0.3776	-0.0785	0.3396	0.09731	0.5599	0.0015	0.1049	0.0
8	S1	standardized	f_small	1126	153	0.02944	0.02901	0.02591	-0.09458	0.149	-0.1265	-0.2969	0.03718	0.1524	-0.04636	0.3646	0.1335	0.1013	0.
8	S1	standardized	f_gly	1126	153	0.01882	0.01906	-0.0755	-0.2266	0.06571	0.08217	-0.07604	0.2354	-0.1577	-0.363	0.05175	0.133	0.1054	0.
8	S1	standardized	f_pro	1126	153	0.01252	0.01307	0.06998	-0.1077	0.2613	0.08477	-0.1023	0.2715	-0.01479	-0.2371	0.2181	0.8985	0.1073	0.
8	S1	standardized	f_ivywrel	1126	153	0.03604	0.03306	-0.05378	-0.2347	0.1376	0.2817	0.1362	0.4312	-0.3355	-0.5446	-0.1097	0	0.1022	
8	S1	standardized	sasa_mean	1126	153	2.72	2.94	0.03866	-0.1166	0.2023	-0.08178	-0.2511	0.09024	0.1204	-0.1147	0.3233	0.1735	0.08851	0.
8	S1	standardized	sasa_total	1126	153	47.68	112.7	0.02996	-0.1074	0.1624	-0.06032	-0.2473	0.1367	0.09028	-0.1613	0.3225	0.4345	0.1116	0.
8	S1	standardized	burial	1126	153	4.484	4.214	-0.002788	-0.1638	0.1397	0.2192	0.069	0.3802	-0.2219	-0.4291	-0.03194	0.0155	0.08937	0.0
8	S1	standardized	rgyr	1126	153	0.1669	0.1868	0.01006	-0.143	0.176	0.008294	-0.1672	0.1947	0.001768	-0.1904	0.1811	0.9805	0.08701	0.
8	S1	standardized	plddt	1126	153	2.659	2.147	0.01671	-0.1471	0.1666	0.1545	-0.0296	0.3534	-0.1377	-0.3693	0.0858	0.1725	0.1001	0.
8	S1	standardized	cavity_open_vol	1126	153	234	-	0.0001851	-0.186	0.2053	-	-	-	-	-	-	0.9975	0.08129	0.
8	S1	standardized	cavity_enclosed_vol	1126	153	32.52	-	0.09562	-0.08705	0.2823	-	-	-	-	-	-	0.21	0.07656	0.
8	S8	standardized	n	6211	959	1.467	2.305	-0.05686	-0.1261	0.008747	-0.01478	-0.08387	0.04997	-0.04209	-0.1124	0.01633	0.06	0.02264	0.
8	S8	standardized	hydropathy	6211	959	0.1734	0.1884	0.0322	-0.02241	0.09033	-0.01192	-0.06825	0.03534	0.04412	-0.02494	0.1238	0.2165	0.03536	0.
8	S8	standardized	resvol	6211	959	2.618	2.52	-0.03982	-0.1098	0.02861	0.1052	0.05166	0.1565	-0.1451	-0.2362	-0.05184	0.0005	0.03829	0.0
8	S8	standardized	f_pos	6211	959	0.01273	0.01538	0.01584	-0.02975	0.06195	0.07245	0.02256	0.1212	-0.05661	-0.1237	0.01045	0.099	0.03445	0.
8	S8	standardized	f_neg	6211	959	0.01737	0.01838	-0.003798	-0.06002	0.05217	0.04769	-0.0104	0.1066	-0.05149	-0.1291	0.02173	0.1305	0.03345	0.
8	S8	standardized	net_charge	6211	959	0.5383	0.9461	0.02085	-0.03234	0.07597	0.01506	-0.04154	0.06878	0.005788	-0.06514	0.07934	0.873	0.03487	0.
8	S8	standardized	f_aromatic	6211	959	0.02104	0.01524	-0.07163	-0.1287	-0.01655	0.02961	-0.03219	0.08859	-0.1012	-0.1871	-0.01058	0.007	0.03641	0.0
8	S8	standardized	f_polar	6211	959	0.03509	0.02943	-0.05599	-0.1049	-0.004837	-0.09522	-0.1489	-0.04182	0.03923	-0.03095	0.112	0.213	0.03185	0.
8	S8	standardized	f_small	6211	959	0.0333	0.02828	0.01019	-0.05562	0.07042	-0.0947	-0.147	-0.04244	0.1049	0.0175	0.1897	0.003	0.03593	0
8	S8	standardized	f_gly	6211	959	0.01712	0.01758	-0.002862	-0.05892	0.04907	-0.05894	-0.1091	-0.009497	0.05608	-0.01347	0.1264	0.1005	0.03392	0.
8	S8	standardized	f_pro	6211	959	0.01027	0.01128	0.004664	-0.05773	0.0642	0.07069	0.02318	0.1193	-0.06602	-0.1523	0.01225	0.05	0.03361	0.
8	S8	standardized	f_ivywrel	6211	959	0.03442	0.03136	0.01319	-0.04167	0.06909	0.174	0.1187	0.2264	-0.1608	-0.2413	-0.07584	0	0.03699	
8	S8	standardized	sasa_mean	6211	959	2.859	3.873	-0.01089	-0.08077	0.05665	0.0556	-0.001383	0.1149	-0.06649	-0.1366	-0.0026	0.0345	0.03119	0.
8	S8	standardized	sasa_total	6211	959	51.21	154.8	-0.05451	-0.1311	0.007926	0.0376	-0.01613	0.08759	-0.09211	-0.1938	-0.004012	0.013	0.03829	0.
8	S8	standardized	burial	6211	959	5.087	5.805	-0.0004706	-0.0761	0.0701	-0.01424	-0.07372	0.03978	0.01377	-0.05157	0.07822	0.6395	0.02966	0.
8	S8	standardized	rgyr	6211	959	0.1985	0.2528	-0.05249	-0.1273	0.01612	-0.02143	-0.075	0.03329	-0.03106	-0.1133	0.04798	0.3425	0.03233	0.
8	S8	standardized	plddt	6211	959	2.306	2.264	0.05187	-0.0103	0.1167	0.05093	-0.001215	0.1023	0.0009456	-0.06658	0.07175	0.979	0.0319	0
8	S8	standardized	cavity_open_vol	6211	959	284.1	-	0.04736	-0.01297	0.1112	-	-	-	-	-	-	0.0785	0.02698	0.

Supplementary Table S4 (continued)

radius	family	adjustment	descriptor	n	clades	within_sd_pocket	within_sd_control	slope_pocket	slope_pocket_lo	slope_pocket_hi	slope_control	slope_control_lo	slope_control_hi	difference	difference_lo	difference_hi	p_perm	null_sd	d
8	S8	standardized	cavity_enclosed_vol	6211	959	39.9	-	-0.001122	-0.05004	0.05037	-	-	-	-	-	-	0.965	0.02512	0
8	S9	standardized	n	3332	374	0.7047	1.159	0.009229	-0.04923	0.06928	0.005175	-0.05275	0.06056	0.004054	-0.06411	0.07542	0.908	0.03475	0
8	S9	standardized	hydropathy	3332	374	0.1305	0.2122	-0.01124	-0.07433	0.05036	-0.03088	-0.09633	0.03144	0.01964	-0.07408	0.1163	0.6285	0.0409	0
8	S9	standardized	resvol	3332	374	2.064	2.299	-0.006451	-0.06228	0.05256	0.08105	0.01617	0.1428	-0.0875	-0.1668	-0.006107	0.0385	0.04247	0.
8	S9	standardized	f_pos	3332	374	0.01329	0.01556	0.02763	-0.03827	0.09696	0.06749	-0.004559	0.1354	-0.03986	-0.1277	0.04811	0.3635	0.04375	0.
8	S9	standardized	f_neg	3332	374	0.01135	0.0183	-0.05337	-0.1238	0.01801	0.1036	0.041	0.162	-0.1569	-0.2549	-0.05447	0.0005	0.04337	0.06
8	S9	standardized	net_charge	3332	374	0.368	0.7872	0.05764	-0.01455	0.1246	-0.03643	-0.1001	0.02819	0.09407	-0.002098	0.1927	0.031	0.04387	0.
8	S9	standardized	f_aromatic	3332	374	0.01645	0.01716	-0.02916	-0.08856	0.0308	0.02208	-0.04563	0.08876	-0.05124	-0.1363	0.03871	0.2365	0.04373	0.
8	S9	standardized	f_polar	3332	374	0.02155	0.03038	0.05569	-0.01892	0.129	-0.1779	-0.2512	-0.1007	0.2336	0.1207	0.3393	0	0.04472	
8	S9	standardized	f_small	3332	374	0.01853	0.02488	-0.01996	-0.07961	0.03642	-0.06944	-0.1413	-0.003143	0.04947	-0.03755	0.1414	0.263	0.04458	0.
8	S9	standardized	f_gly	3332	374	0.01652	0.01478	-0.005192	-0.067	0.05191	-0.01726	-0.08347	0.04684	0.01206	-0.07764	0.09651	0.7815	0.04282	0
8	S9	standardized	f_pro	3332	374	0.007029	0.01199	0.02457	-0.03309	0.08561	0.129	0.06472	0.1934	-0.1044	-0.1863	-0.02433	0.0115	0.04127	0.0
8	S9	standardized	f_ivywrel	3332	374	0.02473	0.0313	0.01279	-0.0503	0.075	0.1753	0.1196	0.233	-0.1625	-0.2462	-0.07876	0	0.04241	
8	S9	standardized	sasa_mean	3332	374	1.968	2.886	-0.01213	-0.06988	0.05242	-0.01885	-0.07601	0.04433	0.006717	-0.06811	0.08549	0.867	0.03923	0
8	S9	standardized	sasa_total	3332	374	43.97	104.3	-0.009356	-0.06803	0.05781	-0.01903	-0.08496	0.04184	0.009673	-0.0698	0.09672	0.8195	0.0424	0
8	S9	standardized	burial	3332	374	2.405	3.897	0.05397	-0.003257	0.1115	0.1031	0.04076	0.1619	-0.04912	-0.1214	0.03112	0.216	0.04009	0.
8	S9	standardized	rgyr	3332	374	0.1768	0.141	0.01091	-0.05734	0.08391	-0.01538	-0.07854	0.04996	0.02629	-0.0607	0.1063	0.483	0.03937	0.
8	S9	standardized	plddt	3332	374	1.043	0.9773	0.02983	-0.03264	0.09272	0.05604	-0.008009	0.1176	-0.02621	-0.09847	0.05159	0.449	0.03476	0.
8	S9	standardized	cavity_open_vol	3332	374	119.6	-	-0.05837	-0.1287	0.001951	-	-	-	-	-	-	0.0975	0.03512	0.
8	S9	standardized	cavity_enclosed_vol	3332	374	33.72	-	0.006986	-0.05227	0.0676	-	-	-	-	-	-	0.8115	0.02965	0

Supplementary Table S5. Descriptor and attribution slopes by distance shell

Within-clade slope on growth temperature for each measure within concentric shells around the catalytic serine. The burial-adjusted pass partials local burial out within each shell; as noted in the Discussion this cannot address a between-shell confound, which is what Supplementary Tables S6 and S7 address instead.

measure	adjustment	shell	shell_lo	n	clades	slope	ci_lo	ci_hi	p_perm	mean_burial
attribution	raw	0-5	0	10992	1768	1.297	0.9973	1.592	0	188.7
attribution	raw	5-8	5	11005	1769	1.45	1.116	1.792	0	191.4
attribution	raw	8-11	8	11001	1770	1.429	1.146	1.736	0	186.9
attribution	raw	11-14	11	11004	1770	1.835	1.498	2.211	0	177.2
attribution	raw	14-18	14	11007	1770	1.957	1.641	2.326	0	164.7
attribution	raw	18-24	18	11007	1770	2.089	1.75	2.493	0	143.2
attribution	raw	24-32	24	11000	1770	2.26	1.823	2.722	0	121.3
attribution	raw	>32	32	10888	1762	2.49	2.016	3.04	0	97.42
resvol	raw	0-5	0	10993	1768	0.1553	0.007075	0.314	0.041	188.7
resvol	raw	5-8	5	11006	1769	-0.2262	-0.4308	-0.02467	0.001	191.4
resvol	raw	8-11	8	11002	1770	0.1871	0.05276	0.3182	0	186.9
resvol	raw	11-14	11	11005	1770	0.05863	-0.0608	0.1742	0.237	177.2
resvol	raw	14-18	14	11007	1770	0.2302	0.1345	0.3145	0	164.7
resvol	raw	18-24	18	11007	1770	0.3133	0.2192	0.4068	0	143.2
resvol	raw	24-32	24	11000	1770	0.3214	0.1803	0.4605	0	121.3
resvol	raw	>32	32	10888	1762	0.192	0.04995	0.3404	0	97.42
f_ivywrel	raw	0-5	0	10993	1768	0.002074	-0.000182	0.004148	0.024	188.7
f_ivywrel	raw	5-8	5	11006	1769	-0.001078	-0.003125	0.0008701	0.207	191.4
f_ivywrel	raw	8-11	8	11002	1770	0.001677	-3.06e-06	0.003364	0.015	186.9
f_ivywrel	raw	11-14	11	11005	1770	0.002029	0.0006109	0.003446	0.004	177.2
f_ivywrel	raw	14-18	14	11007	1770	0.003979	0.002699	0.005208	0	164.7
f_ivywrel	raw	18-24	18	11007	1770	0.006519	0.005253	0.007827	0	143.2
f_ivywrel	raw	24-32	24	11000	1770	0.007704	0.005755	0.009575	0	121.3
f_ivywrel	raw	>32	32	10888	1762	0.006803	0.00494	0.008514	0	97.42
f_polar	raw	0-5	0	10993	1768	-0.0007637	-0.002794	0.001038	0.418	188.7
f_polar	raw	5-8	5	11006	1769	9.82e-05	-0.001955	0.001977	0.904	191.4
f_polar	raw	8-11	8	11002	1770	-0.0003382	-0.001999	0.001407	0.641	186.9
f_polar	raw	11-14	11	11005	1770	-0.001336	-0.002823	0.0003312	0.077	177.2
f_polar	raw	14-18	14	11007	1770	-0.003288	-0.004568	-0.002033	0	164.7
f_polar	raw	18-24	18	11007	1770	-0.004263	-0.005625	-0.002962	0	143.2
f_polar	raw	24-32	24	11000	1770	-0.004971	-0.007217	-0.002702	0	121.3
f_polar	raw	>32	32	10888	1762	-0.005219	-0.00708	-0.003159	0	97.42
hydropathy	raw	0-5	0	10993	1768	0.004812	-0.006118	0.01658	0.318	188.7
hydropathy	raw	5-8	5	11006	1769	-0.002013	-0.01237	0.008082	0.694	191.4
hydropathy	raw	8-11	8	11002	1770	-0.001272	-0.01017	0.006821	0.711	186.9

Supplementary Table S5 (continued)

measure	adjustment	shell	shell_lo	n	clades	slope	ci_lo	ci_hi	p_perm	mean_burial
hydropathy	raw	11-14	11	11005	1770	0.007671	-0.0007034	0.01614	0.044	177.2
hydropathy	raw	14-18	14	11007	1770	0.0009368	-0.006556	0.007835	0.74	164.7
hydropathy	raw	18-24	18	11007	1770	-0.01226	-0.01832	-0.006271	0	143.2
hydropathy	raw	24-32	24	11000	1770	-0.01378	-0.02577	-0.00177	0.002	121.3
hydropathy	raw	>32	32	10888	1762	-0.01046	-0.0194	-0.00129	0.014	97.42
attribution	burial_adj	0-5	0	10992	1768	1.297	0.9724	1.585	0	188.7
attribution	burial_adj	5-8	5	11005	1769	1.444	1.086	1.788	0	191.4
attribution	burial_adj	8-11	8	11001	1770	1.427	1.132	1.749	0	186.9
attribution	burial_adj	11-14	11	11004	1770	1.833	1.497	2.187	0	177.2
attribution	burial_adj	14-18	14	11007	1770	1.951	1.615	2.329	0	164.7
attribution	burial_adj	18-24	18	11007	1770	2.092	1.723	2.45	0	143.2
attribution	burial_adj	24-32	24	11000	1770	2.252	1.821	2.719	0	121.3
attribution	burial_adj	>32	32	10888	1762	2.492	1.998	3.048	0	97.42
resvol	burial_adj	0-5	0	10993	1768	0.1568	0.006036	0.3259	0.04	188.7
resvol	burial_adj	5-8	5	11006	1769	-0.223	-0.4393	-0.02564	0.005	191.4
resvol	burial_adj	8-11	8	11002	1770	0.1878	0.05039	0.32	0.001	186.9
resvol	burial_adj	11-14	11	11005	1770	0.05731	-0.06487	0.1696	0.262	177.2
resvol	burial_adj	14-18	14	11007	1770	0.2097	0.1249	0.2955	0	164.7
resvol	burial_adj	18-24	18	11007	1770	0.2617	0.1641	0.3526	0	143.2
resvol	burial_adj	24-32	24	11000	1770	0.2784	0.1413	0.4121	0	121.3
resvol	burial_adj	>32	32	10888	1762	0.1856	0.03787	0.3218	0.004	97.42
f_ivywrel	burial_adj	0-5	0	10993	1768	0.002092	-3.95e-05	0.004061	0.028	188.7
f_ivywrel	burial_adj	5-8	5	11006	1769	-0.001115	-0.003077	0.000781	0.201	191.4
f_ivywrel	burial_adj	8-11	8	11002	1770	0.001672	-0.0001156	0.003433	0.024	186.9
f_ivywrel	burial_adj	11-14	11	11005	1770	0.001996	0.0006859	0.00342	0.004	177.2
f_ivywrel	burial_adj	14-18	14	11007	1770	0.003838	0.002491	0.005085	0	164.7
f_ivywrel	burial_adj	18-24	18	11007	1770	0.006173	0.004999	0.007465	0	143.2
f_ivywrel	burial_adj	24-32	24	11000	1770	0.007254	0.005334	0.009055	0	121.3
f_ivywrel	burial_adj	>32	32	10888	1762	0.006746	0.004824	0.008743	0	97.42
f_polar	burial_adj	0-5	0	10993	1768	-0.0007763	-0.002654	0.001159	0.391	188.7
f_polar	burial_adj	5-8	5	11006	1769	9.88e-05	-0.001771	0.002224	0.886	191.4
f_polar	burial_adj	8-11	8	11002	1770	-0.0003554	-0.002088	0.001275	0.61	186.9
f_polar	burial_adj	11-14	11	11005	1770	-0.001305	-0.002937	0.000404	0.069	177.2
f_polar	burial_adj	14-18	14	11007	1770	-0.003321	-0.004579	-0.001936	0	164.7
f_polar	burial_adj	18-24	18	11007	1770	-0.004435	-0.005836	-0.002976	0	143.2
f_polar	burial_adj	24-32	24	11000	1770	-0.005003	-0.007205	-0.002646	0	121.3
f_polar	burial_adj	>32	32	10888	1762	-0.005291	-0.007288	-0.003483	0	97.42
hydropathy	burial_adj	0-5	0	10993	1768	0.004928	-0.005398	0.01571	0.269	188.7
hydropathy	burial_adj	5-8	5	11006	1769	-0.001889	-0.01293	0.008118	0.723	191.4
hydropathy	burial_adj	8-11	8	11002	1770	-0.00147	-0.009826	0.00716	0.713	186.9
hydropathy	burial_adj	11-14	11	11005	1770	0.007572	-0.001138	0.01597	0.039	177.2
hydropathy	burial_adj	14-18	14	11007	1770	0.0006179	-0.007243	0.007422	0.838	164.7

Supplementary Table S5 (continued)

measure	adjustment	shell	shell_lo	n	clades	slope	ci_lo	ci_hi	p_perm	mean_burial
hydropathy	burial_adj	18-24	18	11007	1770	-0.01058	-0.01714	-0.004462	0	143.2
hydropathy	burial_adj	24-32	24	11000	1770	-0.01495	-0.02624	-0.002211	0.003	121.3
hydropathy	burial_adj	>32	32	10888	1762	-0.01014	-0.01825	-0.00125	0.022	97.42

Supplementary Table S6. Distance shells stratified by solvent exposure, AlphaFold models

Slopes recomputed inside terciles of solvent exposure taken within each protein, so that every comparison is between residues of comparable exposure. Stratum 0 is the most exposed tercile.

measure	dist_bin	dist_i	burial_stratum	slope	ci_lo	ci_hi	p_perm	n	clades	mean_burial
attribution	0-10	0	0	1.599	-1.004	3.895	0.2062	527	270	105.5
attribution	0-10	0	1	1.431	0.973	1.917	0	10609	1745	146.3
attribution	0-10	0	2	1.449	1.175	1.729	0	10997	1769	202.3
attribution	10-18	1	0	1.976	1.451	2.509	0	10792	1760	100.4
attribution	10-18	1	1	2.311	1.898	2.719	0	11007	1770	144.3
attribution	10-18	1	2	1.655	1.366	2	0	11004	1769	195.8
attribution	18-28	2	0	2.505	2.053	3.029	0	10993	1769	95.44
attribution	18-28	2	1	2.137	1.764	2.589	0	11007	1770	141.6
attribution	18-28	2	2	1.766	1.457	2.106	0	10802	1759	183.3
attribution	>28	3	0	2.699	2.184	3.247	0	10981	1770	76.32
attribution	>28	3	1	2.208	1.766	2.775	0	9601	1606	140
attribution	>28	3	2	2.183	1.663	2.751	0	8026	1402	181
f_ivywrel	0-10	0	0	-0.002329	-0.02853	0.02388	0.875	528	271	105.6
f_ivywrel	0-10	0	1	0.0003832	-0.003798	0.004666	0.8237	10618	1746	146.3
f_ivywrel	0-10	0	2	0.0007585	-0.000411	0.001941	0.1613	10998	1769	202.3
f_ivywrel	10-18	1	0	0.005423	0.001245	0.009688	0.00125	10797	1760	100.4
f_ivywrel	10-18	1	1	0.004899	0.002686	0.006842	0	11007	1770	144.3
f_ivywrel	10-18	1	2	0.00255	0.001542	0.003615	0	11004	1769	195.8
f_ivywrel	18-28	2	0	0.008243	0.006105	0.01029	0	10993	1769	95.44
f_ivywrel	18-28	2	1	0.008349	0.006619	0.0101	0	11007	1770	141.6
f_ivywrel	18-28	2	2	0.00208	0.0001016	0.004061	0.02875	10802	1759	183.3
f_ivywrel	>28	3	0	0.0069	0.005048	0.008772	0	10981	1770	76.32
f_ivywrel	>28	3	1	0.007895	0.005445	0.01035	0	9601	1606	140
f_ivywrel	>28	3	2	0.003944	0.0002116	0.007459	0.0175	8026	1402	181
f_polar	0-10	0	0	0.01795	-0.01275	0.04916	0.335	528	271	105.6
f_polar	0-10	0	1	0.002809	-0.001081	0.006819	0.1288	10618	1746	146.3
f_polar	0-10	0	2	-0.0009927	-0.002328	0.0003012	0.07625	10998	1769	202.3
f_polar	10-18	1	0	-0.001136	-0.006419	0.003217	0.585	10797	1760	100.4
f_polar	10-18	1	1	-0.003773	-0.006014	-0.001354	0	11007	1770	144.3
f_polar	10-18	1	2	-0.002219	-0.003149	-0.001322	0	11004	1769	195.8
f_polar	18-28	2	0	-0.004559	-0.007152	-0.001915	0	10993	1769	95.44
f_polar	18-28	2	1	-0.006294	-0.008218	-0.004279	0	11007	1770	141.6
f_polar	18-28	2	2	-0.002346	-0.004249	-0.0003349	0.00625	10802	1759	183.3
f_polar	>28	3	0	-0.00539	-0.007332	-0.003014	0	10981	1770	76.32
f_polar	>28	3	1	-0.005881	-0.008355	-0.003372	0	9601	1606	140

Supplementary Table S6 (continued)

measure	dist_bin	dist_i	burial_stratum	slope	ci_lo	ci_hi	p_perm	n	clades	mean_burial
f_polar	>28	3	2	-0.002437	-0.005583	0.0004936	0.03625	8026	1402	181
resvol	0-10	0	0	-0.348	-2.551	1.367	0.7812	528	271	105.6
resvol	0-10	0	1	-0.1913	-0.5649	0.2101	0.215	10618	1746	146.3
resvol	0-10	0	2	0.02266	-0.06647	0.1101	0.585	10998	1769	202.3
resvol	10-18	1	0	0.372	0.0328	0.7438	0.015	10797	1760	100.4
resvol	10-18	1	1	0.4664	0.308	0.6389	0	11007	1770	144.3
resvol	10-18	1	2	0.05317	-0.01641	0.1198	0.09875	11004	1769	195.8
resvol	18-28	2	0	0.4049	0.2358	0.5777	0	10993	1769	95.44
resvol	18-28	2	1	0.3334	0.206	0.4565	0	11007	1770	141.6
resvol	18-28	2	2	0.111	-0.04953	0.2683	0.1363	10802	1759	183.3
resvol	>28	3	0	0.2109	0.045	0.3716	0.00875	10981	1770	76.32
resvol	>28	3	1	0.2706	0.08375	0.4468	0	9601	1606	140
resvol	>28	3	2	0.3939	0.1271	0.6649	0.00125	8026	1402	181

Supplementary Table S7. Distance shells stratified by solvent exposure, ESMFold models

As Supplementary Table S6, on the ESMFold structure set. The inner-shell IVYWREL p-values quoted in the Results come from this table.

measure	dist_bin	dist_i	burial_stratum	slope	ci_lo	ci_hi	p_perm	n	clades	mean_burial
attribution	0-10	0	0	1.938	-1.293	4.211	0.1862	344	107	102.4
attribution	0-10	0	1	1.453	0.8872	2.046	0	5454	444	144.6
attribution	0-10	0	2	1.353	0.9046	1.807	0	5506	444	203.2
attribution	10-18	1	0	2.249	1.423	3.089	0	5486	446	93.96
attribution	10-18	1	1	2.127	1.539	2.768	0	5509	446	142.6
attribution	10-18	1	2	1.726	1.242	2.36	0	5508	445	195.2
attribution	18-28	2	0	2.447	1.684	3.402	0	5509	446	90.55
attribution	18-28	2	1	2.309	1.658	3.08	0	5509	446	138.7
attribution	18-28	2	2	1.682	1.061	2.419	0	5269	443	181.1
attribution	>28	3	0	2.774	1.889	3.782	0	5470	442	68.7
attribution	>28	3	1	2.361	1.521	3.446	0	3839	357	137.9
attribution	>28	3	2	2.47	1.226	3.981	0	2462	245	180
f_ivywrel	0-10	0	0	0.0003035	-0.04011	0.05583	0.9925	344	107	102.4
f_ivywrel	0-10	0	1	-0.002496	-0.009773	0.005481	0.295	5454	444	144.6
f_ivywrel	0-10	0	2	0.0007796	-0.0008568	0.002521	0.295	5506	444	203.2
f_ivywrel	10-18	1	0	0.002778	-0.002313	0.008285	0.2188	5486	446	93.96
f_ivywrel	10-18	1	1	0.00474	0.001672	0.007906	0.00125	5509	446	142.6
f_ivywrel	10-18	1	2	0.002295	0.0005844	0.004129	0.00125	5508	445	195.2
f_ivywrel	18-28	2	0	0.008621	0.005398	0.01196	0	5509	446	90.55
f_ivywrel	18-28	2	1	0.006917	0.004146	0.009925	0	5509	446	138.7
f_ivywrel	18-28	2	2	0.002996	-0.001078	0.007043	0.07	5269	443	181.1
f_ivywrel	>28	3	0	0.006157	0.002885	0.009204	0	5470	442	68.7
f_ivywrel	>28	3	1	0.006602	0.001057	0.0123	0.00125	3839	357	137.9
f_ivywrel	>28	3	2	0.00408	-0.003755	0.01295	0.2112	2462	245	180
f_polar	0-10	0	0	0.02448	-0.005843	0.06605	0.1425	344	107	102.4
f_polar	0-10	0	1	0.006632	0.0002018	0.01224	0.0075	5454	444	144.6
f_polar	0-10	0	2	-9.04e-05	-0.002199	0.002064	0.9	5506	444	203.2
f_polar	10-18	1	0	-0.001939	-0.007768	0.004176	0.4838	5486	446	93.96
f_polar	10-18	1	1	-0.002464	-0.005734	0.0008851	0.06625	5509	446	142.6
f_polar	10-18	1	2	-0.00144	-0.002972	0.0002869	0.0225	5508	445	195.2
f_polar	18-28	2	0	-0.006223	-0.01019	-0.002485	0	5509	446	90.55
f_polar	18-28	2	1	-0.00696	-0.01	-0.003462	0	5509	446	138.7
f_polar	18-28	2	2	-0.006059	-0.009565	-0.002348	0	5269	443	181.1
f_polar	>28	3	0	-0.006625	-0.009932	-0.003249	0	5470	442	68.7
f_polar	>28	3	1	-0.007651	-0.01159	-0.003468	0	3839	357	137.9
f_polar	>28	3	2	-0.002331	-0.008865	0.004482	0.2988	2462	245	180

Supplementary Table S7 (continued)

measure	dist_bin	dist_i	burial_stratum	slope	ci_lo	ci_hi	p_perm	n	clades	mean_burial
resvol	0-10	0	0	-1.189	-4.08	1.444	0.4012	344	107	102.4
resvol	0-10	0	1	-0.1383	-0.6253	0.3391	0.5088	5454	444	144.6
resvol	0-10	0	2	-0.09232	-0.2344	0.0533	0.1138	5506	444	203.2
resvol	10-18	1	0	0.1917	-0.2685	0.6091	0.3463	5486	446	93.96
resvol	10-18	1	1	0.3451	0.1219	0.585	0	5509	446	142.6
resvol	10-18	1	2	0.03736	-0.08523	0.1587	0.4238	5508	445	195.2
resvol	18-28	2	0	0.4293	0.1358	0.7475	0	5509	446	90.55
resvol	18-28	2	1	0.2076	0.02259	0.3879	0.0125	5509	446	138.7
resvol	18-28	2	2	0.3419	0.02607	0.6539	0.00375	5269	443	181.1
resvol	>28	3	0	0.1858	-0.06762	0.4306	0.09625	5470	442	68.7
resvol	>28	3	1	0.08347	-0.268	0.4024	0.555	3839	357	137.9
resvol	>28	3	2	0.5059	0.07448	0.899	0.015	2462	245	180

Supplementary Table S8. Exact per-residue attribution by region

Within-clade slope of each region's ESM-2 attribution readout on growth temperature, per +10 °C. The decomposition is exact: per-residue contributions sum to the model's own prediction. The control-minus-pocket row carries the bootstrap interval and permutation p.

region	slope	ci_lo	ci_hi	p_perm
pocket	1.424	-	-	-
control	2.043	-	-	-
whole	2.134	-	-	-
control_minus_pocket	0.6189	0.3534	0.8933	0

Supplementary Table S9. Elastic-network flexibility, AlphaFold models

Within-clade slope of relative Cα fluctuation on growth temperature, pocket against matched control, by family. Fluctuations come from a Gaussian Network Model with a 7.3 Å cutoff, normalised by each protein's own mean; the unnormalised pass is reported alongside.

family	pass	measure	slope_pocket	slope_control	difference	difference_lo	difference_hi	p_perm	n	clades	within_sd_pocket	within_sd_control
all	raw	normalised	-0.0009799	0.001039	-0.002019	-0.006616	0.00245	0.341	10674	1485	0.07195	0.1349
all	raw	unnormalised	-0.001451	0.0001718	-0.001623	-0.007618	0.003898	0.581	10674	1485	0.1381	0.2449
all	standardized	normalised, standardized	-0.01362	0.007705	-0.02132	-0.0631	0.01653	0.2325	10674	1485	0.07195	0.1349
S1	raw	normalised	-0.005263	-0.01944	0.01418	0.003359	0.02434	0.002	1126	153	0.06755	0.1033
S1	raw	unnormalised	0.000424	-0.01414	0.01456	0.003065	0.03305	0.011	1126	153	0.08162	0.1123
S1	standardized	normalised, standardized	-0.07792	-0.1883	0.1103	0.001284	0.2209	0.0165	1126	153	0.06755	0.1033
S8	raw	normalised	0.0008248	0.005787	-0.004962	-0.01252	0.002285	0.166	6216	960	0.07344	0.1566
S8	raw	unnormalised	-0.002929	0.0001711	-0.0031	-0.01248	0.006275	0.532	6216	960	0.1754	0.3142
S8	standardized	normalised, standardized	0.01123	0.03696	-0.02573	-0.08715	0.03042	0.322	6216	960	0.07344	0.1566
S9	raw	normalised	-0.003019	-0.002919	-9.96e-05	-0.003206	0.002974	0.952	3332	374	0.07018	0.09448
S9	raw	unnormalised	0.0007452	0.002813	-0.002068	-0.006427	0.0006298	0.1515	3332	374	0.03711	0.05983
S9	standardized	normalised, standardized	-0.04301	-0.0309	-0.01212	-0.05031	0.02187	0.45	3332	374	0.07018	0.09448

Supplementary Table S10. Elastic-network flexibility, ESMFold models

As Supplementary Table S9, on the ESMFold structure set.

family	pass	measure	slope_pocket	slope_control	difference	difference_lo	difference_hi	p_perm	n	clades	within_sd_pocket	within_sd_control
all	raw	normalised	-0.002329	-0.0003236	-0.002005	-0.008225	0.004457	0.4465	5445	390	0.06693	0.1196
all	raw	unnormalised	0.001569	0.002904	-0.001335	-0.005092	0.002379	0.449	5445	390	0.04851	0.0834
all	standardized	normalised, standardized	-0.0348	-0.002707	-0.03209	-0.0922	0.0264	0.178	5445	390	0.06693	0.1196
S1	raw	normalised	-0.002111	-0.01222	0.0101	-0.0002868	0.02041	0.01	1343	74	0.06734	0.1013
S1	raw	unnormalised	0.002469	-0.005279	0.007748	0.001501	0.01472	0.002	1343	74	0.06954	0.08148
S1	standardized	normalised, standardized	-0.03135	-0.1206	0.08923	-0.003036	0.1868	0.028	1343	74	0.06734	0.1013
S8	raw	normalised	-0.0005587	0.005006	-0.005565	-0.01609	0.004317	0.1665	3147	256	0.0657	0.1352
S8	raw	unnormalised	0.001071	0.004519	-0.003448	-0.009231	0.002358	0.182	3147	256	0.04414	0.09504
S8	standardized	normalised, standardized	-0.008504	0.03702	-0.04552	-0.1321	0.04536	0.1645	3147	256	0.0657	0.1352
S9	raw	normalised	-0.006799	-0.006393	-0.0004057	-0.004547	0.005034	0.832	955	61	0.06992	0.0821
S9	raw	unnormalised	0.002198	0.003655	-0.001457	-0.004821	0.001062	0.1595	955	61	0.01364	0.02309
S9	standardized	normalised, standardized	-0.09724	-0.07787	-0.01937	-0.07091	0.05229	0.4905	955	61	0.06992	0.0821

Supplementary Table S11. Flexibility by distance shell

Mean relative fluctuation and its within-clade slope on growth temperature, by distance shell. The profile is strongly structured while the slopes are null throughout.

shell	shell_i	mean_rel_flex	slope	ci_lo	ci_hi	p_perm	n	clades
0-5	0	0.4173	-0.0002892	-0.003042	0.002483	0.8065	10666	1485
5-8	1	0.4339	-0.001777	-0.004671	0.001148	0.1645	10674	1485
8-11	2	0.4572	-0.0006134	-0.003988	0.002855	0.6585	10671	1485
11-14	3	0.4893	-0.001184	-0.004896	0.00248	0.4505	10674	1485
14-18	4	0.5341	-0.001777	-0.005733	0.001935	0.279	10674	1485
18-24	5	0.6108	-0.002436	-0.00665	0.002038	0.197	10674	1485
24-32	6	0.823	-0.0009794	-0.008517	0.00597	0.768	10674	1485
>32	7	2.44	-0.001247	-0.01873	0.01641	0.876	10571	1481

Supplementary Table S12. Pocket versus control split by domain

The analysis of Supplementary Table S1 repeated within bacteria and within archaea, with an interaction test permuting clade-to-domain assignments. Rows labelled Bacteria_vs_Archaea carry the difference of differences and its permutation p.

descriptor	split	group	slope_pocket	slope_control	difference	difference_lo	difference_hi	p_perm	n	clades
resvol	domain	Bacteria	0.01131	0.248	-0.2367	-0.3581	-0.1139	0	10046	1404
resvol	domain	Archaea	0.06945	0.1325	-0.06303	-0.6575	0.5782	0.8415	628	88
resvol	domain	Bacteria_vs_Archaea	-	-	-0.1737	-	-	0.5335	10674	1485
f_polar	domain	Bacteria	-0.0001679	-0.004308	0.004141	0.002139	0.006147	0	10046	1404
f_polar	domain	Archaea	-0.002542	-0.0002574	-0.002285	-0.008201	0.002886	0.5065	628	88
f_polar	domain	Bacteria_vs_Archaea	-	-	0.006425	-	-	0.1085	10674	1485
f_pro	domain	Bacteria	6.87e-05	0.001108	-0.001039	-0.001748	-0.0003323	0	10046	1404
f_pro	domain	Archaea	-0.0005045	0.001309	-0.001813	-0.004163	0.000694	0.1415	628	88
f_pro	domain	Bacteria_vs_Archaea	-	-	0.0007742	-	-	0.617	10674	1485
f_ivywrel	domain	Bacteria	0.0009906	0.006031	-0.005041	-0.006677	-0.003343	0	10046	1404
f_ivywrel	domain	Archaea	-0.0006617	0.001023	-0.001684	-0.00915	0.005929	0.679	628	88
f_ivywrel	domain	Bacteria_vs_Archaea	-	-	-0.003356	-	-	0.354	10674	1485
f_neg	domain	Bacteria	4.83e-06	0.001571	-0.001566	-0.002556	-0.000554	0	10046	1404
f_neg	domain	Archaea	-0.0002267	-0.002079	0.001852	-0.001459	0.005339	0.3805	628	88
f_neg	domain	Bacteria_vs_Archaea	-	-	-0.003418	-	-	0.1185	10674	1485
hydropathy	domain	Bacteria	-0.003203	-0.004844	0.001641	-0.007981	0.01105	0.718	10046	1404
hydropathy	domain	Archaea	0.003425	0.007624	-0.004199	-0.0376	0.03105	0.8195	628	88
hydropathy	domain	Bacteria_vs_Archaea	-	-	0.00584	-	-	0.7935	10674	1485

Supplementary Table S13. Independent test-set performance

Performance of the ESM-2 ridge regressor on a partition of 30%-identity clusters used in neither model fitting nor penalty selection, against amino-acid composition and a training-mean baseline under the identical protocol. Reported as the journal requires in place of cross-validation averages.

features	partition	n	r2	pearson_r	spearman_rho	mae	rmse	alpha
ESM-2 650M	held-out test	8738	0.4826	0.7002	0.4133	3.794	5.721	3511
AA composition	held-out test	8738	0.02881	0.1849	0.1239	4.894	7.838	1e+04
training mean	held-out test	8738	-0.001018	-	-	4.754	7.957	-
ESM-2 650M	held-out test / S8	3851	0.5628	0.7644	0.4935	4.219	6.573	-
ESM-2 650M	held-out test / S9	4080	0.2744	0.5278	0.3664	3.518	4.991	-
ESM-2 650M	held-out test / S1	807	0.0544	0.3077	0.2078	3.163	4.714	-

Supplementary Table S14. Composition of the training and held-out partitions

Size, cluster count, family breakdown and growth-temperature distribution of the two partitions, showing they are closely matched and share no sequence cluster.

partition	proteins	clusters_30pct	clades_50pct	S8	S9	S1	thermophiles	pct_thermophile	ogt_median	ogt_min	ogt_max
training	36370	2652	8992	15589	17291	3490	1350	3.7	30	4	105
held-out test	8738	665	2256	3851	4080	807	321	3.7	29	5	106